



Topic 123

Classroom Response System

Here we see a picture of this classroom response system. Concerning our earlier problem of change in seasons the students are given a few related questions such as: what happens in summer and winters; why is the sun brighter at midday in comparison to the evening? You see them answering these questions, but observe their happiness and excitement. This is a direct outcome of them being part of an interesting activity and the students not just passive listeners. The students come to the forefront, while the teacher recedes to the background. A junior instructor provides them with two minutes instructions for the first time after which the process becomes automated. The computer marks the answers and the results are instantaneously uploaded to the teacher; providing a mechanism to gauge their prior understanding/knowledge.

As you see in the picture, each child, at first, attempts these questions on their own, without consulting others. Later they are allowed to consult with each other; another interesting and exciting learning experience for the students. The students are allowed to walkabout, argue, at times use the internet and even make calls. More importantly, the underlying commonality among the students is that they are thinking; analyzing and critiquing their own answers and not being lectured. This is the beginning of the learning process. You should also observe that no student is gossiping or wasting their time. People repeatedly point out that classroom discipline is essential. Yet having attempted this collaborative arrangement numerous times, I can attest that nothing happens. The students are there to learn. Turning to an actual example of this the first question the students are asked is why the sun is not as bright at 7 pm in the evening in June, if there is no pollution in the air. After their initial answer, the students are allowed to consult and you see a clear convergence towards one answer based upon argumentation, analysis and

collaboration, collective convergence. We asked another question and you can see that the computer immediately generates a list of all questions for each student. This technology costs little. Infact, in a couple of years the only cell phone available in the market will be smart-phones. So the solution will require little intensive technology.

Circuit Switched

It is a type of network connection which establishes a continuous electrical connection between calling and called users for their exclusive use until the connection is released. Ericsson PBX is a circuit-switched network.

Clickable Image

This may be defined as any image that has instructions embedded in it so that clicking on it initiates some kind of action or result. On a web page, a clickable image is any image that has a URL embedded in it.

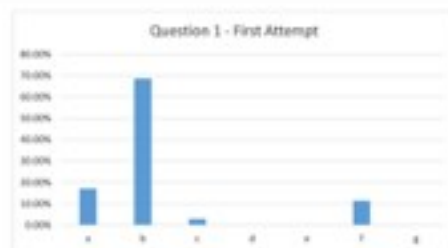




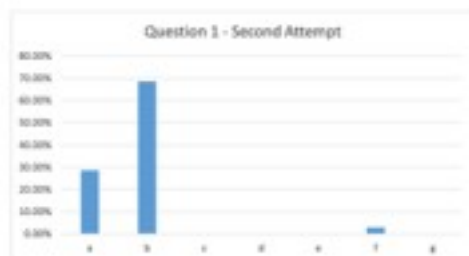
Class Room Response System Testing - Thursday, March 10, 2016, 11:20am in R006 - Course: OOP (A)

Question 1 - First Attempt

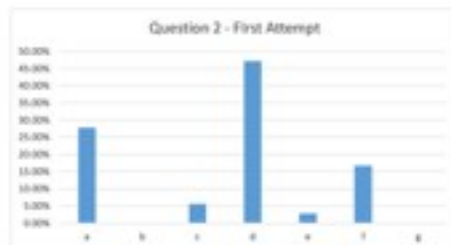
Question: While driving on a bridge at about 6:30pm in the summer of Lahore in a cloud less and almost dust free environment you can see the sun which is not as bright as at the time of noon? Why?			
Key	Model response	Credit	Count
a	The distance of the sun has become larger at 6:30pm than at noon?	0.00%	6
b	Because the sun rays are coming at an angle at 6:30pm while they are coming straight at your head at noon? This is the only reason.	0.00%	24
c	Because there are air particles and more dust at 6:30pm?	0.00%	1
d	Because the pollution has increased at 6:30pm.	0.00%	0
e	Because of the magnetic properties?	0.00%	0
f	None of the options	100.00%	4
g	(No response)	0.00%	0



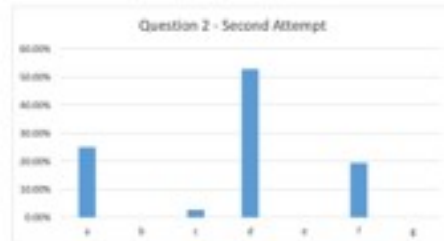
Question 1 - Second Attempt				
Question: While driving on a bridge at about 6:30pm in the summer of Lahore in a cloudless and almost dust free environment you can see the sun which is not as bright as at the time of noon? Why?				
Key	Model response	Credit	Count	Percentage
a	the distance of the sun has become larger at 6:30pm than at noon?	0.00%	10	24.57%
b	because the sun rays are coming at an angle at 6:30pm while they are coming straight at your head at noon? This is the only reason.	0.00%	26	68.57%
c	because there are air particles and more dust at 6:30pm?	0.00%	0	0.00%
d	because the pollution has increased at 6:30pm	0.00%	0	0.00%
e	because of the magnetic properties?	0.00%	0	0.00%
f	none of the options	100.00%	1	2.86%
g	[No response]	0.00%	0	0.00%



Question 2 - First Attempt				
Question: Why there are seasons in Lahore - we have summer in June and winter in January - now the spring is coming? Why?				
Key	Model response	Credit	Count	Percentage
a	Because in winter the earth goes further away while in summer it comes closer to the sun?	0.00%	10	27.78%
b	Because there is the earth magnetic field?	0.00%	0	0.00%
c	Because the earth is surrounded by air and pollution and they increase in winter as compared to summer?	0.00%	2	5.56%
d	Because the rays of light hit the earth with an angle in winter and this is the only reason?	0.00%	17	47.22%
e	Because the sun becomes brighter in summer?	0.00%	1	2.78%
f	none of the options	100.00%	6	16.67%
g	[No response]	0.00%	0	0.00%



Question 2 - Second Attempt				
Question: Why there are seasons in Iqbal - we have summer in June and winter in January - now the spring is coming? Why?				
Key	Model response	Credit	Count	Percentage
a	because in winter the earth goes farther away while in summer it comes closer to the sun?	0.00%	0	0.00%
b	because there is the earth magnetic field?	0.00%	0	0.00%
c	because the earth is surrounded by air and pollution and they increase in winter as compared to summer?	0.00%	1	2.78%
d	because the rays of light hit the earth with an angle in winter and this is the only reason?	0.00%	10	52.78%
e	because the sun becomes brighter in summer?	0.00%	0	0.00%
f	none of the options	100.00%	7	19.44%
g	[No response]	0.00%	0	0.00%



First name	Surname	Question 1 (10 marks)		Question 2 (10 marks)		Question 3 (10 marks)		Total Marks
		First Attempt	Second Attempt	First Attempt	Second Attempt	First Attempt	Second Attempt	
U1F15BCME0011	HASSAN TARIQ	0	0	0	0	0	10	10
U1F15BCME0013	ZULQARNAIN HAIDER	0	0	0	0	10	10	10
U1F15BCME0001	MAHAM SAIED	0	10	10	10	10	10	30
U1F15BCME0002	IQRA MUSTAFA	10	10	0	10	0	0	20
U1F15BCME0004	NOSHEEN GADDER	0	10	0	10	0	0	20
U1F15BCME0005	ABDUL FAHAD FAHEEM	0	0	0	0	10	10	10
U1F15BCME0006	SHAHID MAQSOOD	10	10	10	0	0	10	20
U1F15BCME0009	SHAHRIYAR ASIF	0	10	0	10	0	10	30
U1F15BCME0010	WAFIA JUNAED JALAL	10	10	10	10	0	10	30
U1F15BCME0011	BILAL SHAH	0	0	0	10	0	0	10
U1F15BCME0014	ABDULLAH BUTAHER	0	0	0	0	0	10	10
U1F15BCME0016	ADNAN ARSHAD	10	10	10	10	0	10	30
U1F15BCME0017	MOHAMMAD ABDUL MOEED	0	10	0	10	0	10	30
U1F15BCME0018	SHAHRIYAR NASEEM	10	10	0	10	10	10	30
U1F15BCME0019	ARIYAS MALIK	0	10	0	0	10	10	20
U1F15BCME0020	AKIAD KHALID	0	0	0	10	0	10	20
U1F15BCME0021	MOHAMMAD MAROOF SUBHANI KHAN	10	10	10	10	10	10	30
U1F15BCME0022	KHADJA ARIF	0	10	0	10	10	10	30
U1F15BCME0023	UMME ABRAHMA	0	10	10	10	10	10	30
U1F15BCME0024	BRAXHEM BILAL	10	10	0	10	0	10	30
U1F15BCME0025	FAROHAM NISAN	0	0	0	0	10	10	10
U1F15BCME0026	FAYYAM AHMED KHAN	10	10	0	0	10	10	20
U1F15BCME0027	ALI AHMED	10	10	0	0	10	10	20
U1F15BCME0028	PARVA FARWAD	0	0	0	0	0	10	10
Average		3.75	6.67	2.5	5.83	4.58	8.75	21.25

Topic 124

Outside the Classroom

Now that we have discussed the inside of the classroom, we attempt at taking this technology outside the classroom. We took a class excursion to a remote rural area away from the university. The purpose was to determine if we could provide quality education in this setting. By quality, world class education, we refer to these constitute elements: the availability of instant feedback; awareness of a student's foundation knowledge; their misconceptions, individualized learning; and quality teacher-student communications – all elements which can be addressed through technology. It was decided to commence the classroom in the bus. This entailed setting up of an intranet connection between the teacher's laptop and student android tablets, no electricity or an internet connection is required. Compare the cost of acquiring land and building for constructing a school to a bus, which can be rented at a cost of Rs.25, 000 to Rs.20, 000 for a day. Other factors which need to be considered are; the inordinate amount of time students spent on commuting; endless homework and academies. Could not this time be used more productively?

In comparison look at the class conducted in a field; the student's happiness at being out of the restricting environment of a formal classroom needs to be noted. Here a proper teaching with the inclusion of instant feedback is transpiring; a thing which unfortunately not present in most elite institutions in the country. Despite the fact that each child in all probability posses a smart-phone and have access to the latest computer labs. Throughout the bus journey the learning process is not interrupted; leads us to the conclusion that a classroom can be developed anywhere in the wilderness or on a bus.

Extending this logic can we not have a fleet of bus which can target underprivileged segments of the society? The buses can go to the specified localities and park there. Students, most of whom also work, depending upon the time available can go to the bus, where a tablet is mounted on each seat, log in with their own account, provided to them earlier and upload their homework and download their new assignment. Through technology they will be provided what mistakes and misconceptions they faced also their individualized speed and ability taken into consideration. Naturally, we are assuming that these students do not possess a smart-phone. Those that do can come into the vicinity of the bus and repeat the same process. Here we are talking about a revolutionary concept: a child not going to school.

When a child attends school there are numerous difficulties arise: commuting and associated costs; fixed timing, copies and books and similar requirements. Instead of children going to schools; the school comes to the child. This mitigates the costs and difficulties associated with going to school. A child's daily progress report is also sent (sms-ed) to their parents or guardians. An entirely new paradigm for that can also be applied to existing schools. Also, helpful in assessing if actual learning is transpiring in schools; innovation, knowledge and understanding of new concepts and not confined to just rote learning. There are issues with our paradigm.

Client/Server Relationship

A client application is one that resides on a user's computer, but sends requests to a remote system to execute a designated procedure using arguments supplied by the user. The computer that initiates the request is the client and the computer responding to the request is the server. Many network services follow a client and server protocol.

Clipboard

This is an area used to temporarily store cut or copied information. The Clipboard can store text, graphics, objects, and other data. The Clipboard contents are erased when new information is placed on the Clipboard or when the computer is shut down.



Topic 125

A Multiple Choice Quiz

In context of earlier question of why there is a change in seasons in Lahore; most students who had some prior knowledge of the problem, answered that in the winters the sun rays fell at an angle on the earth and in the summers directly. Before moving on we have to understand the flaw with a multiple choice format if we want to utilize it. In the second attempt by students, where they were allowed the use of the internet, there was greater convergence towards the same answer. Incidentally, this was also the answer given on the internet. There were still some outliers but the majority agreed with the answer. Is the cause of concern or celebration? But what is the right answer? Is the fact that the sun falls at an angle sufficient? Technically, this is the correct answer as provided in books and the internet. But this is only the answer the What – what about the Why, How and Where. Perhaps those students, who did not answer correctly, did not feel this was intuitively the correct answer. “Conventional wisdom tells us that the learners are converging at the right answer, and here lies the root cause of this very serious misconception.”

It is precisely due to this that our question appears so abstruse. At times students provide answers which are available in textbooks but do not relay know the why part. “Multiple choice questions create and reinforce misconceptions sometimes. They may pick the right choice and it may propagate in class during discussion but they may not know why, how, when and where. They may simply seem to know what the correct answer is according to books, internet, etc.” What we did in case of our class was invite groups of two who had answered correctly to the faculty office and asked them to explain their answer. They replied they had read it somewhere but were unable to explain. We also brought up the earlier concept of a person observing the sun

at midday and evening and the difference in brightness. The common answer to which was also that the sun was striking at a different angles. We employed this as counter argument through asking them to remove the earth from the picture. The pertinent point is that despite the convergence upon the right answer the learning misconception persisted.

Clipping

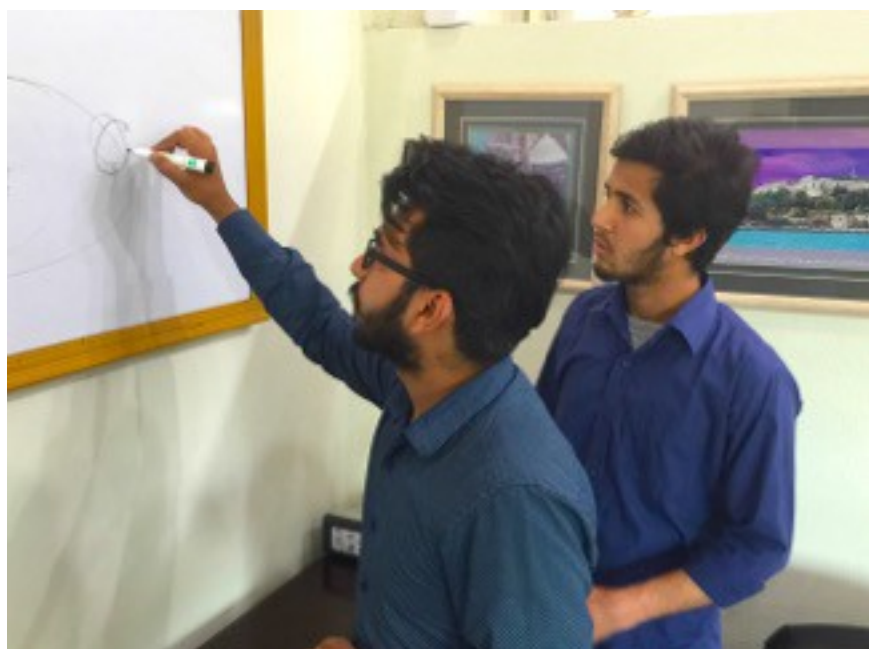
This is the conversion of all tones lighter than a specified grey level to white, or darker than a specified grey level to black, causing loss of detail. This also applies to individual channels in a color image.

CMS

It is color management system. This ensures color uniformity across input and output devices so that final printed results match originals. The characteristics or profiles of devices are normally established by reference to standard color targets.

Learning strategies fostered by the use of ICTs

Research into the types of teaching and learning practices fostered by ICT use (most especially more 'learner centric' pedagogical strategies) and how such pedagogies affect disadvantaged students would also be quite useful.



Topic 126

Elementary Probability 1

We tried another example in class; an elementary mathematical problem of probability. Unfortunately, this is subject which has been traditionally based upon undue formulation. I remember this was how it was taught to us too and this leads to the absence of its real essence and no real intuitive understanding. We asked the students a very basic question; if a coin was tossed what would be probability of heads or tail. The unanimous response was 50%, i.e. $1/2$. But upon being asked what this implied, there was no response. This was surprising since one was not expecting this misconception. The point that probabilities would vary towards one outcome and with greater coin tosses approach towards the defined probability was not clear. On asking; what is the probability of the sun rising tomorrow, the answer was an unequivocal 100%. Technically the probability is closer to 99.99%.

We now commence with our example. There are three boxes, two containing pigeons and one a precious statue. The graphical presentation is to provide visual stimuli in grasping the problem. Next the three boxes are shuffled and we longer know which box contains what. Students are now provided with a single attempt to try locating the statue. For this example we had tried to create an appropriate classroom environment. Outside the classroom the sign say, "Come and discover." Inside the classroom the problem statement is defined as, "probability of finding the statue." In the picture you will see the girl enquiring as to what is the probability. The man's reply is; if you try a 100 times you will be able to locate the statue approximately 33 times. What does this imply? Random selection will yield you the statue 33 times from a 100 attempts and pigeons the remaining 66 times. This appears to quite in line with commonsense. Now how can we increase the probability of finding the statue? Assume the earlier man in the

picture is a magician and will lead you towards this, but which to you may appear counter intuitive.

One box is selected which cannot be opened and is put aside. The selected middle box is set aside. Of the remaining two boxes the magician opens one and a pigeon flies out. Now we are left with two boxes, one containing a pigeon and the other the statue. A choice is given at this stage, to remain with the earlier choice of the box or switching to other remaining box. The story is provided to illustrate the problem more clearly. Now what would be the probability of finding the statue if you remained with your choice or switched? Which would be higher? Intuitively the answer seems to be 50%; think about this.

CMYK

It is cyan, magenta, yellow, and black are the base colors used in printing processes. CMY are the primary colorants of the subtractive color model.

Codec

This is also called a compressor, a compression / decompression software component which translates video between its uncompressed form and the compressed form in which it is stored on media (disk, etc.). The two most commonly used video codecs for CD-ROM video are Cinepak and Indeo.



Topic 127

Elementary Probability 2

When this experiment was conducted in the classroom 70% of students on an individualized basis replied 50%. When the students were allowed discussion between them, the answer converged to 90%. Please, note here the use of the intuitive thinking process and non-reliance upon formulas. Anyway, once the students were allowed access to the internet, they discovered that their answer was incorrect. Again, we need to be careful when employing a multiple choice format for a classroom response system: there is a tendency in converging towards a misconception. The What may be understood and but not the Why. This precludes the application of this knowledge to problems of a similar nature. Suppose that we now have four boxes and one is selected and set aside. Out of the three remaining boxes, one is opened and the pigeon set free. Now what would the probability of finding the statue between the three remaining boxes?

This is a serious flaw with multiple choice format, that the answer may be correct but its understanding absent. Technology is not the cause of it; the problem was there in this format's pedagogy prior to induction of technology. What technology allows us to do is to introduce efficiency in the assessment part, which otherwise a tedious and physically impossible task. "We started using technology to get an idea of severity of student misconceptions. Technology provided instant feedback, which is essential for learning but sometimes flawed. The critics of technology would again promote the role of the instructor; but the instructor can also design and check multiple choice problems and fall in the same trap. The answer lies not in abandoning technology but by using more of it."

“What we require is more intelligent technology.” This is our challenge for the future.

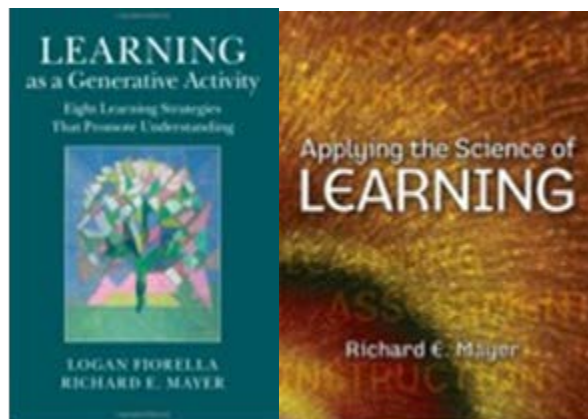
“Sometimes an answer converges towards positive results but with no justification, sometimes an answer converges towards negative results but with no justification. Why learning is not taking place and what should be done.” This leads us again in the role information technology in education. A teacher upon entering a class should be aware of the foundation knowledge of the class; upon which he/she can then build upon, according to a constructivist approach. In actuality this not only needs to be done at the beginning of the class but in the middle and its end. The wrong impression acquired by a teacher, that the majority is correct or wrong needs to be dispelled. What kind of technology is required? How can we get into the mind of a student to generate an understanding of what, why, how and where? We need to revert back to pedagogy. Two books by Richard Mayer: Learning as a generative activity; and Applying the science of learning are useful. Another book, Teaching as a design science is also important. This book is a first of its kind in terms of addressing education as a design science. Some disciplines are by necessity based upon design, i.e. civil engineering and architect. Biology was traditionally an analysis based discipline, prior to us being able to manipulate DNA and create new forms of life, but now is veering towards design. Similarly, chemistry too was a analysis science but now with creation of new composite materials and even elements this is no longer valid. Like these sciences the discipline of Teaching was traditionally looked upon through analysis: what are its problems; how learning takes place; when does it occur; when it doesn't. From analysis we are now turning towards the actual design of education, an interesting and revolutionary change.

Colorimeter

This is a light-sensitive device for measuring colors by filtering their red, green, and blue components, as in the human eye. See also spectro-photo-meter.

Color Cast

This is an overall color imbalance in an image, as if viewed through a colored filter.



Teaching as a Design Science



Building Pedagogical
Patterns for Learning and Technology

Diana Laurillard



Topic 128

Measuring Learning

We begin with a small recap. In the previous module we had discussed a class response system. This was a low cost, technological, solution for gauging targeted student progress. For instance, their foundation knowledge; understanding of a particular concept, after it has been taught to them; and as they finish with a class. The limitations of a classroom response system were also discussed; in relation to the multiple choice format and the tendency of students to converge towards an answer, whether right or wrong. This convergence takes place when students are allowed to discuss a problem in attempting to solve it based on rational logic and innovative thinking. The answer lies not in eliminating technology but the induction of more technology, intelligent technology.

If students are provided with open ended questions, as done previously; why are their days and night? Also the students are allowed to use whatever source they consider appropriate; internet, discussions with friends, their cell phones, etc., what impedes the learning process in their arriving at the right conclusion. We need to understand this process, for deployment of effective and efficient technology in education. The resources book we will utilize, development of the classroom response system, and peer instruction will follow in the next module.

COM1, COM2, etc. -

Most serial ports and internal modems on DOS/WIN PCs can be configured to either COM1 or COM2 in order to accommodate the situation where both may exist. The DOS MODE command is used to change the output direction to such serial devices as modems.

Compression

This may be referred as the reduction in size of an image file. See also lossy and non-lossy.

Topic 129

Peer Instruction 1

On the screen you can see an image of a book; Peer Instruction by Eric Mazur, a professor of physics at Harvard. Comments on the book at Amazon are also displayed. “If you have a class of 30, 40 or 50, you have problems in terms of participation of your students. “ A crucial obstacle in the learning process is, “If your students are not participating in your class or your lessons or if you would like to create useful discussions in class.” Another important point in terms of learning is; the teachers encouraging students in thinking through problems in class. This creates a fun environment which is conducive to learning and discovery. “By which your students can develop their conceptual understanding of physics. This book will serve as a great resource for teacher.”

Similarly, the publisher note states: “Peer Instruction is an interactive teaching style that actively involves students in the learning process by focusing attention on underlying concepts through interactive “Concept Tests”, reading quizzes, and conceptual exam questions. Results, assessed through scores on the Force Concept Inventory and final exams, show that students better understand concepts and perform better on conventional problems in this environment. It can be easily adapted to fit individual lecture styles and used with any textbook. Eric Mazur's Peer Instruction approach has been successfully field-tested in a variety of settings.” The interactivity is critical, since teaching should not be in the form of a monologue. For every ten minutes spent on teaching ten minutes should be set aside for discussion between the students and the teacher.

Computer Search Service

It is Computer Search Service (CSS), a fee-based service, offers access to more than 500 databases, primarily in the sciences and social sciences, from which the Library can create customized bibliographies.

Configuration

These may be defined as the components that make up a computer system (which model and what peripherals). This is the physical arrangement of those components (what's placed and where). It is the software settings that enable two computer components to talk to each other (as in configuring communications software to work with a modem).

Topic 130

Peer Instruction 2

Eriz Mazur in his autobiography, in which he had the conviction to address his weaknesses, had this to say: “I have been teaching an introductory physics course to science majors in Harvard since 1984.” “Until 1990, I taught a conventional course consisting of lectures and classroom demonstrations. I was generally satisfied by my teaching. My students did well on what I considered difficult problems. All seemed right and the students evaluations received were positive.” From a normal perspective, this was Harvard, and learning seemed to be taking place in a rational manner based upon critical thinking. “As far as I knew, there were not many problems in my class. In 1990 however I came across a series of articles by Hestenus and Hallom? That really opened my eyes. A strange thing that these famous professor’s eyes are opened: sadly a fact which doesn’t happen in our case. “As is well known students enter their first physics course, possessing strong beliefs and intuitions, about common physics and common physical phenomena. The notions are derived from personal experience and color student’s interpretation of material presented in the introductory course.” Based on the paper by the two students; instruction does little to change these commonsense beliefs. A popular Harvard professor was challenged by two students. “For example after a couple of months of physics instructions all students can recite Newton’s Law, and most can apply this is in numerical problems.” Again an example of formula based thinking. “A little probing, however quickly shows that many students do not understand the basic law.” The basic essence of the law! The two students provided many examples, in which students asked to compare forces exerted by different objects on one another. “For instance when asked to compare forces in a collision between a heavy truck and a light car; many students firmly believe that the heavy truck

exerts a larger force.” “On reading this my reaction was not my students”, professor Mazur stated. But the seeds of doubt were planted in his head.

Contone (CT)

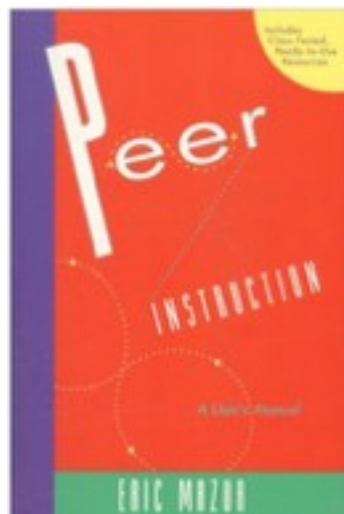
It is an abbreviation for continuous tone. It is a color or grey scale image format capable of illustrating continuously varying tonal ranges, as opposed to line art.

Cookies

It is a file sent to a web browser by a web server that is used to record one's activities.

Emotional, psychological and cultural impacts of ICT on Communities

The impact of ICT use on learners may be most pronounced not on student achievement, but rather on a learner's sense of self and cultural identity.



Topic 131

Peer Instruction 3

I am quoting an interesting story here, “Intrigued, I decided to test my student’s conceptual understanding, as well as that of physics majors at Harvard.” He met with the two students. At the time he was teaching engineering students about circuits. This was rather complex since it entailed nodes, batteries, loops and load for which simultaneous equations needed to be derived. The questions at the back of the text book were all related to this. Setting this aside a simpler question was devised; a battery was connected to two bulbs, with two switches and a two loop circuit. No resistance values, which were the bulbs in fact, were provided; also no value for the battery was provided. On seeing this; the professor enquired if giving them an engineering problem or a simple circuit analysis, which could be handled by an electrician? The problem was now that if one of the switches was closed and other turned on; what impact would this have on the intensity of the bulbs, if lit and what bulb would be lit and other turned off? Eric Mazur was amused and thought this was probably a child’s play. On the question being put to class; a majority failed comprehend the problem. One student said that while being taught circuit analysis, their training entailed solving equations where resistance values were provided, here only bulbs, voltages given and determination of the flow of current was required, not if the bulb was lit or not. How do we start?

Mazur’s eyes were opened and he realized a serious issue existed. The students were just being made into calculating automations. Their thinking process stopped. After this he wrote this book, started experimenting and researching into the problem. The intuitive understanding that teachers cannot explain a lot of things, which may appear simple and commonsensical to them, despite repeating themselves was evident.

In the development of classroom response systems it was discovered that inter-peer exchange was entirely different from how the professor communicated with students. It was easier for students to explain the problem to each other. In comparison the teacher just repeated himself/herself.

Coprocessor

It is a chip designed specifically to handle a particular task, such as math calculations or displaying graphics on-screen. A coprocessor is faster at its specialized function than the main processor is, and it relieves the processor of some work. A coprocessor can reside on the motherboard or be part of an expansion card, as with an accelerator.

Coaxial Cable

It is a type of cable that contains two conductors. The center conductor is surrounded by a layer of insulation, which is then wrapped by a braided-metal conductor and an outer layer of insulation.

Topic 132

Back to the Basics

The only way out is to go back to basics – in terms of learning. Despite the fact that this course is about technology we go back to the basic question of learning: why people learn and why they do not. A new addition to the earlier discussed books, from which you see a model here: on one side of the equation is a student/learner, on the other a teacher/automated learning system. “Information/learning occurs through acquisition or instructional learning through enquiry.” This can be done through various mediums; lectures, videos, demonstrations, etc. In this interaction there is transference of knowledge. But knowledge is not sufficient. Students have to do something actively to understand, as propounded in constructivist theory. It could be in the form of a thought experiment, an experiment or some activity by hand. This leads to a modification in thinking processes (assimilation); generation of new connections, disassociating from earlier associations (integration) and the formulation of new questions. Based on these questions the student revisits the original problem. This forms a closed loop/cycle. This is an internal and individual process. Ultimately through this rational and intuitive basis he arrives at a conclusion, which may be wrong or right.

Moving ahead in the next scenario the teacher, whether a human being or intelligent technological program, is interacting with the student – the feedback loop is completed. The teacher is aware of how the student arrived at a conclusion, wrong or right. On this basis the conceptual understanding can be modified. This is entirely the function of this course, i.e. how technology can alleviate the misunderstanding in learning. Earlier we had tackled questions of difference in day and night, summer and winters as examples. Through technology a simulation can be presented, which actually illustrates the problem. From this the earlier knowledge, based

upon preconceived notions, can be altered. Here we do not say that a person is wrong. Rather new evidence can be presented which provides new insights and challenges our earlier assumptions: sending us in new directions which were earlier closed. Through this simulation, which is also fun, you can interactively play and change various variables to see the actual effects upon the results, i.e. the tilt of the earth, its orbit, etc. This allows for the integration of knowledge intuitively upon which both hemispheres of your brains agree. This innovation through which new knowledge is acquired is essential for learning. Rote learning is not done as in the past. Technology allow for the realization of the objective: one student one teacher.

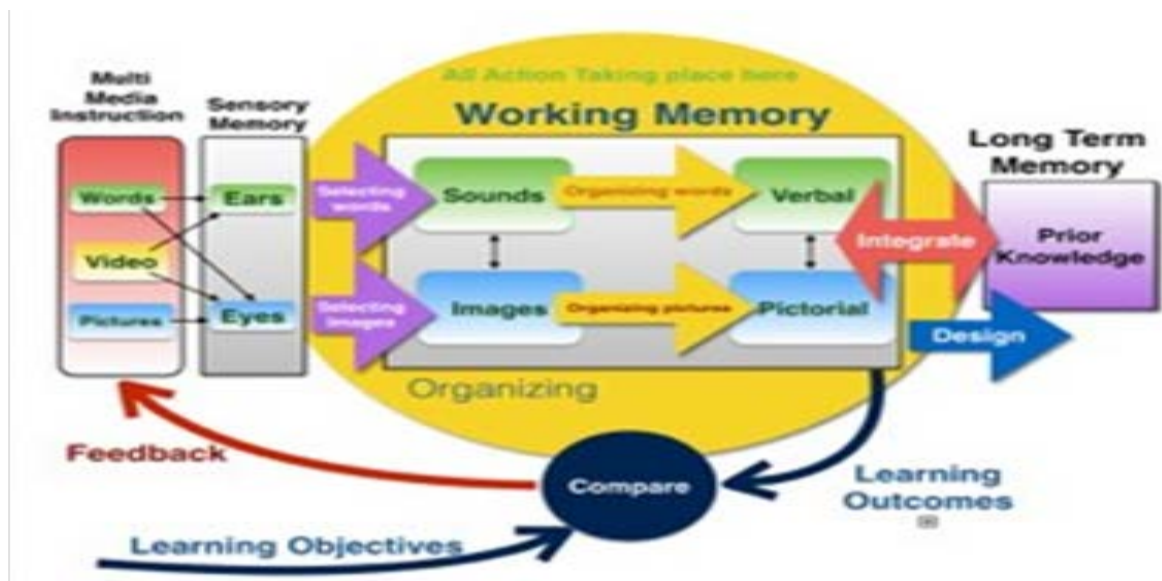
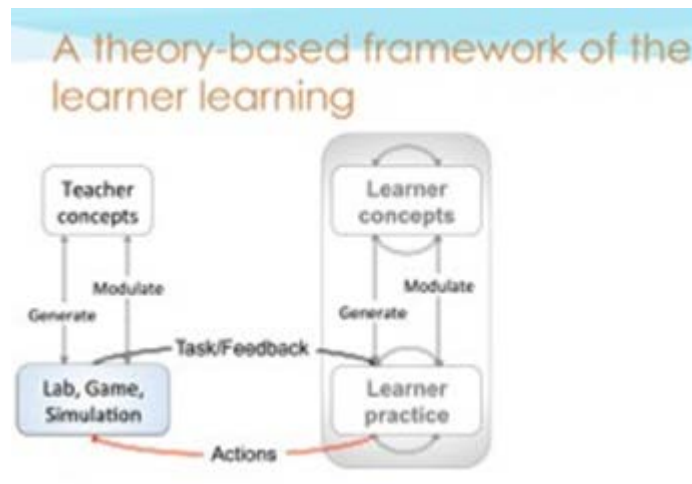
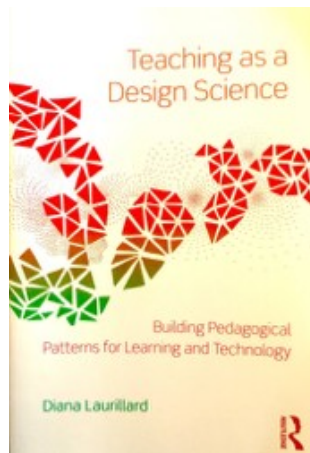
How the teacher enters the students mind and gaining an understanding of what prevents a student from learning and its rectification is critical. As with any individual no two students are alike. Each has an own unique learning style; visual, auditory, verbal, etc. These needs to be understood: the task is not daunting as appears, since for a 1,000 students there would only be 10 to 12 learning obstacles/styles. The greatest defect with conventional schooling is the assumption that one size fits all. Teachers due to the paucity of time cannot pay individualized attention to each student- whatever the institution.

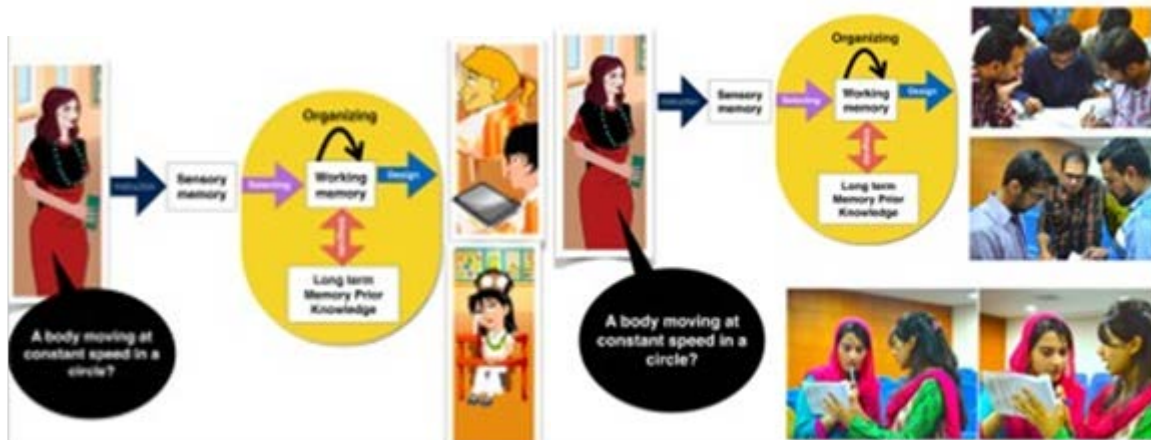
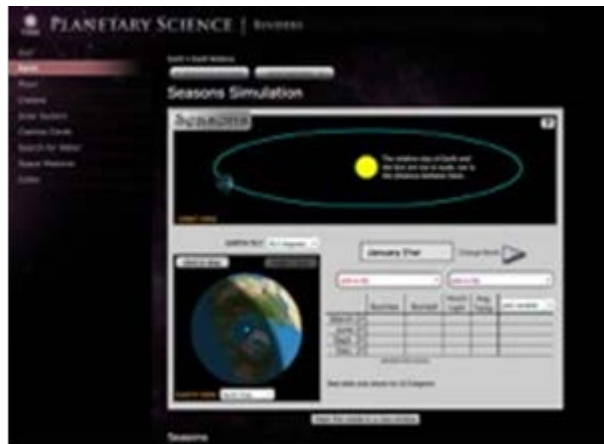
CPU

This is Central Processing Unit; the brains of the computer. The CPU interprets and executes the actual computing tasks.

Crash

It is a problem (often caused by a bug) that causes a program, or the entire operating system, to unexpectedly stop working.





Topic 133

Learning Styles 1

In the earlier picture, shown here; a student attempts to discover, question, sort out his contradictions and confusion on his own. Whether it is through books, web or simulations: in this he might be successful or not. The second model, in comparison, involved the use of an intelligent learning application, which is compatible with his unique learning style. The onus of learning is not on the student, it is the teacher's duty through the use of technology to understand a student's learning roadblocks. Importantly, the student is not contradicted rather guided and led to modify his thinking processes.

Here we present four different models from four different book; that illustrate how we can affect a child's thinking. In the first scenario a fashionable teachers poses a problem on the blackboard, concerning a body moving in a circle at constant speed. While the student's sensory memory absorbs this input, his working memory starts arriving at conclusion, in the background. The new picture painted, if based upon a rational and logical process passes to the long term memory. But this leaves large gaps and loopholes in the newly acquired knowledge. In the second model, Eric Mazur, inter-peer discussion occurs. Even in this model the teacher or a learning application; still not aware of what is transpiring in a student's mind. For instance when a girl hears the problem, what comes to her mind? The girl only picks up on the words; body, moving, and constant speed. An activity occurs in her mind in an attempt to organize this information. The conclusion reinforced and arrived at; that is subsequently passed to the long term memory is that of zero acceleration. Is this what the teacher was trying to communicate? This is a common occurrence; students during lectures miss out on numerous minor details

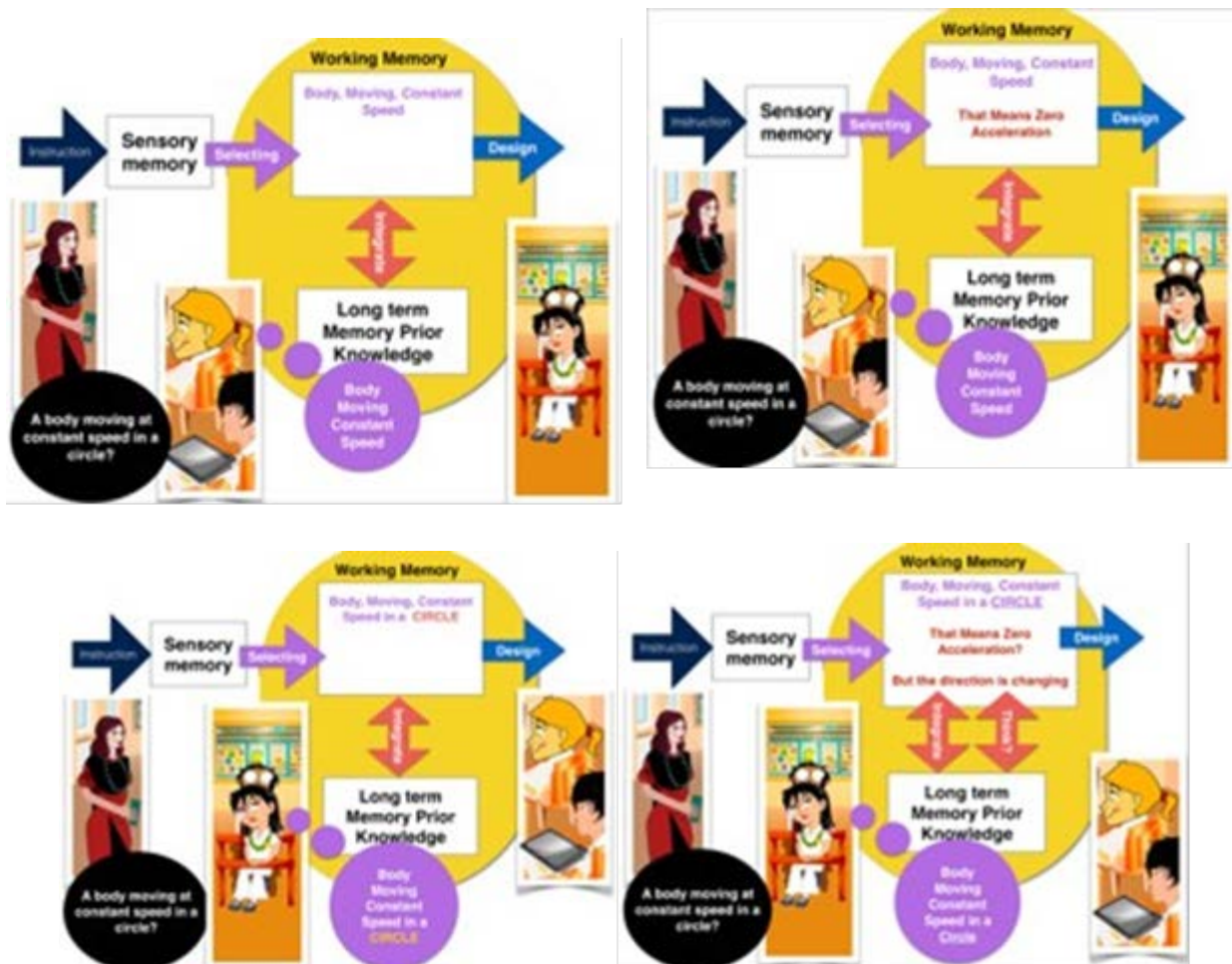
which result in major misunderstandings. Even illustrating a problem is better than pure verbal communication.

CREN

It is abbreviation of Corporation for Research and Educational Networking. This organization was formed in October 1989, when BITNET and CSNET (Computer + Science Network) were combined under one administrative authority. CSNET is no longer operational, but CREN still runs BITNET. See BITNET.

Cross Platform

It refers to software (or anything else) that will work on more than one platform (type of computer).



Topic 134

Learning Styles 2

Another student upon hearing the same problem also incorporates the word circle in his short terms memory. Here based upon his earlier experiences his thinking is modulated and a new picture painted. He too arrives at conclusion of zero acceleration, upon attempting to integrate this into his long term memory a conflict arises. His earlier knowledge, that the body is changing directions continually, creates this conflict. A learning, that a teacher or a learning application needs to highlight, i.e. remembers the body is moving in a circle. Due to this conflict in the students mind, arrives at a different conclusion. Here the answer lies in not providing an answer but repeatedly addressing the factors which create this misunderstanding. Perhaps it might be shown by an illustrative drawings or a concept map.

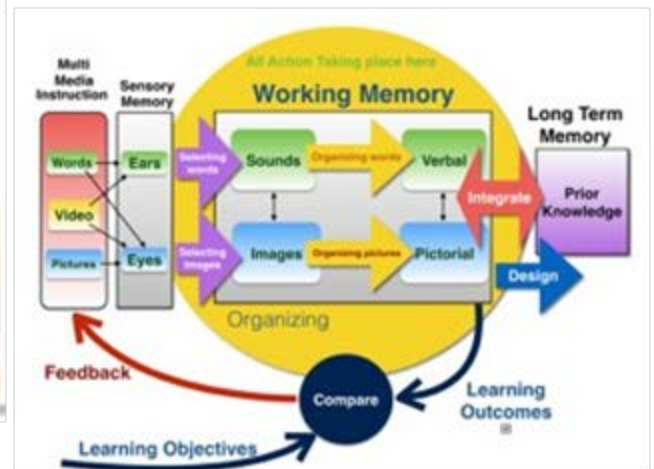
Moving to the next model, the internal processes within a student's mind, short term memory, are described in details in the boxes seen. Naturally, this has greater utility in designing a learning application. In terms of presentation sequentially a multimedia presentation would be preferred. If solely listening to a lecture some information missed. If a picture or diagram is presented it is better and finally the form of a movie would be more effective. Through these media all this sensory information; sound converted to verbal data and images, would be integrated through linkages to prior knowledge. An important consideration that in this system there is a built in feedback loop, which allows you to gauge student misunderstandings, through the use of technology. Is there a discrepancy between, learning objectives and outcomes? Once the information is passed to the long terms memory, a student's behavior would change, when tackling new problems. If the desired outcome is still not achieved, feedback is gathered and the mode of presentation of the problem altered.

CSUNET

It is California State University Network. This is a packet-switched network that connects the 22 campuses of the CSU system.

Cursor

It is the representation of the mouse on the screen. It may take many different shapes. Example: I-beam, arrow pointer, and hand.



Topic 135

Another Quiz

Now we share a quiz, which was recently taken for computer science university students. The questions asked were rather simple. The answers, while not wrong - a separate debate, were really eye opening. The entire exercise learning was experience for me. The objective of the exercise was to understand limitation and strengths of thinking, classroom response systems and peer- discussion. As mentioned earlier the Harvard Gazette, under the headline which stated: Teaching teachers what kids are not learning, it goes on to mention: “that if you graduated from Harvard do you think you would know why it is warmer in the summer than in the winter? Educators who surveyed Harvard students, on their graduation day in 1986, discovered that most of them could not answer this question correctly.” Not because the students were not intelligent or their professors. Rather it was a case of how students think. Our initial and common assumption that students do not want to learn, is totally wrong. The failure is on our part in not assisting them in learning. When you arrive at a conclusion in your mind there are repercussions, are you ready to deal with these. For instance of our earlier example; a body moving in a circle at constant speed and acceleration is zero. If the repercussion negates your understanding, it needs to be reconciled and your hypothesis rethought. We have never attempted this, to see if there is a flaw in their thinking. The process and voyage of thought not undertaken. Initially they need to be guided through this process, much like a child learns to walk holding a parents hand. Later he can walk, run and jump on his own.

The following quiz problems were given, which were conceptual in nature and linked to Newton’s laws, quite commonly known – or so assumed by people. But the litmus test is of the ability to apply these laws.

1. A car is moving in a straight line on the motorway, with the speedometer reading a constant 40km/h. No mention of velocity. Will there be any force needed or fuel consumed to keep the car moving?

Now Newton's laws contain no mention of fuel consumption also its applicability in real life is not that simple. If the speed is constant is velocity also constant? Straight line, constant speed and other variables are all churning a person's mind – working memory. An assumption would be that if the speed on the speedometer is constant then speed should also be constant. If moving in one direction then velocity should also be constant; would this imply acceleration also constant? No fuel being consumed? Would acceleration be constant or zero? if zero then no fuel consumed. It would appear so but in reality fuel is being consumed. Can these contradictions be resolved: these needs to be checked?

2. A car is turning with the speedometer reading a constant 40km/h. Would there be any fuel consumption? A situation which happens when you take an exit from the motorway, which are quite circuitous. Again the question is whether there will be acceleration or not. If you observe these problems are descriptive in nature and no calculations are required.

Now if we introduce two new variables that the road is frictionless and there is no resistance for the two earlier problems; will there be acceleration and fuel consumption? Are Newton's laws applicable for all or none; or apply in a particular case? This type of questioning generates a lot of turmoil in a student's mind. This curiosity leads him to revisit the earlier problem and compare his earlier assumptions and answers. Generation of this curiosity needs to be part of any learning applications we design.

The last question involved equations. All these university students had handled equations, very complex ones, earlier. First they were shown this equation $a^2+b^2=c^2$. What picture came to

their minds? Next they were shown $x^2+y^2=r^2$ and finally $a^2-b^2=-c^2$. Again the question was asked, as to what came to their minds? What was their long term memory communicating to their working memory? These problems are also posted on my Facebook page. As you would realize these types of problems cannot be used for classroom response systems; which is based upon a multiple choice format. The advantage of descriptive problems is that, since you have to provide a rationale, you begin to see a picture of what is in a student's mind. This can be followed up with detail meetings. The results which we will share with you were very interesting. After forty years of doing this, I can still be surprised. The requirement of technology, for determining individual thinking styles: become apparent and an imperative. The journey of these students to where they now stand needs to be analyzed.

Cyberspace

It is a term used to refer to the electronic universe of information available through the Internet.

DAT

This is abbreviation of Digital Audio Tape. It is the most common type of tape backup.

Differential impacts of ICT on education

It is well established that ICTs used in education can have differential impact on boys and girls. However, such research has neglected to explore the possible impact of ICTs on certain sub-groups of girls (and boys), and the resulting implications for impact of ICT in education on different types of girls and boys.

May 29, 1997

THE HARVARD
UNIVERSITY

Gazette

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Video Revelations

Teaching teachers what kids are not learning

By William J. Cromie

Gazette Staff

If you graduated from Harvard, do you think you would know why it is warmer in summer than in winter? Educators who surveyed Harvard students on their graduation day in 1986 discovered that most of them could not correctly answer this question.

Descriptive Problems

Assume there is a frictionless surface on the motor way and the air resistance is almost zero?

- A car is moving in a straight line on with speedometer reading constant equal to 40km/h. Will there be any force needed or fuel consumed?
- A car is turning in a circle with a speedometer reading of 40km/h? Will there be any fuel consumed?

Descriptive Problems

When you look at the formula $a^2 + b^2 = c^2$ - what picture comes in your mind?

When you look at the formula $x^2 + y^2 = r^2$ - what picture comes in your mind?

When you look at the formula $-a^2 - b^2 = -c^2$ what picture comes in your mind?

When you look at the formula $a^2 - b^2 = -c^2$ what picture comes in your mind?

Topic 136

Teaching Teachers 1

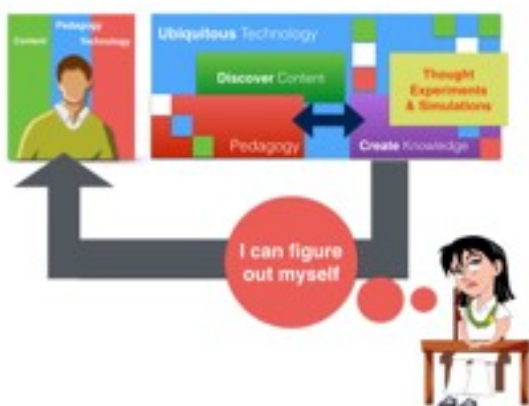
Last time we had discussed a current course which I am teaching on thinking. We had given certain descriptive questions to university students. The results were surprising to say the least. Any weaknesses were entirely due to the system and not a personal failing. We will look at each answer, to the earlier questions posed, to diagnose where learning breakdown was occurring. This analysis is essential for any educational design; whether a lecture or a learning application. The basic answers were in a yes or no format and the students allowed to discuss the problem with their peers, use the web, book, or whatever resource required. The answers not only to what but critically why is essential. These descriptive answers are useful in developing any learning application. The task is not as taxing since, as mentioned earlier, from a sample of a 1000 students the standard learning styles or common mistakes would fall under 10 to 12 categories.

Daughterboard

This is a board that attaches to (rides piggyback on) another board, such as the motherboard or an expansion card. For example, you can often add a daughter card containing additional memory to an accelerator card.

Database

It is a file created by a database manager that contains a collection of information organized into records, each of which contains labeled categories (called fields).



Topic 137

Teaching Teachers 2

You have seen this picture earlier. On being given the quiz and have gone through it, selective verbal cues/words came to the student's mind: a car; 40 km/h; constant speed. With these words a picture is formed in their working memory. Will fuel be used and force applied? Linkages are created in the student's mind based upon prior knowledge. The creation of this feedback loop is essential; without which learning cannot take place. This point needs to be retained and will greatly assist in educational design.

In the first slide a student's response is as follows: utilizing conventional wisdom arrives at the conclusion; no acceleration and consequently no fuel consumed. This appears to be contrary to conventional wisdom, i.e. fuel consumed. Also there seems to be confusion in terms of constant velocity and speed as shown by the speedometer. If no acceleration then $F=ma$ and force exerted is zero. The belief, that if no acceleration no force, is a common part of physics conventional wisdom. This point needs to be put across to the students repeatedly. In reality while driving on the motorway at a constant speed, fuel is consumed. This creates a conflict in terms of observation compared to what science taught; cognitive disequilibrium occurs. A desirable state, in terms of learning, and if it does not occur has to be induced artificially.

The student also links acceleration to the application of car brakes, i.e. no brakes are applied. Extending this logic he states that no acceleration while the car turns: turning at a constant speed hence no acceleration. This is a worrying thought for the teacher. The picture in the student's minds shows he equates acceleration with the screeching of tires and the use of brakes.

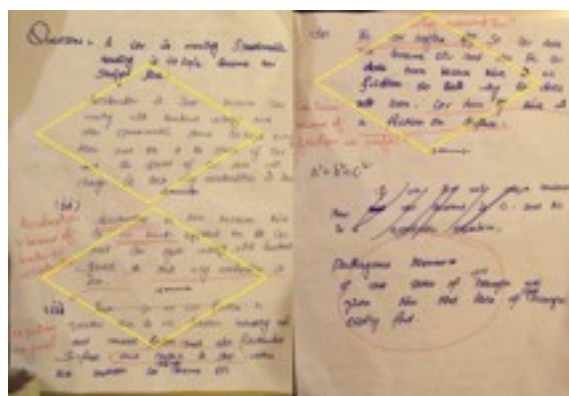
There is another issue with problem formulation; one is not able to think of certain things, that student's come up with: a fact by which their intelligence can be gauged. A point in case; when a car is turning or traveling in a straight line on a frictionless surface, it wheels should not be moving. Without friction and air resistance this is technically correct; the car would be sliding/gliding. Also, in the scenario if you tried turning the car, using the steering wheel, it would not turn. Despite the fact that the students made some basic mistakes, this point was one which personally one could not think of. From this quiz it was evident that there were flaws in how they were taught, but they are still using their brains. The fault lies with us and not them.

DCS

This is abbreviation of Desktop Color Separation. It is an image format consisting of four separate CMYK PostScript files at full resolution, together with a fifth EPS master for placement in documents.

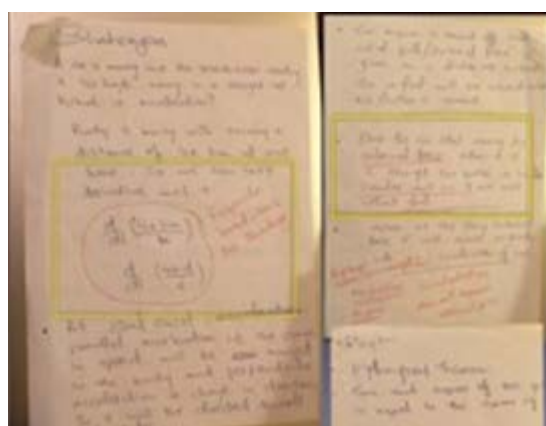
DDN

This is abbreviation of Defense Data Network. A global communications network serving the US Department of Defense composed of MILNET, other portions of the Internet, and classified networks which are not part of the Internet. The DDN is used to connect military installations and is managed by the Defense Information Systems Agency (DISA).



Slide No. 1

- Conventional wisdom - no friction - no acceleration - no fuel consumed. This is Contrary to normal observation?
- No acceleration because no brakes applied as if turning does not involve acceleration?
- Can not differentiate between speed and velocity?
- Car will not turn in a friction less surface? **Very Intelligent?**



Slide 4

- Formula based plug and chug without thinking? Formula is not being used as a tool - but the learner is becoming un necessary dependent and helpless **because the formula is not providing sense - Over Dependence on Formulas and unnecessary use of calculations? It becomes a source of confusion?**
- Car will not turn - an intelligent idea?

Topic 138

Teaching Teachers 3

In the second slide acceleration is zero because velocity is constant. This answer was apparently based on intuition, since no applicable formula used. This illustrates the lack of cognitive disequilibrium, no conflict apparent, something which might have to be induced artificially. In the case where the car was turning, the acceleration was assumed to be zero, since constant velocity. The same mistake is done as for the earlier example; perhaps confusion between speed and velocity. Here velocity is not constant, which varies, rather speed is constant. Another background factor, in his mind, could be that turning does not involve acceleration. Another mistake, in the third problem, where the car is travelling in a straight line and there is no friction or air resistance, is that there would be no opposing forces. This naturally implies no friction, no acceleration and consequently no fuel consumption. The student also states that the car will not turn on a frictionless surface, an intelligent observation. Based on the answer, we presume, that the student is thinking; on a frictionless surface where there is no air resistance, a car can travel in a straight line but cannot turn.

The point, that there are no opposing forces, as in space where a vehicle can travel in a straight line but would require force to turn, is quite startling. Discussing this with other professors they were equally surprised. Conventional wisdom dictates that upon turning, force comes into play. But what is the opposing force in turning which we need to overcome, and which cannot be seen? While friction can be seen, and requires force to overcome it.

In the third slide we find a demonstration of conventional wisdom, where a formula based interpretation is attempted without the requisite thought process. A formula only helps when a model based on rational thought is developed earlier. No formulas are required for

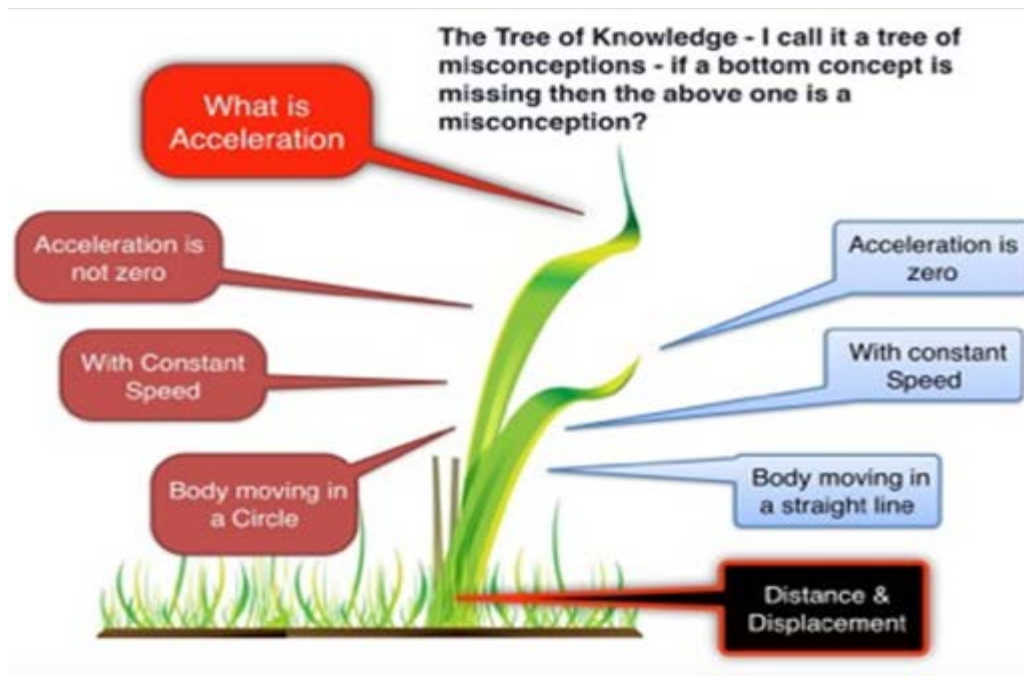
understanding the basic problem; here an inverse approach was taken. Naturally, this approach leads to confusion at a later stage. The differential arrived at, 40km/h, for calculating acceleration apparently derives from this approach. Assuming velocity at 40km/ hr, differentiating for acceleration, they arrive at $40 \cdot d/t$, which is entirely meaningless. A significant majority took this approach of hiding behind a formula. Once the car starts moving, by external forces, either in a straight line or circular it will move without consuming fuel. Even if there is a frictionless environment we need to differentiate between straight line motion and circular – a serious misconception. When a car is moving due to external force, in a straight line or a circular motion, it will not consume any fuel, since no resistant opposing force. As the earth orbits the sun, does it require any force? In space there is a vacuum and no friction or air, so no force required. Similarly, for the earth to spin on its axis, is any external force required?

DEC

This is abbreviation of Digital Equipment Corporation.

Decompression

It is the expansion of compressed image files.



Topic 139

Teaching Teachers 4

With slide four, diagrams pertaining to how learning occurs are also shown and need to be remembered as reference. Here the student again resorts to formulaic learning, but in this instance could not apply the formula correctly. He states that acceleration will be constant in a straight line, while when turning acceleration decreases. Here the formula provides him with the wrong conclusion. From his answer it is also apparent that there is no internal conflict in his mind – formula versus logic. The student was called in for an interview. He explained that he wanted to write speed instead of acceleration. His logic was based on his parents continually telling him to slow down while driving and taking turns. A fact which was ingrained in his mind and when he was noting the problem had difficulty in understanding – mental training.

A car is turning at 40km/h, without air or friction, the assumption that a car can only go in a straight line and not turn, since wheels are not moving: much like space where an object will continue in a straight line at constant speed and can only turn if force is applied – a technically correct and an intelligent observation. This is an example of formula based, plug and chug thinking; where the formula is not being used as a tool but the learner is becoming unnecessarily dependent and helpless because of over-reliance on its use. An over-dependence upon formulas and unnecessary use of calculations becomes a source of confusion. That the car will not turn is an intelligent insight. Not sure if the thought process behind this is valid since the student applies this even when a force is present.

In this new slide a case of inability to solve a contradiction. What problem of learning does this imply? Do we train them to resolve contradictions? The use of the formula $\Delta V/\Delta T$ is right, but since not aware of definition of acceleration, states that it is decreasing. But if the

formula had been used correctly he would have derived a value of zero for acceleration, which would be correct. Instead he based his answer on car turning upon intuition. Since the quiz was based on open textbook format, he could have gotten this information. Or for that matter from other students since discussion was allowed. In this case he the educational system had equipped him with the right tools but his application, a small step, was lacking.

Pedagogically he should have gotten a hint when employing the two methodologies; some dichotomy since two different answers. It is critical to try solving a problem through a multipronged approach. If there is a conflict, in terms of answers, a red flag should be raised and revisiting of the problem required in determining where the problem lies. This is a great flaw in our educational system. If we want to create learning designs, applications, lectures or blended form; training in utilizing a multipronged approach is required. Finding a contradiction is a step forward, in fact the first step in learning.

Dedicated line

It is a telephone or data line that is always available. For example, a leased telephone line can be dedicated for computer data communications. This line is not used by other computers or individuals, is available 24 hours a day, and is never disconnected.

Default Route

This is a routing table entry which is used to direct packets addressed to networks not explicitly listed in the routing table.

Topic 140

Teaching Teachers 5

Contradiction, friction, conflict, cognitive disequilibrium all lead to meaningful learning. Resolving a contradiction is ultimate means to learn. It is important for us to allow students to explore several paths towards the same problem – rote learning has no place. If this is the case then the students' needs to be provoked in finding or towards a new path, that intuitively appears correct but is in reality not. The underlying point is to force a student to appreciate and understand the rationale behind his answer.

Earlier we had discussed the case of students whose mind was so conditioned, decreasing acceleration when turning, that his working memory fails to register what is being asked. Similarly, in this another case the student assumes, if velocity is constant then acceleration too is constant. A common mistake, most students assumes that when dealing with linear values, applicable across the board. Also if constant then all values constant; based on this his assumption that while turning a car must slow down. Prior experience of actual driving overrides his understanding of the problem statement.

In the next slide we again see a case of formula plug and chug, without thinking: leading to results which are absurd. The student does not understand differentiation of a constant value. He might be aware of differentiation and can probably solve complex problems, but fails to understand what differentiating a constant implies; an approach reflective of overdependence upon formulas. In the second part concerning frictionless surface: again hindered by the use of a formula. For acceleration a formula based on Newton's law, $F=MA$ is used. The other conclusion, intuition based, derived is that no fuel would be consumed since, no opposing force. Both approaches are contradictory since if force acceleration can be calculated – a slave to

formula. As Einstein said, the greatest hindrance to his discoveries was his education. This is one of the major flaws in our education system, which impedes the thinking and understanding faculty of the students. On a personal level I have found that one has to make students de-learn for further progress.

In the next example, the student uses the formula $s = \frac{1}{2}at^2$. Since distance is not known assumes 40 km and time as one hour. This is another common mistake where the applicability of the formula is not understood. Under what conditions and case does this formula apply? Another interesting approach, which is quite prevalent, used by students, entails use of Pythagoras. The formula $a^2 + b^2 = c^2$ is used. Where a is assumed to be the base, b the height and c the hypotenuse, for a right angled triangle. The c is then replaced by r (radius) to derive an equation of the circle, concerning circular motion. The variable utilized is different, but the same approach that is utilized by many students. What they fail to understand is r constant, a cause of concern???

Moving to the next slide, eight, we see a case of using the wrong the formula but arriving at the right answer. The formula in question is $a = v/t$, i.e. of velocity is constant so is acceleration. The student fails to comprehend differential over time. An error: arising out of earlier faulty understanding in mathematics or calculus. The contradiction in his answers is apparent. The derivation of acceleration as zero is based on intuition and not the formula used. If the formulaic approach had been carried out to its conclusion, acceleration would have been finite. The anomaly between the formula and answer should have caused a red flag in his mind. A common misconception is that no fuel is used since frictionless surface. Why is that we require force to overcome friction but not for the car to turn?

Densitometer

It is a measuring instrument that registers the density of transparent or reflective materials.

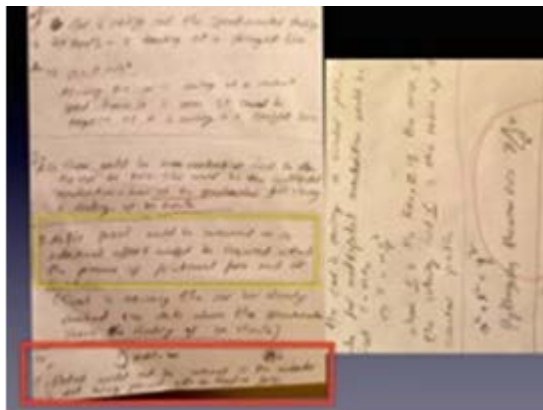
Colors are read as tonal information.

Density

Density is a brightness control to lighten or darken a printout to more closely reflect its screen appearance and to compensate for deficiencies in toner or paper quality.

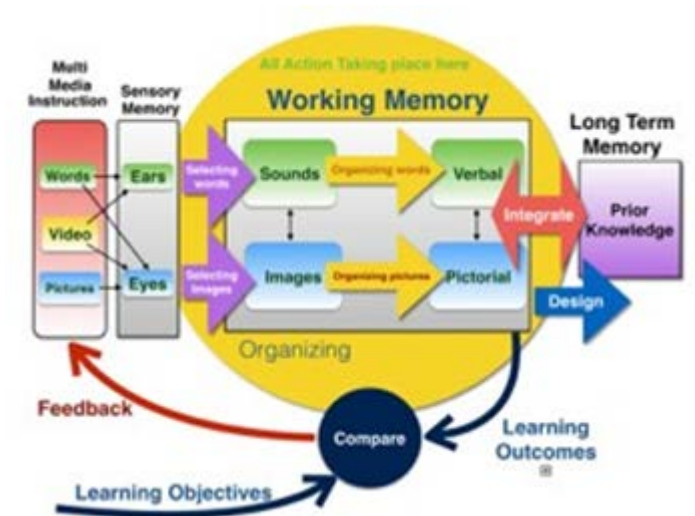
Use of ICTs to teach illiterate youth

Issues of equity are central to the discussion of the education-related Millennium Development Goals. For a variety of reasons and in a variety of ways, the introduction of ICTs to benefit education can have profound impact on issues related to Education For All. While for many ICTs offer a promise at greater inclusion of previously marginalized groups (whether marginalized by gender, disability, distance, language, culture, race, age or economic status), their use also brings with it very real dangers of increasing the marginalization of such groups inside the education system.



Slide 14

- If there is no friction then there is no acceleration?
- **A Serious Misconception? Implies that turning without friction is possible without force?**



Topic 141

Teaching Teachers 6

The assumption is understandable since if a car travelling in straight line and no friction or air resistance and hence no opposing force, why should it be different while turning. A similar problem to why earth orbits around the sun. Gravity needs to be taken into account. For students the inherent ability to think about consequence of their statements is missing. This common and prevalent misconception needs to be addressed when designing interactive learning applications. The assumption of constant velocity and hence no acceleration by the student in comparison to his real life observation to the existence of friction and air resistance shows that he is not entirely relying on what is taught. Unfortunately, whether teaching Newton's laws, kinesics, the periodic table or cell structure, no effort is made to connect this new learning to everyday experiences.

This linkage to compare and apply needs to be encouraged. If a contradiction occurs the thinking process starts. The conclusion is derived that force is applied yet no acceleration is a failure of not understanding Newton's laws. The student is not to blame and in the fact that if force, acceleration also present, he is partially correct. Fuel will be consumed at a constant velocity of 40km/hr. The fact that Newton's laws are so counterintuitive, demonstrates Newton's insight in arriving at them. In a laboratory environment it is impossible to create a frictionless surface and to negate air resistance a vacuum is required.

De-screening

It may be defined as removal of halftone dot patterns during or after scanning printed matter by defocusing the image. This avoids more patterning and color shifts during subsequent halftone reprinting.

Dialog box

It is a window that displays additional options or questions when a command is chosen.

Topic 142

Teaching Teachers 7

A most serious pedagogical problem remains the ability of conflict resolution among students. Based on the previous examples we have seen that despite the availability of several methodologies, personal intuition and real life experiences; in using these and arriving at different conclusion the students are still unable to resolve this conflict. In any learning design and applications we need to introduce this conflict, provide student with various approaches, show how these are resolved and what were the wrong answers and why. Having undergone this process the student is not only aware of the right answer but the rationale behind it.

Dial-up line

It is a communication connection from your computer to a host computer over standard phone lines. Unlike a dedicated line, you must dial the host computer in order to establish a connection. Dial-up line is currently the most popular form of Net connection for the home user.

Dichroic Mirror

This is a special type of interference filter, which reflects a specific part of the spectrum, whilst transmitting the rest. It is used in scanners to split a beam of light into RGB components.

This is where technology can be
most effective?

Conflict Resolution?

How to do it - is the
objective of this course?

Topic 143

Analyzing the Learning Process

A small interlude before today's lecture: One way in which the East differs from the West is the strength of our relationships. A popular song, the link of which is provided, is as follows:

Jidhar bhi yae dekhaye,
jidhar bhi yae jaye
Tujhe hi dhonti hai yae pagal nighanye.
Tu jo nahi hai, kuch bhi nahi hai,
Ye mana kay mehfil jawan hai, haseen hai.
Main zinda hon lakin, kahan zindagi hai
Meri zindagi to kahan kho gayi hai.
Whatever they see, wherever they turn
These troubled eyes search only for you
If you aren't present nothing is
Agreed the gathering yet young and beauteous.
That I am alive, yet what life
Wonder where my life lost

We will discuss how this linked to our lecture later. In the earlier lectures we had discussed a quiz which was given to students to assess their learning. A car is travelling in a straight line and later turning at a constant 40km/hr. There is no friction or air resistance. The question then became; would there be any fuel consumption. Earlier we looked at some of the results. The purpose of this in-depth analysis was to diagnosis the breakdown in the learning

process. In terms of our course and the use of technology, it is imperative that before its deployment, we have to understand a child's learning process. How they learn and how they fail to learn. We begin with a brief overview of the earlier results.

- I. No one tried to visualize the problem and failed to grasp the bigger picture. No one attempted to depict the problem through a graph or a concept map.
- II. Most of the student faced warning signals that their thinking was flawed. A single most common problem; yet these signals were unheeded. As is the case with aircrafts there is a warning alarm when they get too close to the ground, warning them to pull up. Failing which they plane could potentially crash. Similarly, failing to heed their internal warnings students' crash out of the educational system.
- III. If they are able to register these warning signals, their inability to resolve these flaws is absent. A good example of this relationships, linked to our earlier rendition of a popular song: That I am alive, yet what life – wonder where my life lost.

We now turn to an excerpt from Zia Mohiuddin, a famous orator. On hearing the passage, if you pay attention, you will observe numerous warning signals. A lot it you will find hard to swallow and improbable. A doctor to the insane:

“The doctor for the insane asylum was examining new inmates. One of the patients appeared to well and in his senses.” Upto this point there is nothing in the narrative which challenges your mind or creates a contradiction. “The doctor asked the patient, what the reason was for him being admitted since he appeared well. The patient replied he was healthy but recently got married.” At this point you might think this might have been the cause, but marriage can only drive you half insane. “The woman I married had an eighteen year old daughter and she too started living with us.” At this point some eyebrows maybe raised. “My father took a liking

to this girl and got married.” Things appear to be clear to this point but the next part of the story is guaranteed to create incredulity in your minds. The startled neurons firing will point to something being wrong. The same applies to studies where mostly these signals are by students, leading to breakdown in learning.

Digital

It may be defined as data or voltages consisting of discrete steps or levels, as opposed to continuously variable analog data.

Digitizers

It is a machine which converts analog data into digital data on a computer (such as a scanner digitizing pictures or text).

Initial SUMMARY

- No one tried to visualize it - have drawn a graph or picture? Of course no concept maps?
- Most of them got warnings or clues that what they are thinking is wrong - but the warning switch was turned off? **Just like the warning signal that the airplane has come too close to ground?**
- Even if they have not ignored the warning - they do not know how to handle it? **They were not trained to do it?**

پانگلوں کا ڈاکٹر

پانگل خاتون کا ڈاکٹر تھے داخلہ ہونے والی مریضوں کا معائنہ کر رہا تھا۔ ایک پانگل اُسے سنچیدہ اور بےوش مند نظر آیا۔ ڈاکٹر نے اسے پوچھا تمہیں یہاں کیوں لایا گیا ہے؟ تم تو اچھے خاتون ہو۔ پانگل نے جواب دیا۔ جناب میں صحت مند ہوں۔ ٹھیک ہوں۔ دراصل ہوا ہے کہ میں نے ایک عورت سے شادی کر لی جس کی ایک اتھارہ سال کی جوان لڑکی تھی۔ وہ بھی اپنی ماں کی ساتھ میرے گھر میں رہنے لگی۔ انتقال سے وہ لڑکی میرے باپ کو پسند آ گئی اور اُس نے اُس سے نکاح کر لیا۔ اُس دن میری بیوی میرے باپ کی ساس بن گئی۔ کچھ عرصہ کے بعد میری بیوی کی لڑکی جو میرے باپ کی بیوی تھی کے ماں آواک ہوئی۔ یہ کہ رشتہ کے لحاظ سے میرا بھائی ہوا کیوں کہ وہ میرے باپ کا لڑکا ہے۔ مگر اُسے میری بیوی کا نواسہ بھی ہوا۔ گویا میں نے اپنی بھائی کا ناتا بن گیا۔ کچھ دنوں بعد میری بیوی کے ماں نے بھی بچہ ہوا اُس دن سے میرے باپ کی بیوی بھائی کی بن ہو گئی مگر وہ اُس کی داری بھی ہو گئی۔ کیوں کہ وہ اُس کی شوہر یعنی میرے باپ کا پوتا تھا اسی طرح میرا بچہ اپنی داری کا بھائی بن گیا۔ اب لڑا سوچئے! ڈاکٹر صاحب میری سونپلی ماں یعنی میری بیوی کی لڑکی میری بیٹی بن گئی ہو گئی تو میرا بیٹا میرا ماموں بن گیا اور میں اُس کا بھائی بن گیا۔ جبکہ میں خود اُس کا ناتا بھی ہوں اور میرے باپ کا لڑکا جو میری بیوی کی لڑکی کا بیٹا ہے وہ میرا بھائی بھی ہے اور میرا نواسہ بھی۔ ڈاکٹر نے کہا خدا کے واسطے بندہ گرو ورنہ میں پانگل ہو جاؤں گا۔

Topic 144

Family Ties

Since that day my wife became my father's mother in-law" – surprising since not a normal relationship. "After sometime my wife's daughter, who was also my father's daughter had a child. According to relations my brother, since my father's child but it is also my wife's grandson. This was confusing since also my brother's maternal uncle." Try to understand this strand difficult without the help of a pedagogical aid is impossible. So here it gets even more complex. "In a few days my wife too had a son. From that day my father's wife became a sister to step-brother, but also a paternal grandmother – confusing. Because of her husband, my father's grandson, my son, became a brother to his parental grandmother. Now understand doctors, my stepmother, my wife's daughter became my son's sister; my son became my uncle and I his nephew. Wouldn't this drive you crazy too? Since I am also his maternal grandfather, my father's son, my wife's son, is my brother and grandson. The doctor exclaimed I too will go crazy."

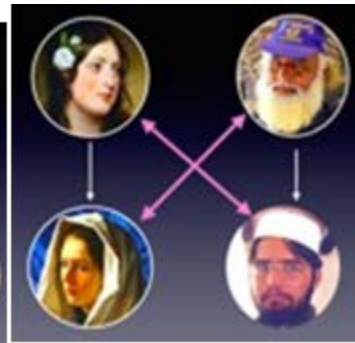
This story should provide a parallel how a teacher teaches in class. The students if not taught through learning applications or a video have difficulty in matching the lecturers speed, in terms of assimilation of knowledge. The story also provides many warning signals signifying conflict. For example to understand a maternal uncle's relationship; the sister has to be married and have a child. But a relationship like my wife my father's in-law, seems hard to fathom. Despite its complexity a visual aid, graphical representation can perhaps simplify things to an extent. The complexity of the subject otherwise would lead to great agitation, friction and conflict in the mind which cannot be resolved.

DIP switches

It is Dual Interface Poll switches allow for either an ON or OFF setting with any number of circuits. DIP switches commonly allow you to change the configuration of a circuit board to suit your particular computer.

Direct connection

This is a permanent communication connection between your computer system (either a single CPU or a LAN) and the Internet. This is also called a leased line connection because you are leasing the telephone connection from the phone company. A direct connection is in contrast to a SLIP/PPP or dial-up connection.



Topic 145

Doctor to the Mad

Here we attempt to capture the complexity of “the doctor to the mad,” through a graphical representation. Initially the relationships are quite simple. On one side is the picture of the father and the son, a natural relationship, on the other side a mother and her daughter; again a natural relationship. The man marries the divorced woman, as indicated by the arrow and who also has an eighteen year daughter from the previous marriage. The father of the man takes a liking to the daughter and they too get married. Things seem quite clear upto this point. That I and my wife become my father’s in-laws: things start becoming confusing. When the man’s father has a son, who now the man’s brother and his wife’s grandson, things become murkier. I also became a maternal grandfather to my brother. How is this possible? At this point there is friction and confusion in your mind. Continuing the man too has a son; now my father’s wife becomes a stepsister to my son and also his parental grandmother. The brain tends to start losing grip; yet the graphical representation/pictures still provide a modicum of understanding.

We now turn toward our earlier dissection of student quizzes; concerning common misconceptions, problems of learning and why these occur. The assumption that acceleration is zero while turning, with friction or without, i.e. force is not required is a common misconception. For this to be resolved there need to be conflict artificially induced in the students mind. The problem is that the students fails to observe the holistic picture and does not ask the right questions.

In another answer, acceleration is still zero, since brakes are not used. At constant speed there is no acceleration: again a widespread confusion that turning requires no force. Technology can be utilized to artificially create a conflict in an attempt to surmount this misunderstanding.

With the aid of technology we will develop scenarios which would help in this. Why is this problem widespread; having diagnosed the problem we now need to look at its cure. Through many animations we can provide a signal to their brain that there is something inherently wrong with the answer.

In another slide the plug and chug approach is seen; the wrong formula is used despite being an open textbook exam and discussions allowed. The remaining, more difficult problems are correct despite of not knowing the correct formula for acceleration. If enough educational research was done and these problems identified, these problems would not be so prevalent. The most widespread misconceptions are: circular motion without friction requires no acceleration; association of frictionless surface with no acceleration; and linked to this, no friction no opposing force, even in circular motion. Pedagogical theories tell us that if 10,000 children undergo an exam, there will be around ten or so common mistakes. If we are able to capture these mistakes and create the requisite conflict and have students resolve it, we can tackle the problems of learning. The students are not that far off since they cannot visualize force or intuitively justify it. Especially, relevant in case where turning, since there is no opposing force. There confusion is genuine, for example if a body is moving in space at constant speed no force is required since no gravity. Only if you try to stop or provide acceleration to the body will force be required. “You have not seen it or experienced it, especially in power steered cars: although the car has a tendency to come out of the circle and move in a tangent to it.”

From the descriptive problems provided in the earlier discussed quiz; most people aware of formulas and also possess common sense, yet they fail to understand the problem or try to visualize the big picture. In terms of technology we have to ensure understanding: what inputs and outputs; appropriate methodologies from different angles and intuitive understanding.

Direct-To-Plate

It is direct exposure of image data onto printing plates, without the intermediate use of film.

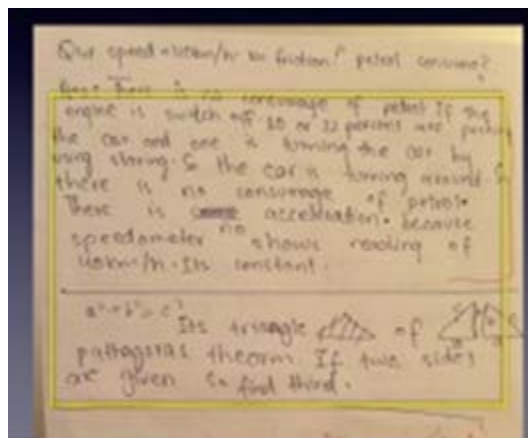
Direct-To-Press

This is elimination of intermediate film and printing plates by the direct transfer of image data to printing cylinders in the press.

Successful and relevant strategies for using ICTs

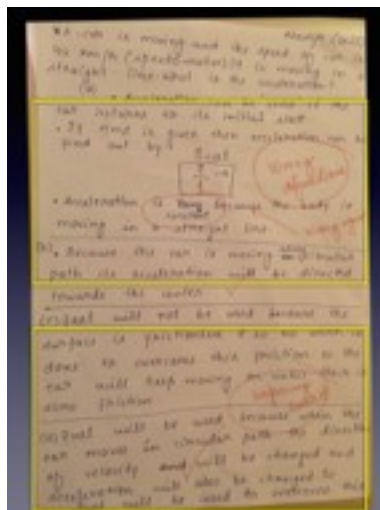
The development of models and case studies on successful strategies for using ICTs to change pedagogical practices would be useful. Especially useful would be case studies and examples of so-called "multi-channel learning" practices, which focus on enriching the environment by engaging the resources that are available to help effect incremental change by coordinating the various ways to connect learners with information, knowledge, and stimulation, and to mediate those interactions. Such practices may be especially relevant in countries seeking to utilize ICTs to help meet education-related MDGs.

The extent to which teacher training is currently being done with ICTs to meet these challenges is not well documented, especially at the in-service level. It is widely believed that interactive radio may be a cost-effective way of providing regular outreach and support to teachers, especially as it relates to (a) teacher subject knowledge and (b) teacher pedagogical practices. In many countries, radio networks exist that serve non-education sectors (for example, in agriculture or community development), but that could be used to benefit education. In addition, where interactive radio is being used, many of them are part of pilot projects, and it is unclear to what extent such initiatives can be mainstreamed and sustained by the education system after the pilot project is completed.



Slide 17

- People are turning the car - no acceleration?
- A widespread confusion - turning does not require force? **Why this is so wide spread? Must be a solid reason behind it? Diagnose the disease and then we can cure it?**
- No Acceleration: speedometer shows constant reading?



Slide 19

- Formula plug and chug - wrong formula used in spite of the fact it was open book and open discussion?
- The rest of the more difficult problems are correct in spite of not knowing the formula of acceleration?



$$a^2 + b^2 = c^2$$

- This remind us the Pythagorus theorem?
- This becomes equal of a circle when $c=r$ because then r would be the radius of the circle?
- C can not be the radius of the circle?

Topic 146

Considering Feedback

We again start with the same statement; a car moving in a straight line at a constant speed with the speedometer showing 40km/hr. This time we highlight the constant speed part. What picture now comes to your mind? Turning towards another book, we see a comprehensive model with the addition of feedback. By feedback here we refer to a result, which you have arrived and seeing if it fits in with picture you have developed. Does it make any sense? Here is another picture containing feedback. Either the feedback loop is distorted in the students, switched off to not account for conflicting warning signals or not taught in terms of conflict resolution. The harmonizing of intuition, observation and formulas is a key necessity and which needs to be taught. Another picture and another example of the feedback being blocked; in any organization there are measures of efficiency. Feedback from consumers is taken for each product and modifications made. Without this consumers would stop purchasing the product and the organization would go out of business.

Summarizing our earlier quiz findings; we see failure of visualizing the bigger picture and warning signals not heeded. If in terms of technology, if we are able to capture the common mistakes and the reasoning behind them understood we would be optimally placed to design the right learning aid. Also, these categorized mistakes needs to be discerned from a large sample to cover the maximum possible misconceptions. A golden rule for initial homework for learning application design, blended learning design or in a teacher centric classroom: if we know the mistakes we can deliberately design to provoke the students in making these mistakes. This allows them to discover on their own, where they went wrong. Once having undergone this process they will never repeat these mistakes again.

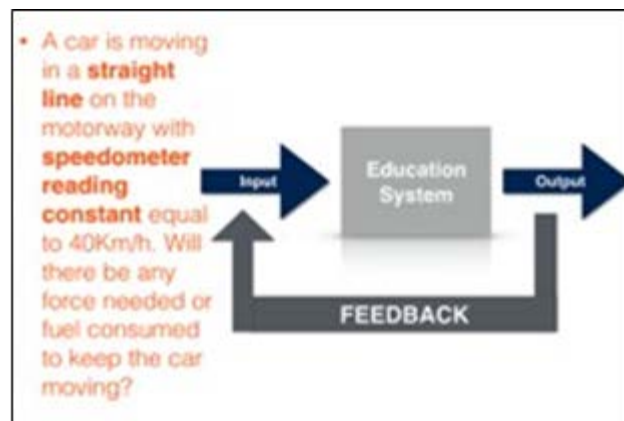
Agreed that you would design these applications separately to cater to various disparate disciplines; but once done you would not have to update them on a daily basis. Perhaps an annual revision may be required to cater to the possibility of a new mistake arising, for which the application should have provision to incorporate the new mistake, i.e. self-correcting. With technology this is all possible.

Directory

It is a system that your computer uses to organize files on the basis of specific information.

Disk Defragmenter

This arranges the blocks of information for a file into adjacent blocks on your hard drive, which may significantly improve the file access times.





Topic 147

Why, When & Where

In the last module we had discussed the collection of large sample data to cover all types of mistakes. This information provides us with the basic tools required for learning. There is more; despite the fact that we may understand a misconception, i.e. force and turning in Physics, do we know why? This is the basic requirement; what, why, when and where. “If the number of mistakes committed by a 1000 students is less than a dozen, then why they commit these mistakes would perhaps be even less.” If we know this our job of designing learning applications becomes simplified. “If somehow we know why they committed these mistakes or how they would have corrected these mistakes by themselves or avoided these mistakes: a more effective use of technology can be made. Knowing the correct answer is one thing, this could be by chance, luck or rote learning.” For real learning to transpire we require a definition, a major milestone in our course: Real learning is not only knowing the correct answer but also the why, how, when and where and what would be the formal, intuitive or formula driven answers and why they are the wrong answers. Can we derive a general strategy to find the right answers, and avoid the wrong ones? If we could do this we would be home. Even if our learner goes astray we should be able to guide him back. “It would also be great if the strategy is problem and content driven addresses all disciplines.”

The investigation of the slide tells us what the mistakes are, but also something more subtle. The students were not dumb but intelligent, they knew the subject but despite this they made serious mistakes. They were fully equipped to avoid these mistakes or correct them but they did not know how to use or resolve the conflict which was arising in their minds. It is also not the fault of the system. The system provided them with tools, to heed the clues, warning

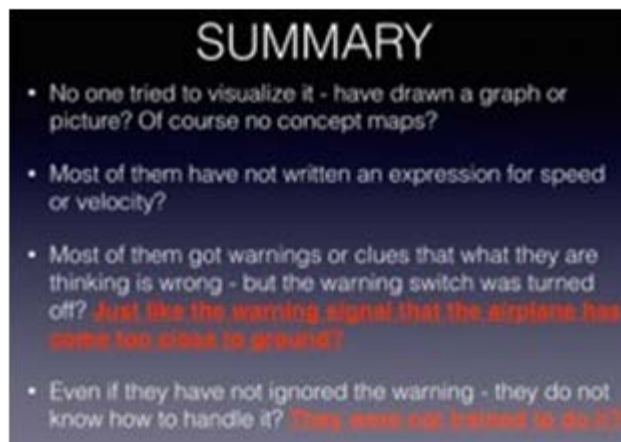
signals which should have made them correct these mistakes. Their training was missing; the ability to approach a problem from various angles, visualization of the big picture, intuitive feel and the ability to not follow formulas blindly. These signals need to be heeded; for example while driving if you see a truck in the middle of the road, it needs to be avoided, otherwise result in a serious accident. Sometimes the disequilibrium is too strong and we do not know how to arrive at equilibrium. This is a serious flaw in our educational system and is not content specific applies across the board.

DMAX

It is the point of maximum density in an image or an original.

DMIN

It is the point of minimum density in an image or an original.



Topic 148

Incorporating Error into Interactivity

An interesting thing about this flaw; if we remove teacher-centric learning with interactive discovery based applications, addressing this flaw becomes simpler. We need to teach the students how to think; where the feedback loop is strong that whenever a conflict arises, alarms go off. This missing element is perhaps one of the greatest mistakes of which I too was a part in my 40 years of teaching. If we can impart this conflict resolution training or build it into our learning applications, we can bring fresh impetus to learning. The system should not only be applicable but also recordable to cover any new misconceptions. Naturally, this does not apply non-conceptual subjects and where only memorization is required. Next we turn to past examples, earlier lectures, where the conflict arose either explicitly or implicitly.

DNS

This is called Domain Name System. A general purpose distributed, replicated, data query service. Its principal use is the lookup of host IP addresses based on host names. The host names are also known as domain names. Some important domains are: .COM (commercial), .EDU (educational), .NET (network operations), .GOV (U.S. government), and .MIL (U.S. military). Most countries also have a domain. For example, .US (United States), .UK (United Kingdom), .AU (Australia).

Domain Name Server

It may be defined as, a computer that converts host names, such as rohan.sdsu.edu to its corresponding IP Address, such as 191.130.1.10. An SDSU computer provides this service any time mail is sent or received and permits users to use TELNET and FTP between SDSU and other sites.

Categorize Mistakes

- We have categorized mistakes - we can and should do it for a large sample?
- If 1000 students solve a problem - the mistakes that they commit are not 1000? More more than a dozen? If we know those mistakes we can design an application which can provoke all students to commit those mistakes and learn from them? A golden rule for initial home work for a learning application design - in dependent - blended - or even teacher centric - which I do not like?

Topic 149

Back to School

If you remember when a student in Muslim High School in Multan; a student in a mathematics quiz, instead of proving a theorem disproved it. To be honest I was not intelligent enough to make such an intelligent mistake. But making mistakes can be beneficial. The subtle mistake by the student created a strong conflict in my mind; the reconciliation of which required great intellectual effort and engagement on my part. Through this process my understanding of that particular theorem and others became absolute. This is also a basic principle of science; you develop a theory, test it and if any contradiction go back and modify it until validated. Another vital point in conflict resolution is using collaboration to artificially induce conflict. If class has ten students and they share their mistakes the final clarity for all students much clearer. Not the correct answers but what mistakes. As it is collaboration is usually associated with problem solving, i.e. at the workplace. Creating conflict which some might refer to as confusion is quite correct. Once this is reconciled and integrated with prior knowledge the repetition of the mistake becomes unlikely. If you have never faced conflict you have not really understood.

Another example from our high school: our teacher was explaining evaporation to us. In those days water was kept in earthen jars. When water evaporates from the porous surface of these clay vessels the water is cooled through evaporation; a natural method of cooling and one that does not utilize electricity. The professor was repeating the fact that if there was more evaporation there would be more cooling. An intelligent student created a conflict at this point based upon the flaw in the statement of the professor. He said that heating the container would create more evaporation and consequently more cooling. A unique conflict; that if we place the vessels on a cooker the water would be cooled. Based on this the realization that the professor's

statement needed modification, a conflict arose. The modification needed should have stated that the heat generated should be within the vessel. This type of statement is known as conditional. If you would have picked up only part of the information, your answer would have wrong.

DOS

This is called as Disk Operating System. The operating system used on IBM personal computers and compatible machines.

Dotted Decimal Notation

It is the convention for writing 32-bit IP Addresses as a set of four 8-bit numbers written in base 10 with periods separating them.

Topic 150

Conflict Resolution

The study of 'Role of Information Technology in Education,' is currently the most popular and in-demand fields in Pakistan and globally. In Pakistan it is critical since lack of resources for construction of typical brick and mortar schools in comparison to the West. Technology allows us to create a parallel system where world class education can be provided to anyone, anytime and anywhere. The level of education which can be delivered would be far superior to existing standards. We begin today with Mullah Nasaruddin, a hugely popular Turkish and central Asian character, whose stories, humor and tales, containing great wisdom, are known to all. In this series of cartoons; in the first depiction the Mullah is arriving at a mosque, after a rather prolonged absence. The people are startled and excited: thinking what the special occasion is. The Mullah addresses the gathered crowd and enquires, if they know what he is going to narrate? The crowd expresses its ignorance: to which Mullah replies if you don't know this then there is no point in telling you.

Turning to the second cartoon, the people appear quite despondent, believing they had disappointed the Mullah. Again after a long absence he comes back and enquires of the crowd what he wants to convey. This time they all respond in the affirmative. The mullah responds if you already know the answer what use of my asking? Think about this since it is simply not a witticism. In terms of education if you do not know what you want to learn, i.e. there is no prior knowledge, nothing can be taught. Similarly, if you know it all, you cannot be taught. After a more protracted absence the Mullah returned. This time the people prepared. Upon Mullah's enquiry, half said that they knew, while the other half that they did not. So Mullah told those that knew to tell those that did not. He also said that you can solve your own problems, why do you need

me? He also illustrated the point that with the right pedagogy you can ask the right question and arrive at the right answer.

In terms of conflict resolution, when danger signals are observed and if a collaborative consensus can be arrived at; leading to further insights, that can only be arrived at through collective wisdom. A fact illustrated by the Mullah. These days this is referred to 'crowd sourcing;' something known by the Mullah hundreds of years ago.

Moving ahead with collaborative conflict resolution, we will look at some more examples. If we are able to incorporate this system, solving all the problems that arise, we arrive at true knowledge. With the right mix of pedagogy and technology the teacher becomes redundant.

Down-Sampling

It is the reduction in resolution of an image, necessitating a loss in detail.

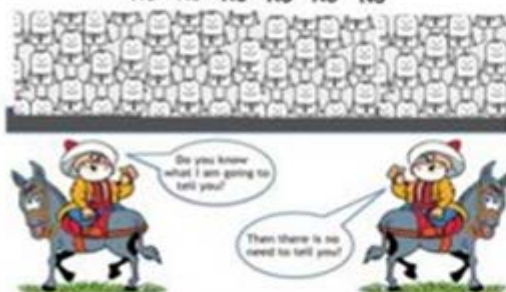
Download

It is a process to retrieve a file from another computer using a modem.

Costs/benefits of situating ICTs in schools

Cost-benefit analyses of situating computers outside of classrooms, but rather in school libraries, teacher training institutions and community tele-centres (including those that are school-based), would be quite useful.

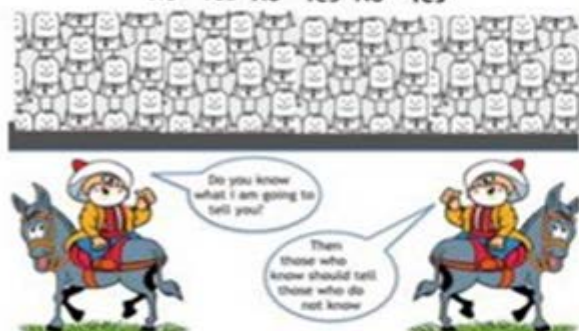
No No No No No No



Yes Yes Yes Yes Yes Yes



No Yes No Yes No Yes



Topic 151

Student Politics

A few days earlier, I met up with a colleague from the University of Engineering and Technology, after a about 20 years or so. Our talk turned to the past when we were teaching there. In those days politics was rife on campus. Student political organizations were supported by main stream political parties and some even provided arms. Complete lawlessness prevailed. Students would at time make their own and exams and conduct them. My attitude was rather strict: my stance that we should not yield to student pressure under any circumstances. Having said that if the student is wielding a Kalashnikov, one is left with little choice. I was advised by other teachers to desist since I had been warned by the students and had little children. One was torn between concern for my children and student's injustice. Human beings are built by the creator with an inherent mechanism which signals between right and wrong. Similarly, in education while solving a problem through a formula, despite the answer, your intuition at times signals that there is something wrong with the answer. These warnings should never be ignored. I too paid heed to my conciseness and disregarded student threats. These internal conflicts exist in everyone's life.

These days I am associated with a university where the inverse is true. Students are completely subjugated to the faculty. While reminiscing about old times, with my friend, and my strict stance against the student body in UET; he enquired if it had changed at all. I told him it had undergone a 360 degree turn. Now I always sided with the students, whenever there is an attempt to throttle their demands. The prevailing consensus, in the university is that, if students are leeway they will take undue advantage. My position is that they are intelligent people who are there to acquire education and their opinions should be considered. Observe how

circumstances and conflict resolution changes. Life too, in a large part, constitutes of conflict resolution. Here a poetical interlude: what do we mean by life?

Zindagi ka jawaz kiya hai	what justification for life?
Is bandage ka hisab kiya hai	what account of this devotion?
Hamaray dil may khalish hai kis ki	what agitation exists in the heart?
Hamaray dil me malal kiya hai	what regrets present in the heart?

This captures the discontent when one not at peace with oneself. Life continues in tandem with learning. One does not exist without the other. From the day a person is born to the day he dies, learning continues and with it conflict resolution.

In February 1986, I was with NASA at their research center in Langley. It was snowing, which was rare, and I was trying to get my car out to get to work. While attempting this I turned on the radio and heard the tragic news of the space shuttle Columbia exploding. Since I could not take out the car, I went inside and tried calling the office. It was the first time in my stay in the States that I did not get a dial tone. The lines were overloaded by people calling to enquire of the tragedy. Another aspect to this tragedy was that a female school teacher was also part of the crew and the launch was being broadcasted live in all schools. To investigate the incident a commission was formed and according to their findings NASA was entirely responsible. The space shuttle contained rings which distorted in the winters. There were clear instructions that the shuttle should not be launched in the winter. Junior engineers had warned the administration about this, yet their warnings were disregarded. There are many stories as to why this was done. One states that the President was conducting his state of the union address and wanted to mention the launch. But the relevant things here is that despite warning signals being present and which were ultimately disregarded, the tragedy occurred. The young school children witnessing

the tragedy required counseling. An apt lesson in not heeding warning signals. Here you see a picture of those brave astronauts.

Dpi

It is defined as Dots Per Inch. It is a measure of the resolution of a printer, scanner, or monitor. It refers to the number of dots in a one-inch line. If it will be more dots per inch, then higher the resolution is.

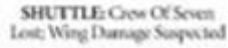
Driver

It is a piece of software that tells the computer how to operate an external device, such as a printer, hard disk, CD-ROM drive, or scanner. For instance, you can't print unless you have a printer driver. Hard disk drivers are invisible files that are loaded into memory when you start the computer, while scanner drivers are usually plug-ins accessed from within a particular application.



LAST MOMENTS: A Sonar-suit In Space And A Wizard Of Checklist Entries

the 1990s, the number of people who have been infected with HIV has increased significantly. In the United States, the number of people who have been infected with HIV has increased from about 100,000 in 1980 to over 1 million in 1995. In the United Kingdom, the number of people who have been infected with HIV has increased from about 10,000 in 1980 to over 100,000 in 1995. In the United States, the number of people who have died from AIDS has increased from about 10,000 in 1980 to over 100,000 in 1995. In the United Kingdom, the number of people who have died from AIDS has increased from about 1,000 in 1980 to over 10,000 in 1995.

[illegible]

Topic 152

Kinesthesia

Turning to another example from Kinesthesia, which was presented in various schools and colleges, a car about to move on a road is stationary. This implies a starting time of zero, distance covered zero, speed zero and consequently zero acceleration. As the distance covered is zero, differentiating for speed (speed differentiated to find distance) we get zero. What mistake am I making? With speed zero we also get zero acceleration, which implies the car cannot move. In this problem we are not provided with any numbers, rather it needs to be solved through logical thought. Any car is stationary prior to moving, but this does not imply that it will not move, since it takes you from point to another. How do we reconcile this conflict? If we are able to successfully do this we would achieve a higher level of learning.

Another conflict developed by a philosopher. A common methodology in which paradoxes provided, that can be solved with the tools available. They are designed to make you think and not used a formula based approach, which does not lead to deep and meaningful learning. Those were the days of philosophers and arrows, not bullets. In a related paradox the question was: “if you have an arrow and in order to make it cover a distance of 100 yards, what is required?” For it to cover the 100 yards, first it must cover the initial 50 yards. If covering of the first 50 essential; prior to that it should cover 25 yards, 12 and $\frac{1}{2}$ yards and so on – this leads to infinite numbers. Since you cannot cover infinite numbers in infinite time, the arrow cannot cover the 100 yards. This does not tally with our observation, which leads to generation of warning signals forcing you to think. Learning, technology, education all have their biggest and singular purpose to make you think; an objective which can only be achieved by creating a dichotomy between results and observation. This is known as a triggering event: an event which

causes you to be startled and your mind to be agitated. A triggering event creates warning signals, plays a critical role in learning. With technology, through animation we can use this very effectively.

Drum Scanner

Early drum scanners separated scans into CMYK data, recording these directly onto film held on a second rotating drum.

DTP

It is called Desktop Publishing.

Topic 153

Multan & My High School

Back to Muslim High School, Multan; it must seem that I cannot forget my school. But this place shaped me and my beliefs. If you remember there were a few examples of conflict in my school. The incident of our teacher telling us that evaporations causes water to cool in an earthen container; the faster the evaporation the greater the cooling. Extending this logic one of the student said that if we heat the container the water would get even cooler. This goes against common sense and experience. The conflict between logic and observation needs resolution for learning.

Another example was of an intelligent class fellow of mine, who managed to disapprove a theorem, through a subtle mistake rather than proving it. I was asked to check for his mistake. The exercise in conflict resolution, trying to find his mistake, provided me with mastery over the subject.

Perhaps you have noticed that collaboration is usually used in conjunction with conflict resolution and problem solving. Creating a conflict, which some people label as ‘creating confusion,’ is an essential part of learning. In a collaborative setting you encourage each individual child create unique, fun mistakes. When you come out of so many unique confusions, real learning takes place and you become a real learner.

Another example was when my mother’s pretty neighbor glanced and smiled at me while leaving her house. My wife was rather jealous and had assumed the worst. This only lasted by the time we closed the car door. The triggering event was the damage done to my car by the pretty lady. The bubble of confusion in my wife’s head burst and her mental conflict resolved.

Concept maps help in charting where confusion arising from. We will talk about them in detail later on.

Dump

It is Back up of data.

Duplex (Full, Half)

Full duplex is data is flowing in both directions at the same time. When Remote echo is ON communication is occurring in full duplex. Half Duplex has data moving in only one direction at a time (Local echo is ON).



Topic 154

Introductory Functions

We now look at introductory functions, which constitute the basics of calculus. What these are; what is required to understand them; and what conflict needs to be generated in the student's mind. In the slide in front of you five different functions – y as a function of x – are presented. If no questions arise in your mind, or emotional response evoked you do not really know functions. If no passion exists in your learning and not aware of its pitfalls no warnings will exist when attempting to solve. What pictures arise in your mind if we now we define function y as distance which a car covers in time x ? With the addition of this descriptive element, do any questions, conflicts, and agitation arise in your mind? If there is no reaction you have not studied functions properly. Today we look at how these should be taught to create a passionate association. First we shift the y value to kilometers and the x to hours? Does this make a difference? When being taught functions did your teachers, just jot them down on the board or tried to create stories around them? If you create a story the associations are tagged in your mind. When I should you the functions, if any linkage existed, these stories would come back to you. Storytelling is an effective tool in teaching, with or without technology. With technology now it is possible to create the ultimate stories through animations and movies. For their utility as essential linkages x and y should not just be symbols, instead characters of a story.

Secondly, did the teacher allow you a visualization of the problem; through drawings and graphs? Without this the pictures associated with the story would not be available to you. Our minds are structured visually to effectively and efficiently manipulate images and not equations. Easier for us to understand and analyze images. Computers are much more effective in manipulating and analyzing equations. Even when primitive human beings dwelled in caves and

relied on hunting for food, they were painting images and not developing equations. Steeping out of their caves, even the slightest rustle of leaves would point to a danger signal against which they could take defensive measures. From this example it should be apparent that how life and learning were always closely intertwined. Also how critical a role warning signals play.

An interesting dilemma with computer science students; when designing an algorithm – which entails y steps in solving a problem: when a student is confronted with $y=10$ power x , where y number of steps and x is the size of the problem, which due to its exponential nature relationship is very large. Conversely if $y=x$ squared he would be quite satisfied. And if $y=\log$ to power $x+1$, he would be overjoyed. Please, note how emotional linkages are created in relation to these problems. If these exist, future problem solving would have inherent generation of warning signals capability. A story needs to be associated with the problem. The actual impact of $y=10$ power x will be looked at later in detail. Why does it cause so much anxiety; why are PhD's made and break on this premises?

Let us try to visualize these to understand this emotional response. A small example, also available on Wikipedia, involves the number of school going population. The function $y=10$ power x or x power e (exponential) define how this population increases – exponentially. Comparatively, the population of trained instructors does not even grow linearly, let alone through a square function. This is because most governments face budgetary constraints and do not allocate this quantum of funding to education. This creates a significant gap between school going population and the availability of qualified teachers. A trend which extrapolated will lead to the demise of conventional teaching. With this you can understand how the importance of our course would increase over time. The brick and mortar schools will no longer exist and technology would be solely utilized in the delivery of education - perhaps not in our lifetimes but

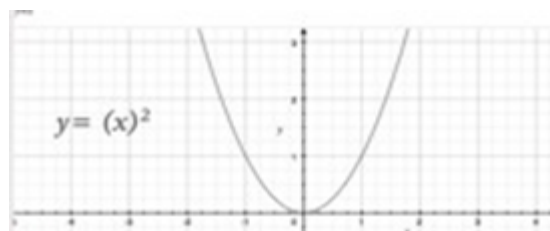
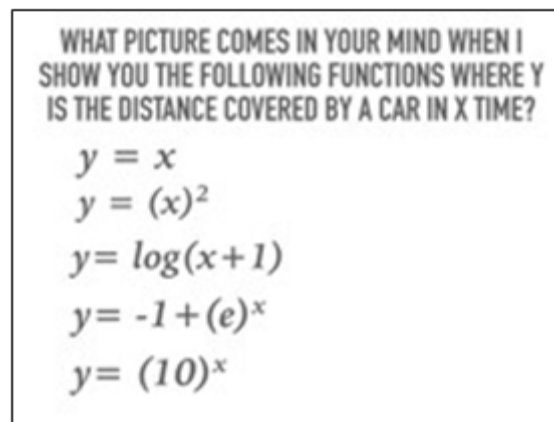
soon. With this example you can see the utility of function, linear or polynomial and how an example/story can create emotional affinity with teaching.

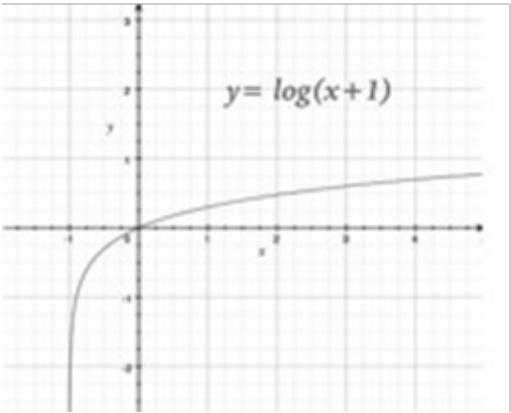
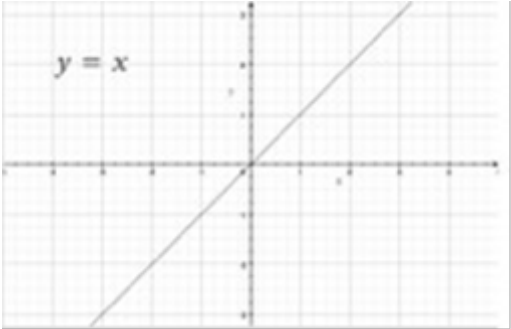
DXF

Drawing Interchange Format used for Macintosh graphic files. It is the standard file-exchange format for 3-D and CAD programs.

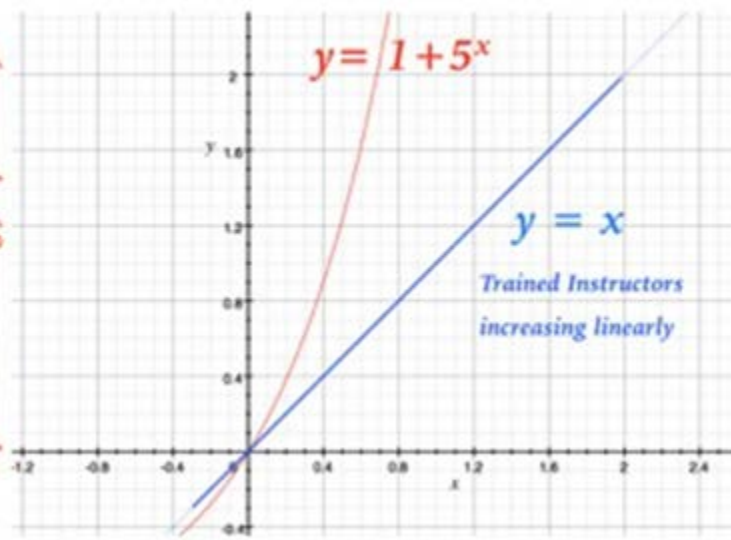
Dye Sublimation

This is a printing process using small heating elements to evaporate pigments from a carrier film, depositing these smoothly onto a substrate.





Population increasing exponentially?



Topic 155

Describing Equations 1

Here you see a face; faces are easier to remember than names and for that matter equations. These linear polynomial functions are graphically plotted. What is so critical with these polynomial factors that makes PhD students quake, when confronted with this time complexity in their algorithms? Compare the rise of these exponentials in comparison to log functions. Population growth is also exponential and can be captured quite dramatically through animation. A good question to ask students would be if they can narrate any underlying stories behind these functions. If these linkages do not work; no intuitive or real understanding would be present. In terms of layman the concept of power usually denotes strength;; the ability to do anything one wishes. How does this compare with 10 power x ? To understand its magnitude we turn towards an example.

Chicago is usually cold and during summers people take full advantage of the weather. Here we see a couple with their child picnicking. The family is photographed from 1 meter up, i.e. 10 to power 0. We move to 10 power 1 and the picnic cloth appears to be small patch. Moving up sequentially you can see the edges of Lake Michigan. In another step you see the entire lake. By the time you reach 10 power 7 you are in space and can see the entire globe. Moving up to 10 to power 15 you are out in the solar system by power 19 near the Milky Way and by 20 beyond. With this you realize the immensity of this exponential growth. Most movies only go up to power 35, where you are dealing with the universes. If this seen in a movie form you would never forget exponentials and also understand the dilemma of computer science students confronting this in their algorithms. Next the movie goes in reserve and at 10 cm you can see the details of their skins. 10 power -5, we get blood cells, at power DNA, our basic

building block. At -8, DNA strands start opening up; by -9 molecules and at -13 atoms and electrons. These number 35 and -20 are small number but here sufficient to define existence. This underlines the importance of stories and how they assist learning. Turning back to comparative graph, it hopefully makes much more sense now, in context of how school going population (exponential growth) will quickly outstrip trained teachers (linear growth).

EARN

It is abbreviation of European Academic Research Network.

Echo (Local, Remote)

It may be defined as Local echo ON causes all transmitted data to be sent to the screen of the sending computer. Remote echo ON causes everything that the remote computer (the one you are communicating with) transmits to be duplicated on your computer's screen.

Public private partnerships in cutting ‘cut costs’ for ICT

Significant work needs to be done related to the costs of ICT in education. All of the claims listed in the related “Knowledge Map” pertaining to costs found in current literature deserve additional scrutiny.

There is little credible data related to the costs of using ICTs to support education in developing countries. Few good, reliable cost studies of ICT in education implementations exist. Those that do exist measure different things, and thus are not easily comparable. While the impact of ICT interventions on student achievement may be difficult to measure, costs explicit and implicit costs of ICT-related educational activities should be measurable.

POWERS OF TEN



Topic 156

Describing Equations 2

Earlier we were discussing functions, and this slide containing five functions. The pertinent point was to see if these function elicited any emotional response. We also tried to demonstrate visualization through a problem of involving population of school going children, which was growing exponentially, in comparison to the availability of trained teacher which was growing linearly. When we observe a scene of torture or conversely a flower blooming, both evoke strong emotional responses. We need to create these linkages in education. We now turn to a game designed by us. The key elements in game should be its ability to first create conflict and surprises for the user and then maintain the level of excitement and immersion. In the light of strong emotions, a linkage is created, which leads to retention. This is the function of all games, immersive and interactive; to focus your attention in entirety and to challenge you so you are completely immersed in the experience. This needs to be replicated for learning. The reason children play with these games for hours in comparison to passive toys, is challenge involved. This game was designed in collaboration with one of my students, Sajid. The thought process, if not equally critical, is more important than implementation of the game. By this we imply all the elements within the game which constitute its challenge.

In the game, 'educational drive,' (created by one of my students with my light provokation) again graphs of functions; linear, log and exponential are shown; the point here is to refresh your prior knowledge. Squares and polynomial 1 functions are also shown, these are for reference and we will discuss them later, since part of the surprising element in the game. Next there are slides which contain definitions for displacement vectors, vectors and acceleration, vectors and velocity, etc. Now the underlying problem in the game involves a new

car, moving on a road, with the odometer showing 30km, which you have driven out of the showroom. On a straight road you increase in a manner that speed becomes 30km/h. The displacement and speed are equal. Observe that here we have not discussed any specific mathematical formulation rather we create an actual example which will be later linked to mathematics – this lead to understanding. The fact that displacement and speed are equal is mathematically interesting. As I had told you earlier understanding the question is critical, if you are able to do this you have solved 60% of the problem. In terms of the problem 30km on the odometer and 30km/h on the speedometer; if speed remains constant your odometer reading remains constant but displacement increases. If speed too increases, speed and distance increases. If speed gradually decreases and direction remains constant; speed decline but odometer value increases. Now in the problem is how we accelerate to ensure that both values, of the odometer and speedometer, remain the same? For example if speedometer value goes to 40km/h, odometer reading should also increase to 40km. The second question is that if we think in terms of function, displacement increase and is equal to speed; what does this imply? Finally, if this is done for a period of ten hours, how far would you get from say Lahore?

EBCDIC

It is abbreviation of Extended Binary-Coded Decimal Interchange Code. It is a standard character-to-number encoding used primarily by IBM computer systems.

E-mail

It is called Electronic Mail. Private messages sent between users on different computers, either over a network or via a modem connection to an on-line service or BBS.

SENTIMENTS SHOULD BE ASSOCIATED WITH EACH EQUATION
- ONLY THEN DANGER SIGNALS WILL WARN YOU?

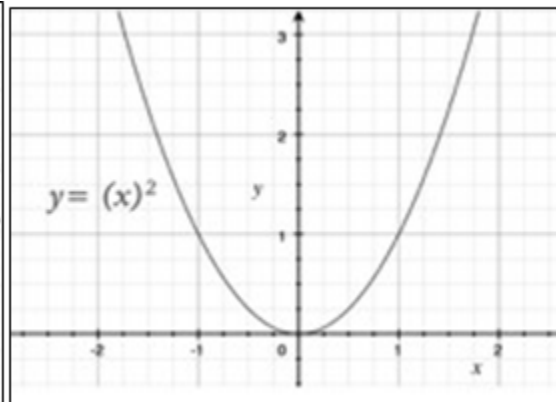
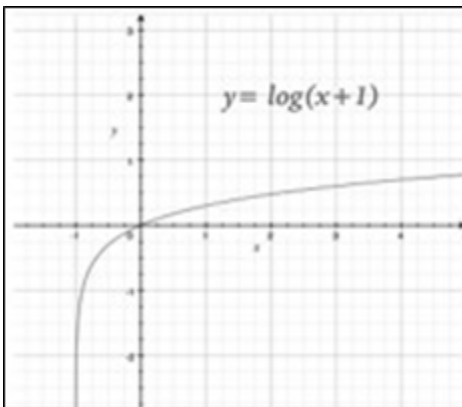
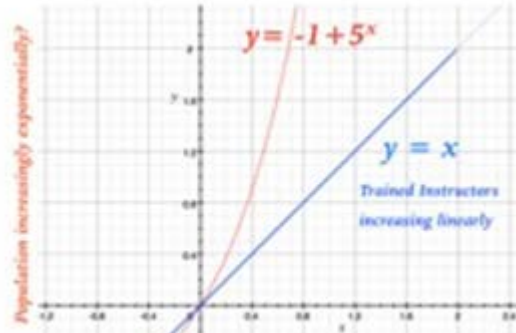
$$y = x$$

$$y = (x)^2$$

$$y = \log(x+1)$$

$$y = -1 + (e)^x$$

$$y = (10)^x$$



Introduction (Velocity & Acceleration)

$v(t) = \frac{dx}{dt}$

$a(t) = \frac{dv}{dt}$

Velocity is defined as the rate of change of displacement(x) with respect to time, and acceleration is defined as the rate of change of velocity(v) with respect to time.

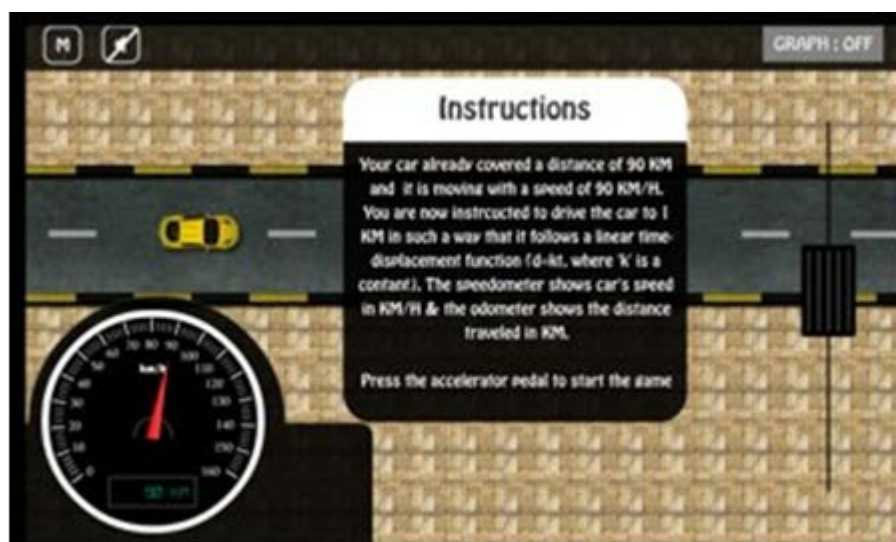
Page (3/3)

Introduction (Displacement)

A diagram showing a curved dashed path from an 'Initial' position to a 'Final' position. A straight green arrow points directly from the initial position to the final position, representing the displacement vector.

A displacement is the shortest distance from the initial to the final position. A 'displacement vector' represents the length and direction of that straight path.

Page (2/3)



Topic 157

Describing Equations 3

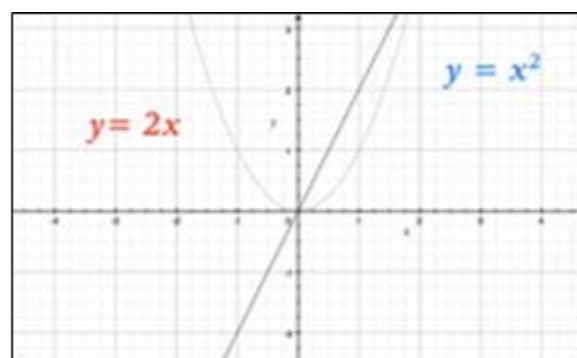
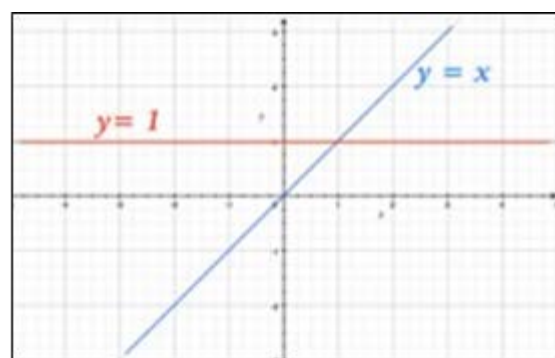
Now that the problem is in front of us, i.e. odometer and speedometer reading should be the same; implying that speed and displacement to be equal. What is the relationship between speed and displacement? Here your prior knowledge is needed. If we differentiate displacement, we should arrive at the function of speed. We are now transforming the problem into mathematical formulation: a y function is required, which differentiated with respect to time, provide the same y function. A real life problem has now been turned into a mathematical one. We need to remember which function reflective of this relationship. If this is not remembered, we will arrive at the right function.

Encoding

It is the file transfer formatting that enables encrypted, compressed or binary files to be transferred without corruption or loss of data.

Encryption

It is a way of coding information in a file or e-mail message so that if it is intercepted by a third party as it travels over a network it cannot be read.



Topic 158

Describing Equations 4

We need to try different functions to arrive at the appropriate relationship. The first function, which was previously shown, is linear. Differentiating $y=x$ ($x=\text{time}$), we can see that displacement rises with time. If we differentiate for speed we arrive at the value of 1, which is constant. As with all linear relationships the two values will never be the same. An important mathematical concept associated with functions, which you have learned.

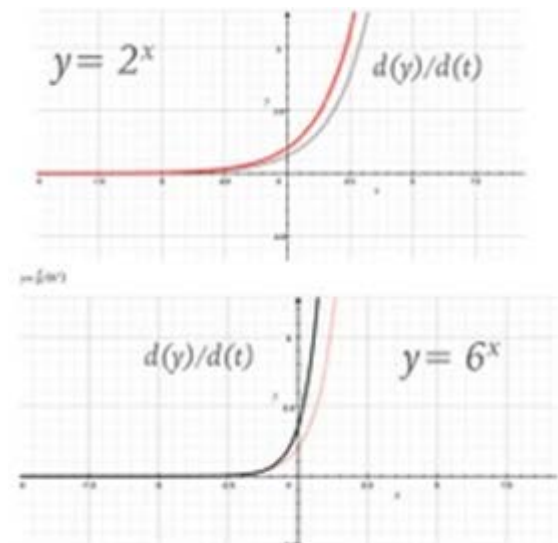
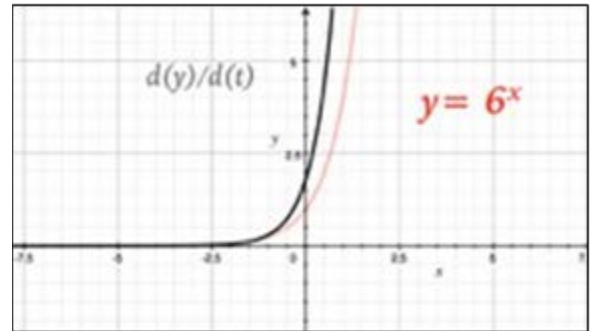
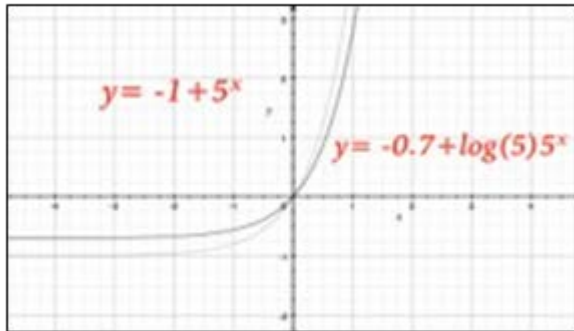
If $y=x$ squared, with respect to time; differentiating we arrive at $y=2x$, which is shown on the graph in front of you. The two values seem to approximate, yet are not equal. You can try the value of x cubed, on your own, and you will see that it is even a closer approximation but still not equal. We now turn to $y=6x$, an exponential function. As in earlier graphs the distance is plotted in black, while the function in red. Again, you see that the two are quite close, but not equal. Speed, in red, is increasing faster than the displacement in black. Time would be e to power x . E is a special number and the only one possessing property for allowing speed and displacement to be equal. Here we have introduced this special relationship, in mathematical terms, through a real life example of a car.

EPS

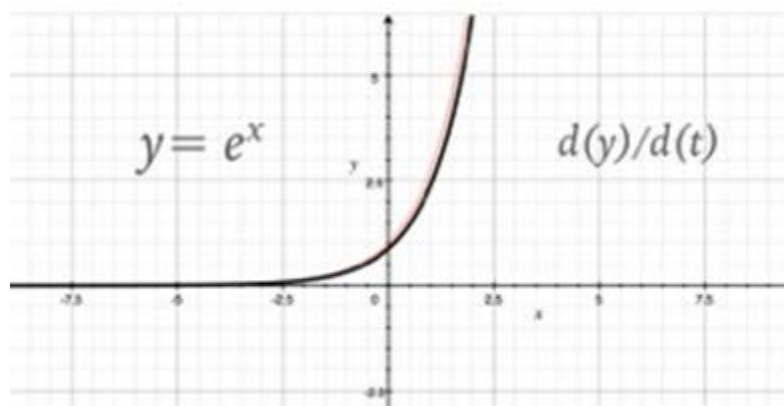
This is abbreviation of Encapsulated PostScript. An EPS file usually has two parts: a PostScript (text) description that tells a PostScript printer how to output the resolution-independent image, and (optionally) a bit-mapped PICT image for on-screen previews. (EPS files without a PICT preview is usually displayed as a gray rectangle.) EPS files generally can't be edited, even by the program that created them (Illustrator files are exceptions).

EPS 5

It is called another term used for DCS.



There is only One Such Function?



Topic 159

Describing Equations 5

Now, we are turning to game, a truly interesting one, you see three cars racing. You instructions tell you that the yellow car represents linear relationship, blue exponential and green log. You can specify how they will be plotted, i.e. displacement with respect to time. At this stage you should have acquired some familiarity with exponential, linear and log growths. In the game you now have an option to bet upon a particular car in the race, i.e. select one. If the game animation was available, you could visualize it better. Yet, here I will try to explain how it works. In the opening screen you can see two buttons. One is for turning on/off the graph plotting function. While other is of 'luck.' What does this imply?

Suppose your friend bets on an exponential functional. Most people with a background in working with algorithms would choose this option. Unfortunately, they forget a certain condition applies for this rapid growth. One the race commences you see you see your car lose. You have an option to begin again. Seeing that the green car (log function) won in the last race, you select that particular car. Again you lose. This continues to transpire, due to which your attention is drawn to the luck button. What variable does this control? As you turn the luck switch to on, your car starts winning. You are convinced by now that something is being manipulated. This is not actually so rather, a linkage and an introduction to a critical property developed through the luck function. Exponential growth is large, but only if the x value is great. Similarly, log function increases slowly if the value of x is great. In a linear relationship both variable grow linearly.

Thus, if time is reduced or distance shortened, a variable which is being manipulated. Much like a marathon runner in comparison to short distance sprinter. In the earlier instance, i.e. when luck was off, this is what was taking place. One the button is turned on, time and

distance/displacement would be increased, leading to converse results. The point of the game is to clarify many misconceptions which arise due to functions in terms of time and distance. Based on this the answer to the question of how far you would be from Lahore if speed and displacement were kept constant over ten hours: this is an exponential function and you would probably end up well outside in the Earth in space. Another point in terms of graphs, if we magnify the image we can see that the small difference which are hardly noticeable, acquire a significant magnitude. We hope this game proved useful in terms of functions.

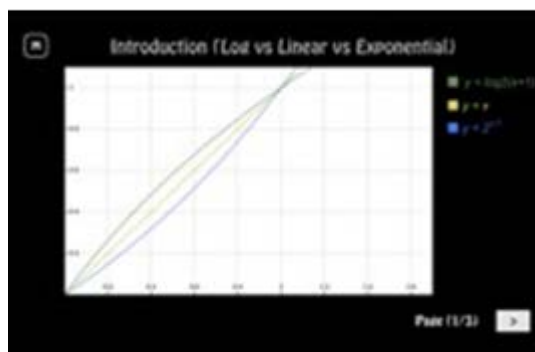
ERIC

It is an index to educational journal articles and documents collected by national clearing houses for educational information. It provides summaries of the articles and documents, and contains citations back to 1966.

Ethernet

This is an IEEE 802.3 standard data link layer which can operate over several different media including fiber optic, coaxial cable and twisted-pair cable. This 10 million-bit-per-second networking scheme is widely used on campus because it can network a wide variety of computers; it is not proprietary; and components are widely available from many commercial sources.





Topic 160

Describing Equations 6

As mentioned earlier, learning entails heeding any danger signals which arise and how this conflict is resolved; leading to new thinking patterns and new neural configuration through processing of new concept/data. As promised we will now look at problem which was presented as a midterm question to first year students of computer sciences and electrical engineering at a university. The problem was designed with artificially introduced warning signals, at multiple stages, to see if students would heed them and manage conflict resolution. The inability to do so a major hurdle in terms of learning. Also the ability to think logically and critically, very important on its own, in no way undermines this leading to good grades. If thinking is not translated into good grades, there is something lacking in the learning process. Some advice which needs to be incorporated and disseminated between students to allow them to design interactive learning applications, is the necessity of incorporating warning signals, artificially, to keep students on the right path. Much like warning signs on a dangerous road.

Incidentally, the problem which we will dissect was also given to LUMS student during their PhD entrance interviews. Here we are talking about students with sixteen years of schooling and significant number with mathematical backgrounds. We will compare the two. It has to be understood that this learning process begins in school, if not integrated at that level, would imply it not being present in college. In a certain sense it is quite simplistic, when you meet a person for the first time, how do you interpret signals from the other person, which allow you to trust him. So from that perspective it is common sense.

Now to the problem a car which is traveling in one direction in a straight line from Lahore (no vectors required). The displacement/distance is plotted against time graphically as

presented here. Assuming starting time to be 0 hours; in the first hour there is no change in displacement, i.e. car is stationary. From one to two hours linear speed; at 2 hours displacement is 200; a fact which can also be used to calculate speed. In the second and third hour, displacement remains at 200. What does it mean? In the third and fourth hour; same displacement remains as second hour. From the graph of a function equations for each stage of the journey can be derived. At y value 0, time 0; a straight line equation can be derived. Similarly, equations for 2 to 4 hours constant and displacement equal to 200. What does this mean? From fourth to fifth hour distance decreases. What does this mean? At the fifth hour 100; what story is being told here? As discussed earlier in context of the “Harvard Calculus,” a hidden story in every graph; linguistic, descriptive, visual aspect. If equations were solely provided, an analytical expression, it would be probably hard for significant majority. People think differently and require an appropriate representation which fits with their style – these needs to be accounted for. For instance this information could also be provided in a tabular format. There are basically five or six main methodologies for presentation of data. The important point is the story they are telling, which can be captured in an analytical, tabular and other frameworks.

Meeting with a group of school principals, discussed the same problem, i.e. what distances they were covering. The response was that this question was out of course. They were only teaching them how to draw graphs, not interpret or read them or understand what story they were telling. This is not learning. A student should be able to one form of expression, presentation of data, to another, even if not all: if they understand the underlying descriptive aspect.

Finally to the actual question, which is also seen here, a car leaves Lahore for Islamabad through the motorway, starting at time 0. The magnitude of displacement or distance is plotted

on a graph. Before turning to the questions, try to really listen to them; which are the real questions, which designed to induce warning signals.

Find the distance of the car from Lahore in half an hour? This only entails reading the graph, starting from time 0. This illustrates the importance of reading the question. Also find the distance at two hours. A question to which, as mentioned earlier, the principals had replied that they did not know how to read the graph. Most of you must have visited a hospital, where nurses, who are not well educated, plot your temperature graphs. Upon query by the doctor they can tell particular reading and any observable patterns. Again, distance from Lahore at four hours? These answers are already provided in the graph.

In the next questions find the distance of the speedometer at 1.5 hours and 3 hours, some manipulation is required; function needs to be differentiated. This is followed by a yes and no question; did the car reach Islamabad within four hours? The next questions is that if someone is drinking coffee at 1.5 hours, is there a possibility of them spilling it due to high speed. You might have realized that this question is designed to generate a warning signal. Similarly, if coffee being drunk at three hours, spillage due to high speeds? Finally, how far is the car away from Lahore after five hours? These constituted the basic questions which were asked.

The initial questions were solely designed to see if students understood how to read a graph. No formulas were required and all that was required was commonsense counting. At 1.5 hours and three hours some manipulation was required but if one knew one answer no manipulation was required for the second. Reading of speedometer and coffee question together you would realize that since there was no displacement; how can coffee be spilled? The car was stationary at that point, which you can see from the graph. We next turn to actual mistakes committed by student with these problems.

Executable File

It refers to a file that is a program. It is the executable in DOS and Windows usually have an .exe or a .com extension. In UNIX and Macintosh environments, executable files can have any name.

External Viewer

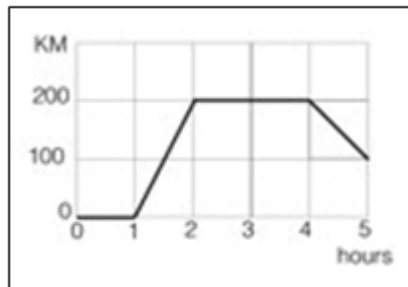
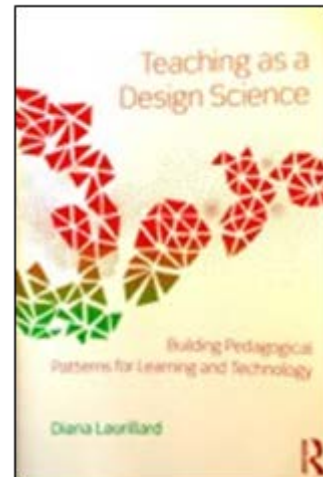
Program used for presenting graphics, audio and movies while browsing World Wide Web pages via a Web client program. Helper applications are another term for these external programs.

Components of ICT in education

If the presence, magnitude and nature of ICT investments in education in developing countries is of interest to donors, standard definitions of ICTs in education must be developed, agreed to, disseminated and utilized, and changes must be made to current coding practices of donor-supported education projects. To this end, the formation of a multi-donor task force to study and this issue and agree upon certain common definitions and coding standards could be useful. Such standards and definitions need not (initially) be comprehensive, but should not focus exclusively on hardware indicators. A comprehensive study of a small number of donor-supported education projects to identify all uses of ICTs components could be a useful tool to support the effort to articulate and agree upon common definitions and standards.

Conflict Resolution

- How technology can help?
- Game Design
- Functions - the
- game of three cars
- Other examples - PhD interviews - interpreting a graph?



Problem No. 4: A car was supposed to start from Lahore on the motor way and move towards Islamabad. It was supposed to start at 0 hour? A graph of the magnitude of displacement of the car from Lahore is plotted in the diagram as shown?

- Find the distance of the car after half an hour from Lahore - half an hour starting from 0 hour?
- Find the distance of the car from Lahore at 2 hours?
- Find the distance of the car from Lahore at 4 hours?
- Find the reading of the speedometer at 1.5 hours?
- Find the reading of the speedometer at 3 hours?
- Did the car reach Islamabad within 5 hours?
- If some one drinking coffee at 1.5 hours is there a possibility of the coffee spilling over because of high speed of the car?
- If some one drinking coffee at 3 hours is there a possibility of the coffee spilling over because of high speed of the car?
- How far away the car is from Lahore at 5 hours?

Topic 161

Educational Technology

The technology, focal area of our course, is continually evolving. Keeping up with new developments is imperative. Coming across a new book, I immediately ordered it since one felt it would have utility for you, in terms of this course. The book by J. M. Spector, *Foundations of Educational Technology: Integrative Approaches and Interdisciplinary Perspectives*, published in 2016; is one book captures these changes. In terms of educational technology, while underlying foundation definition remains the same, yet practical definitions change.

Phillip Harris, Ed.D., Executive Director of AECT, has this to say about the book: "For serious scholars, researchers, and practitioners in the field of educational technology, this is the book to own. It makes reading about the field exciting as well as informative and provides the reader with insights and understandings by one of the field's leading authorities. This may be the first book of its kind that is a true page-turner."

Similarly, Dr. Monica Tracey, Wayne State University says: "J. M. Spector's book, *Foundations of Educational Technology: Integrative Approaches and Interdisciplinary Perspectives*, alleviates the fear that teaching novices the basics in the field of educational technology must be reduced to simplified models and procedures. This textbook manages to embrace the breadth and depth of educational technology while providing understandable definitions, examples and exercises."

Despite its cost the book is well worth it. Immediately, upon opening it you notice fresh insights into education, its definitions, technology and educational technology. As you are probably aware I rely immensely upon social media, particularly Facebook. Having purchased the book, I took no time in declaring its utility to friends and associates. My only fear was that

they would ask for a photocopy of the book. In ten minutes, I was told that a copy of book had been acquired; not from Amazon but downloaded from somewhere. A good example of how technology has progressed. While it took me a week to acquire the book, others took only ten minutes – how's that for technology? It is another thing that they would not access to reader reviews and rating, if not purchased through Amazon.

The book commences with a quote; “that we shape our tools and afterwards our tools shape us” - Marshall McLuhan from Understanding Media. The same holds true for technology, which we initially create and later it defines our lives; an unending cycle. The book's new insights and perspective into education will be appreciated by the layman; but particularly by you who by now have greater familiarity with the subject.

Another quotation which I want to narrate to you from chapter nine of the book; Theories of Information Communications: “When I use a word,’ Humpty Dumpty said in rather a scornful tone, ‘it means just what I choose it to mean — neither more nor less.’” - Lewis Carroll, Through the Looking Glass. The book is as interesting, from an educational perspective, as Alice in Wonderland. Before discussing the book in detail we look at an analysis of the midterm, as promised earlier. The purpose of the exam was to see if students can heed their intuition, warning signals and the ability to engage in effective conflict resolution. We will focus on one question and its interesting outcomes.

FAQ

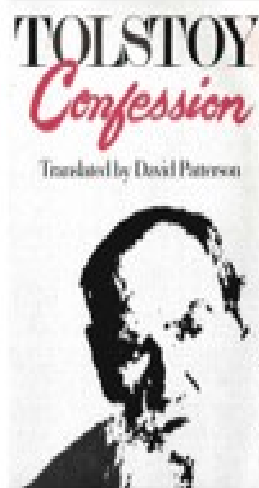
This is abbreviation of Frequently Asked Questions. It is a document that covers a topic of general concern to many users. FAQs are a good way for new users to get information on various aspects of the Internet.

FDDI

It is abbreviation of Fiber Distributed Data Interface. It is an emerging standard for network technology based on fiber optics that has been established by the American National Standards Institute (ANSI). FDDI specifies a 100-million bit per second data rate.



INTRODUCTION



IN THE FALL of 1879 the fifty-one-year-old author of *War and Peace* (1869) and *Anna Karenina* (1877) came to believe that he had accomplished nothing in life and that his life was meaningless. Either of these works would have assured him a permanent place in the annals of world literature; both testified to the depth of his genius and creativity. If artistic achievement of this magnitude cannot instill life with meaning, then where is meaning to be found? Such is the "question of life" that Tolstoy addresses in his *Confession*, a question as timeless as the spirit.

Ernest J. Simmons has described the *Confession* as "one of the noblest and most courageous utterances of man, the outpourings of a soul perplexed in the extreme by life's great problem—the relation of man to the infinite—yet executed with complete sincerity and high art." It is a tale of middle spiritual crisis, the ingredients of which had been fermenting in the man since his youth. As such, the *Confession* marks a turning point in Tolstoy's concern as an author, and after 1880 his attention was concentrated quite explicitly and almost exclusively on the religious life that he believed to be idealized in the peasant.

Although there are parallels between the torments of Levin in *Anna Karenina* and Tolstoy's own conflicts in the *Confession*,

Topic 162

War & Peace 1

From the book we will discuss an interesting story. In front of you is a picture of an immensely famous author, philosopher – Leo Tolstoy. Much has been written about him, his books have been adapted for movies and among his famous works is Anna Karenina. In terms of education, the heeding of warning signals and their reconciliation, which in some cases not possible, an insight can be found in the works of Tolstoy. He understood this pedagogical phenomenon and underwent a change through it. “The story is taken from Leo Tolstoy’s Confessions, which is based on his journal and personal reflections. Confession was written in 1879, when Tolstoy was 51 after he was well known and highly respected for War and Peace and Anna Karenina; it was first published in Russia in 1882.” I remember watching War and Peace, with my mother and aunts in Multan. At that time I was in the sixth standard in Muslim High School Multan, a reference to my alma mater is necessary. Multan was covered in movie poster and we saw the movie either at Starlight or Zenith. “Tolstoy already had all the fame and fortune, and was enjoying the benefits of his success when he happened to visit Paris, France, around the year 1865, when these events occurred.

At that time, Paris was regarded by Tolstoy and others as the intellectual center of the universe and a showcase for progress in civilized society. Tolstoy was regarded as one of the leading lights of modern civilization.” In the background you can see some pictures from War and Peace. “He had gone to Paris to meet friends and have a good time. While cavorting in Paris, he chanced upon a public execution. At the time, France applied the death penalty for serious crimes and the form of execution was the guillotine. Executions were public, perhaps based on

the belief that the horror of such executions would deter crime.” As you know we have repeatedly discussed danger signals, conflict resolution and triggering events.

Imagine the event which mentally shook the great author. “Tolstoy records in his journal something like the following (loosely translated): ‘When I saw the heads divided from the bodies and heard the sounds with which they fell separately into a box, I knew with my whole being that this was a bad thing.’ Tolstoy was shattered. ‘This unexpected and unplanned event marked a turning point in Tolstoy’s life. It is marked by the following words recorded in his journal (again the same rough translation): When I saw the heads divided from the bodies and heard the sounds with which they fell separately into a box, I knew with my whole being that this was a bad thing. (The repetition is intentional.) He goes onto reflect about his prior belief with regards to civilized, with regard to Paris as the cradle of civilization, with regard to progress proceeding from Paris throughout the rest of the world, and so on. He writes that his belief in progress included a faith that society was moving in a positive direction and that Paris was leading this movement.’ Consider here the earlier discussed prior beliefs put across by Paget in terms of pedagogical learning.

‘Then he recalls the: “When I saw the heads divided from the bodies and heard the sounds with which they fell separately into a box, I knew with my whole being that this was a bad thing. (The repetition is intentional.)’ He could not shake off the event and it kept turning in his mind. ‘He knew with whole being – he was fully engaged in this moment.’ Despite the fact this is a gruesome example you will notice many parallels with pedagogy. Here he saw, heard and thought; observed, absorbed and integrated. ‘He saw, he heard and he thought. As a result, he could no longer hold onto his prior belief in progress and faith that Paris was the cradle of civilization. He could no longer regard himself as one of the leaders of civilized society if this is

what civilized society represented.’ Again implications in terms of design of education, which need to account for prior knowledge and how it needs to be reshaped. In this instance, stark example, the entire foundation (belief system) shattered. The inhumanity displayed collapses the belief in civilization.

File

It is a collection of information on a disk, usually a document or a program, that's lumped together and called by one name.

File Permissions

When you place files on a UNIX system you can assign the files various levels of permission, specifying who can access them, and what type of access they can have.



This story is taken from Leo Tolstoy's *Confession*, which is based on his journal and personal reflections. *Confession* was written in 1879 when Tolstoy was 51 — after he was already well known and highly respected for *War and Peace* and *Anna Karenina*; it was first published in Russian in 1882. Tolstoy already had fame and fortune, and was enjoying the benefits of his success when he happened to visit Paris, France, around the year 1865, when these events occurred. At the time, Paris was regarded by Tolstoy and others as the intellectual center of the universe and a showcase for progress in a civilized society. Tolstoy was regarded as one of the leading lights of modern civilization.

In any case, Tolstoy records in his journal something like the following (loosely translated): "When I saw the heads divided from the bodies and heard the sound with which they fell separately into the box, I knew with my whole being that this was a bad thing."

This unplanned and unexpected event marked a turning point in Tolstoy's life. It is marked by the following words recorded in his journal (again the same rough translation): "When I saw the heads divided from the bodies and heard the sound with which they fell separately into the box, I knew with my whole being that this was a bad thing." (The repetition is intentional.) He goes on to reflect about his prior beliefs with regard to civilized society, with regard to Paris as the cradle of civilization, with regard to progress proceeding from Paris throughout the rest of the world, and so on. He writes that his belief in progress included a faith that society was moving in a positive direction and that Paris was leading this movement.

Then he recalls the event: "When I saw the heads divided from the bodies and heard the sound with which they fell separately into the box, I knew with my whole being that this was a bad thing." (The repetition is intentional.) He knew with his whole being — he was fully engaged in this moment. **He saw, he heard, he thought.** As a result, he could no longer hold on to his prior belief in progress and faith that Paris was the cradle of civilization. He could no longer regard himself as one of the leaders of civilized society if this was what civilized society represented.

Topic 163

War & Peace 2

We are continuing with the story of Tolstoy: “His faith in progress was shattered. He changed his life entirely based on this event. He gave away his fortune. He declared all of his writing public property. He began writing short moral tales for the common people rather than writing for the educated elite. He began advocating for the reform of peasant schools. He changed his life on account of having witnessed by chance a public execution in Paris.” This dichotomy between the educated elite in our country and their proclamation of being civilized, merit consideration, in light of this story. Another point which needs to be thought about is the Author’s changed focus on schools for the poor. Something we seriously require in this country and one of the purposes of this course; developing technologies which can provide world class education to them.

“He learned something that day in Paris. That he learned something is marked by the fact that change occurred that persisted through other events and for the remainder of his life. These changes included how he thought as well as what he did; the changes are observable in his subsequent writings and in how he lived his life after witnessing the execution. What he learned was not planned. He was not explicitly directed towards a goal nor guided a plan to achieve a goal. What he learned might be characterized as incidental to his having witnessed the execution.” This is the learning we have attempted to discuss earlier and what is required, in terms of triggering events.

“By contrast, the kinds of education primarily discussed in this volume involve intentional learning where a particular goal is involved. It is worth noticing that the goals of a learner might not always coincide with that of the instructor. In formal learning environments,

which generally involve intentional learning in school settings, it is optimal when the learner's goals and those of the instructor are closely aligned.”

What did Tolstoy learn? Do you recall the sentence from his journal that was repeated three times? Why do you think I repeated it three times?) “When I saw the heads divided from the bodies and heard the sounds with which they fell separately into a box, I knew with my whole being that this was a bad thing.” Perhaps this is difficult to say without knowing more. He learned that not all social practices in Paris were good. He learned that he could not accept a society that publicly executed criminals as a civilized society – this reflects a value perspective, which is essential in education and in the application of technology to support education, according to the framework presented in this volume. He probably learned a lot more, and that learning event might well be regarded as a process that only began that day in Paris in 1865; perhaps it began earlier.

This story provides a concrete way to think about learning – learning is marked by stable and persistent changes in attitudes, behaviors, beliefs, knowledge, mental models, and skills and so on. Learning may be planned or unintentional. Learning in which an individual is fully engaged is especially effective. Full engagement often involves perception, cognition and emotions. Technology can be especially useful in promoting active engagement.

Did I change? A question we will address in the future. What caused this and how my philosophy and thinking changed? Naturally, I am no Tolstoy, but I believe every person goes through this change, to some extent or other.

File Server

It is a computer that shares its resources, such as printers and files, with other computers on the network. An example of this is a Novell NetWare Server which shares its disk space with a workstation that does not have a disk drive of its own.

Film Recorder

It is used in reference to color transparency recording devices, and sometimes also to image setters.

What did Tolstoy learn? (Do you recall the sentence from his journal that was repeated three times? Why do you suppose I repeated it three times?) Perhaps this is difficult to say without knowing more. He learned that not all social practices in Paris were good. He learned that he could not accept a society that publicly executed criminals as a civilized society—this reflects a values perspective, which is essential in education and in the application of technology to support education, according to the framework presented in this volume. He probably learned a lot more, and that learning event might well be regarded as a process that only began that day in Paris in 1865; perhaps it began earlier.

This story provides a concrete way to think about learning—learning is marked by stable and persisting changes in attitudes, behaviors, beliefs, knowledge, mental models, skills, and so on. Learning may be planned or unintentional. Learning in which an individual is fully engaged is especially effective. Full engagement often involves perception, cognition, and emotions. Technology can be especially useful in promoting active engagement.

Topic 164

Recognizing Danger Signals

We are turning back to the midterm problem, in which warning signals were artificially induced. To create the requisite conflict to ensure a corrective mechanism in terms of the given questions. The problem was given to both, first year university students, in one college and PhD aspirants, during their interview in another famous university. The underlying rationale for the exercise was to see if they possessed the ability for conflict resolution and if they were trained to do so. If this was not the case, could technology alleviate this situation? First we turn towards the first year students. You are already familiar with the question, i.e. a car is traveling, in a straight line, from Lahore and its displacement/distance is plotted over various hours starting from time 0 on a graph.

The first question was what would be the displacement in half an hour? This was simple question which only entailed reading of data from the graph or from the table. No calculations were required and consequently little marks associated with the question. No special consideration went in the question. Despite this most students resorted to a formulaic approach. Why do they think everything is formula driven? It is commonly observed that they always turn to chug and plug approach and have stopped thinking.

In the results shown in this snapshot; the answer provided is 0 distance. While this is correct its derivation, again through a formula, is entirely wrong. The student assumes velocity to be zero and multiplies this with distance. No value for velocity was provided, still used. Similarly, in snapshots three and four you see the same mistake, velocity is unnecessarily introduced. Looking at it from a student's perspective this is how they have been trained; to look

for velocity (formula) while ignoring the obvious. A fact and prevalent practice of which, earlier, we had no inkling.

Again a correct answers, but cannot read the graph. Another formulaic approach but the wrong formulas used: reflective of either the inability to read graphs or a lack of understanding. The same pattern, was repeated, despite the fact it was an open book problem. Most students knew the formulas, or at least memorized, but did not understand this approach was not required. A problem which was probably acquired in schools and now an inherent part of their thinking process: a fact which makes our task harder since have to make them unlearn this. Currently, in a course on thinking which I am teaching this is what we are trying to teach. Whenever faced with problem in college, job or for that matter real life; first look at the inputs provided and then what outputs required. If these equate there is no need to do anything. If not that we have to understand the relationship between the two.

As teachers we thought, or rather took for granted, that reading of graphs would be trivial. When designing educational technology or pedagogy if we had gone with this assumption, an assumption which most teachers would have agreed with, we would be entirely wrong. We can probably see how critical since if incorporated we would fail bitterly. Although, the results were rather disappointing, tremendous learning was acquired. Useful information for educational design gleaned. This clearly illustrates the importance of assumptions on prior knowledge: teaching teachers what students do not know (Harvard Gazette). Till we address this there is no effective learning.

Filter

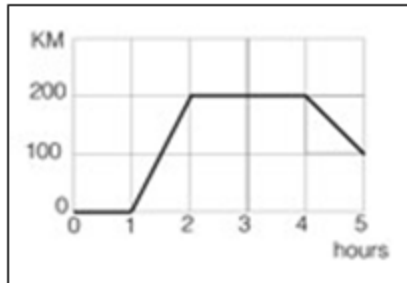
It is a piece of software that an application uses for file-format conversion or special effects.

PageMaker, for example, has a filter that lets it import Microsoft Word files, while Photoshop

has dozens of filters for special effects (such as image blurring). Filters can be part of the main application or external programs called plug-ins.

Finger

It is a program that displays information about someone on the internet.



Problem No. 4: A car was supposed to start from Lahore on the motor way and move towards Islamabad. It was supposed to start at 0 hour? A graph of the magnitude of displacement of the car from Lahore is plotted in the diagram as shown?

- (a) Find the distance of the car after half an hour from Lahore - half an hour starting from 0 hour?
- (b) Find the distance of the car from Lahore at 2 hours?
- (c) Find the distance of the car from Lahore at 4 hours?
- (d) Find the reading of the speedometer at 1.5 hours?
- (e) Find the reading of the speedometer at 3 hours?
- (f) Did the car reach Islamabad within 5 hours?
- (g) If some one drinking coffee at 1.5 hours is there a possibility of the coffee spilling over because of high speed of the car?
- (h) If some one drinking coffee at 3 hours is there a possibility of the coffee spilling over because of high speed of the car?
- (i) How far away the car is from Lahore at 5 hours?

Topic 165

Further Learning's From the Mid Term 1

Earlier we were discussing a customized midterm which we had given to gauge student response in terms of conflict resolution and their thinking processes. The questions contained deliberately designed warning signals. It was observed that most students either ignored these or were unable to resolve the resultant conflict. These signals constitute a critical part of learning. A baby when learning to walk stumbles. The body sends warning signals that, toppling over; immediately the cognitive feedback mechanism re-calibrates balance known as cognitive equilibrium. In this manner a baby learns to walk, an inherent learning process for which it requires no teaching. Similarly, in a murder mystery a detective looks for clues at the crime scene, from which he pieces together and solves the case. Researchers too look at all past research and try to analyze the reasons for previous researcher's inability to progress further. Through this process all new knowledge is created which is also essentially learning. Warning/danger signals form an essential part of this cognitive process – academically or in real life.

In the news you currently see a lot of comparison between democracy and authoritarian rule. Most analysts tell you about the weaknesses of democracy and how decision making is much more effective under a dictatorship. This is entirely wrong. If you look at all major wars, you would notice that; ultimately won by democratic countries in comparison to authoritarian regimes. Why did this happen? The basic reason is that in a democracy all voices are heeded and their perspective incorporated in the decision making process, unlike a dictatorship. The inability in conflict resolution results in wars when nations involved. This conflict exists at all levels; from nations to family and finally at the individual level. Without this ability in conflict

resolution no learning or progress. Listening to another perspective essential, the reason democracy more suitable. Consider the examples of Libya and Iraq, two very prosperous nations, where citizen were looked after completely. Free education, healthcare and job were provided for all. Yet due to their toleration structure, the West was able to leverage dissent and ultimately leading to toppling of the regime. The scenario was played out in Iraq. Closer to home, the same applies to education. It is said that in a university the quality of education is inversely proportional to how powerful is the vice-chancellor. If powerful usually the singular perspective applies, conversely collective wisdom prevails. Also, bureaucratic red tape slows down decision making in universities, unlike say a speedometer which immediately indicates your speed and consequently warns you of over speeding. This type of instant feedback is not possible in education without technology as well as conflict resolution.

Firewall

It is a mechanism that isolates a network from the rest of the Internet, permitting only specific traffic to pass in and out.

Flaming/Flame War

This is reacting to someone's newsgroup posting or e-mail in a hostile manner by publicly chastising the person or bombarding the person with nasty e-mail.

Access and use of ICTs outside school

The relationship between out of school use of ICTs by students, teachers and administrators to in-school use of ICTs in LDCs should be explored in greater depth. Given that confidence in ICT use and ICT-related abilities in school tend to correlate with access to ICTs outside of school, studies examining the differential impact (on achievement, uses and attitudes)

of ICT use in schools in rural and urban settings could provide an important contribution to understandings of equity issues in the uses of ICTs in education.

Topic 166

Further Learning's From the Mid Term 2

Turning to slides containing midterm answers we can look at an analysis of these. In the first slide, warning signals were ignored as well as the input and required output. The same thing occurred here and problem not understood correctly. It has communicated to these students that in the next exam there were be immobilized for half an hour, so that they can use their minds and not their hands. In the third slide unable to read graph and blind reliance upon formulas. This is not their mistake, since they know no other way .They have not been taught to think; according to teachers out of course. The difficult task of undoing is required. Continuing with the exercise the same mistakes are being repeated. If input is equivalent to output, nothing needs to be done. If not, what approach is required and attempting of multiple approaches to discern what discrepancies – heeding of warning signals. Moving on the pattern continues; greater reliance upon formulas than intuition. For example while driving, mostly rely upon intuitions. In cases of avoiding collisions, scientists have discovered that while braking suddenly, you body responds prior to you mind. If our reliance is so great in real life why not in education or for that matter exams. This is because our educational system has deliberately managed to suppress our intuition. Returning to the problem, some students have no idea what they are doing. What is the problem? This requires a detailed analysis. Looking at further analysis we see that the answer is correct, while the explanation wrong. If the students had tallied the two he would have noticed the discrepancy. Some answers, as you see, are completely correct. These students are the exception and require little help from the teacher, pedagogy or technology. The challenges of our course not for these students, rather those who require help. Other examples follow. From a total of 21 respondents, we see that these exceptions only number four. Those that had failed to heed

warning signals, or not having read the problem correctly, i.e. what is provided and what is required, number 16 or slightly over 70%. In our university about 700 students apply for the computer science/engineering course. After the first year two semesters 70%, find the work too hard and dropout. In other universities and other courses the dropout rate is not so high. It is not that these students are not intelligent or that they cannot do it; but their thinking systems are so distorted that they have a terribly hard time. This clearly illustrates the importance of technology required to transform their thinking.

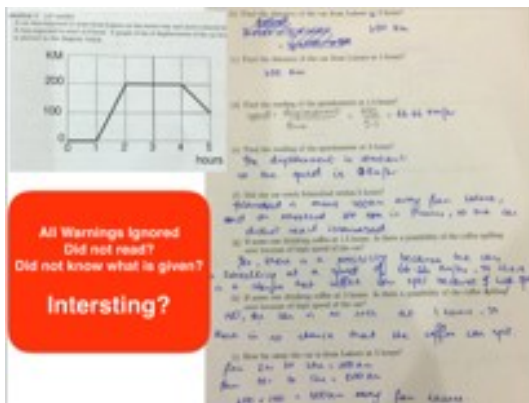
On this graph you see two peaks, usually a bell curve for results. Incidentally, a class fellow of mine in UET, once analyzed the results of final year students in engineering through histograms; this was on an IBM 1103, on which cards were used. He discovered that the results were not in the form of a bell curve. Mathematics, unlike other disciplines, is not proportionate the amount of time you study. There is a certain barrier in terms of understanding which only arrived at once you understand a problem, associated warning signals and conflict resolution. Of the two peaks you see one represent the students who have managed to cross this barrier and are conversant in the subject, while the other of careless students. These are not unintelligent people, some of my class fellows who at the time were careless are now business leaders. First twelve years of education critical in this respect.

Flatbed Scanner

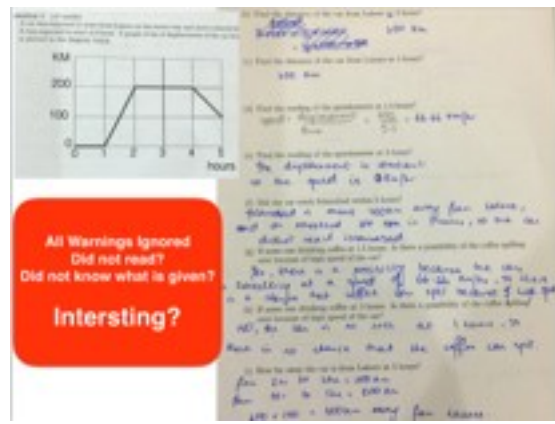
It may be defined as any scanning device that incorporates a flat transparent plate, on which original images are placed for scanning. The scanning process is linear rather than rotational.

Floating-point Processor

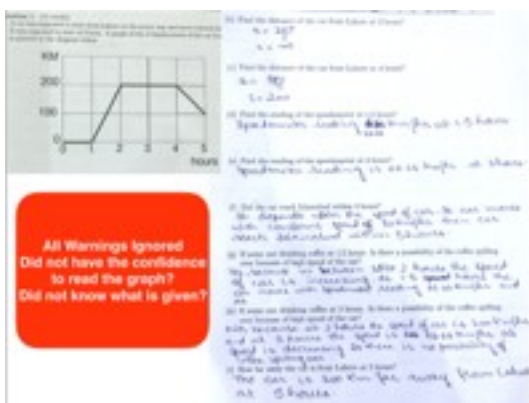
It is a special chip that handles sophisticated calculations, such as those used in spreadsheets, CAD, and scientific programs.



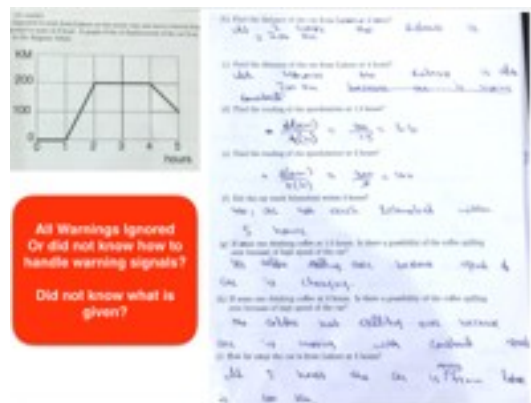
All Warnings Ignored
Did not read?
Interesting?



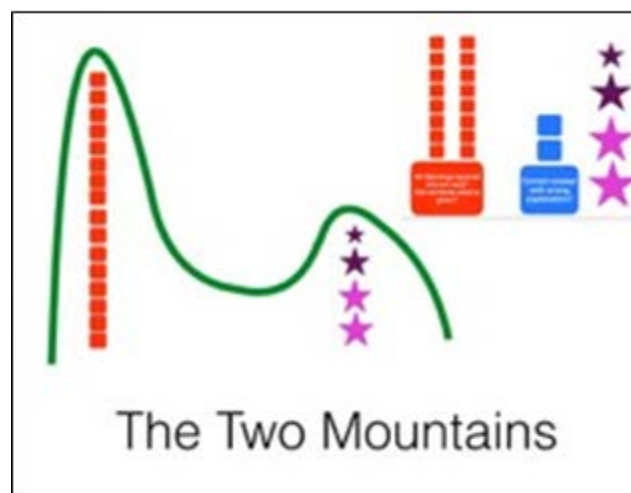
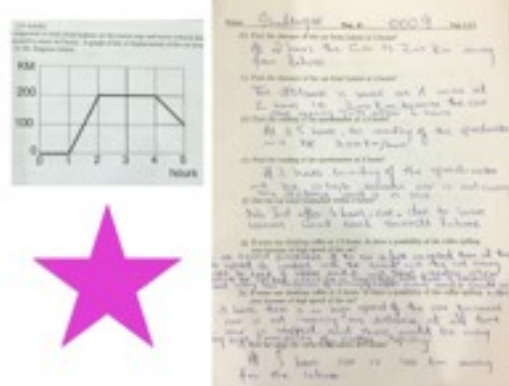
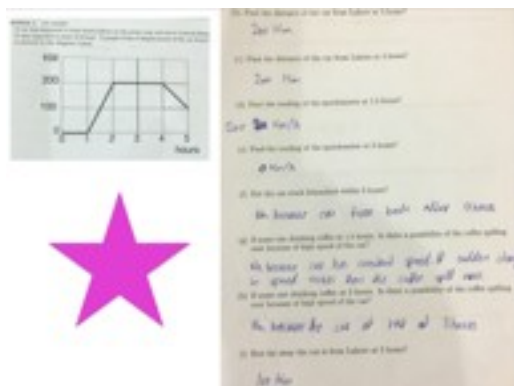
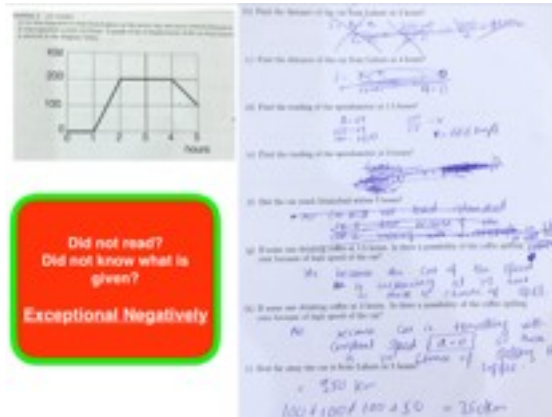
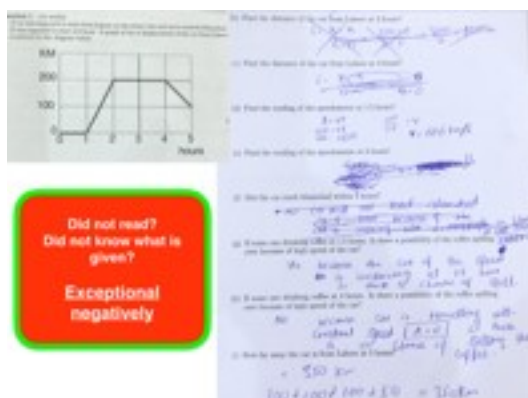
All Warnings Ignored
Did not read?
Interesting?



All Warnings Ignored
Did not have the confidence to read the graph?
Did not know what is given?



All Warnings Ignored
Or did not know how to handle warning signals?
Did not know what is given?



Topic 167

Further Learning's From the Mid Term 3

We now look at similar question which were used for PhD applicants for LUMS computer sciences program, which was inceptioned in 2001. We had numerous applications from varied backgrounds. The main interest was that at the time we were offering HEC scholarships, which paid a stipend of Rs.20, 000 and fees and books were free. The stipend amount in those days was equal to what the industry was offering. Admission interviews were recorded and cross questions put to the students.

The question was something along these lines; a question which should have been easily solvable by O and A level students. A car traveling from Lahore to Islamabad is delayed by one hour. At origin the time is 0. The displacement of the car is plotted with time. The question then becomes; describe the story behind the graph? If you remember from the Harvard calculus, it was discovered that there were many approaches to framing a problem; linguistic, graphical, analytical and tabular. Any teacher or a learning application needs to account for these, to allow students to learn in their own style. The problem with engineering and math's students was that they had never heard a question like this. No stories were told to them while learning and could not narrate any. Most answered that no displacement took place in the first hour; which was correct. The problem arose as the displacement increased, in the first hour 200km, in the second 200. Apparently, they were having difficulties reading the graph. We had plotted the graph of displacement from origin and not hourly displacement. There assumption was wrong. At the end of the graph we see a decline; what does this imply? Has the speed been reduced or distance decreasing, i.e. moving back towards Lahore. The answer was that it was confusing. This was perplexing since graphical interpretation required. The displacement in the fifth hour was asked.

Questions deliberately designed to induce warning signals. There we no time pressure and question repeated in simpler terms for them to grasp the concept. Further still the question was broken down into parts, i.e. what distance covered in two hours. The answer 200km was correct. Yet problems further on. Form this you see that a teacher can never address all students in a one to one situation. This is only possible through technology.

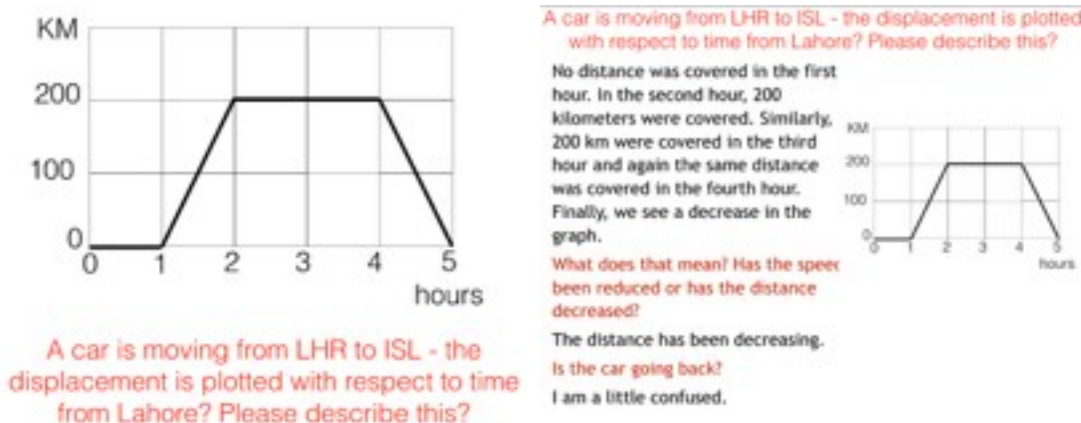
We now move to the written part of question , but before that two quotes might be appropriate: Gorge Polya, a Hungarian mathematician said, “yes the solution seems to work, it appears to be correct; but how is it possible to invent such a solution – how would I invent or discover such things by myself. Another: “In today’s Information society, the future of economic well-being rests on the effectiveness of schools and corporations to empower their people to be more effective learners and knowledge creators - Joseph D. Novak.

Folder

It is an object that can hold other objects, such as other folders and files.

Font

It is the software that creates a typeface on a computer screen.



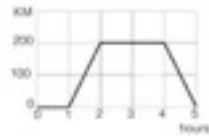
What is the confusion? This is the distance from Lahore to Islamabad.

In the fifth hour, the car did not cover as much distance.

How much distance was covered?
This is a graph of distance, not of speed. How much distance was covered in five hours?

I didn't understand.

This is a graph of kilometers.
Distance is increasing with time.
Study the graph and tell me the total distance.



What is the distance after two hours?

200 kilometers.

How much distance has been covered after three hours?

400 kilometers.

Someone is recording the distance that has been covered after every hour on the motorway while going from Lahore to Islamabad and it has been plotted on this graph. Now tell me how much distance has been covered after two hours.

200 km.

After three hours?

Again 200 km.

After four hours?

Again 200 km.

After five hours?

Zero.

Topic 168

Danger Signals & Higher Education

The written question given to PhD students was similar to the earlier question. Displacement of a car traveling to Islamabad from Lahore was measured over time, in graphical form. The questions asked were the measurement of instantaneous speed; this was simplified for clearer understanding, i.e. speedometer reading. Distance at 2.5 hours and various other time intervals. The only different question was the calculation of speed and its graphical representation. 25 HEC scholars were asked to provide distances at various intervals. Only three answered correctly on their first attempt – a cause for concern. Concerning instantaneous speed, four partially correct answers provided. Finally, in drawing of a graph, 18 did this incorrectly. How do we generate, handle and reflect upon this? We will look at this in detail. Yet, we do learn an important lesson in terms of learning and passing exams from these examples. A subject needs to be taught in a manner that it remains memorable. The book shared with you earlier; *Foundations of Educational Technology: Integrative Approaches and Interdisciplinary Perspectives* by, J. M. Spector, contained one such event in the life of Tolstoy. While what he witnessed was not planned, we deliberately attempt at creating this triggering event.

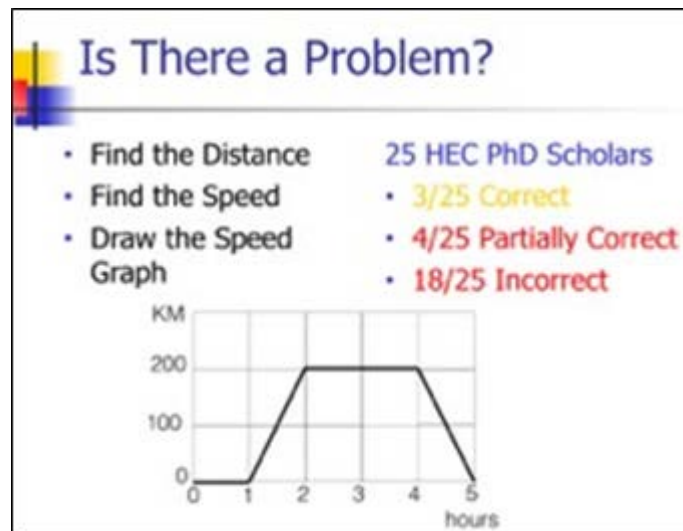
In the first chapter, defining educational technology, in the book there are some new definitions on education and technology. You must be wondering, why we are looking at these definitions at this stage of the lecture. This course, like its subject, is not static. Technology evolves at every instance and consequently the need for new definitions and the latest in books which I try to acquire for you.

Format

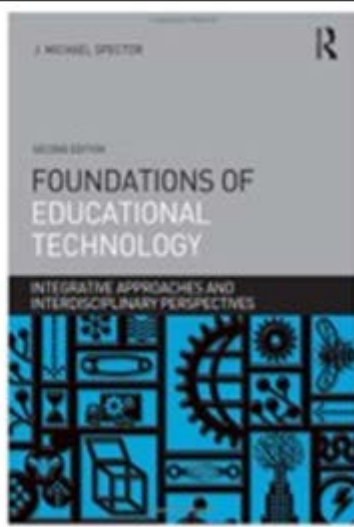
It is a process to initialize a disk to prepare it for use. The disk is checked for errors and organized so that data can be recorded and retrieved. Formatting a used disk erases any previously stored information.

FPO

It is abbreviation of For Position Only. A low resolution image placed in a document to indicate where the final version is to be positioned.



LOOK AT THIS LATEST
BOOK WITH A NEW
DEFINITION OF
EDUCATION AND
TECHNOLOGY
PUBLISHED IN 2016



Topic 169

Technology Implies Change

Technology simply is change. Change in this case is considered valid if effects society, otherwise not considered change. How do we cope with this change in terms of its conflicting requirements? I will try to explain through my story.

The story begins in the heart of Multan, At Pul Mauj Daray, where I resided. It was in the 60s and I was around six or seven. At the time we used sleep outside, no other option in the sizzling summers of Multan. In the evening the ground would be sprinkled with water and everyone would move outside. It was the way to cope with the heat in an era, where no air-conditioners, coolers and even at time electricity. It was very pleasant outdoors, since no pollution, vehicle or dense buildings. From this you probably realize that there are positives and negatives to technology. Sleeping outside would mean that we would get up early, due to the sun. Yet, I felt very fresh. I walked to school as did most people. No student came in a car. Our headmistress would come by bicycle. I would pack my books in a cloth sack, since I did not take my school bag. The reason was that I was bought a very nice school bag and when I took it to school all the children, who were rather poor, gathered around to gawk. Things have changed these days it is the rich who gawk if you bring something inexpensive. Even little children have pencil boxes which cost a couple of thousand. Anyway, I felt terribly embarrassed and told my mother that I would not take the bag to school. She enquired if it was not to my liking since I had chosen it. I told her I liked it but would not take it to school.

In those days we wrote on tablets (takhtis) to improve our handwriting. An old servant, Babaji, was responsible for shopping; whether vegetables, meat or other grocery items his mode of transport a bicycle. Food and other perishable were stored in a ventilated larder (ganjina).

Cooled water was available through clay containers (surahis), ice bought and stored for guests in thermoses. The supervising of cooking of food and maintenance of the household was my grandmother's responsibility, since my mother and aunts all taught. They traveled on a horse drawn buggy (tonga) which was dedicated for their use, a rare thing since most were shared. No petrol or gas was used. Compare this to how technology has impacted our lives and society. Not all good. We had fresh milk, butter and lassi (milk beverage). In terms of media our sole technology was a Marconi radio, still with us to this day. From the limited number of newspapers, we subscribed to the Pakistan Times. Incidentally, my grandfather, my aunts and I were regular contributors to this: despite the fact that technology was primitive, greater awareness common.

I remember distinctly my first letter which I contributed. In our college during our FSc, the course for Physics, Chemistry and Mathematics was revised. While there were no issues with the physics and mathematics books, we could not figure out the chemistry textbook. Upon enquiry with the teachers, they told us still coming to grips with the new course. I wrote to the paper, anonymously and pointed out what we could not understand. This started a debate and lots of people wrote to provide the answer. That this was included shows the difference in level, as to what content contained these days. Later we found out that while the physics and mathematics textbook were copied exactly from foreign textbooks, consequently no issues, the chemistry book was a hash of local and foreign content. The issue of pedagogy, a term and concept, of which I was not aware at that time was highlighted by me in the paper. Finally, I asked a friend and was delivered the original textbook, bought from Mirza Book Agency and couriered to me through PIA – the sole courier service in those days.

In those days the use of electricity in schools and home was minimal. In schools we only had a ceiling solitary fan. In the 50 centigrade heat of Multan we studied in the heat. Despite being drenched in sweat we had no issues. Petrol and gas were not used as fuel. For cooking we used deadwood found in our house, which had a large garden. As time lapsed and population grew, a wood vendor (Pathan) established his shop across the road and we used to purchase wood from him. Later this was replaced by wood shavings. Next came coal; a big leap in technology. Coal was specially used for making tea since it was faster. The people living in our quarters would use cow dung in the form of patties, a cheaper form of fuel – technology. When the water bubbled away for tea and the steam caused the rattling of lid, I could envision the technology behind the steam engines. Afternoon was a treat, a time of family gathering after a hard day's work and rest: something which I cherish and will remember forever.

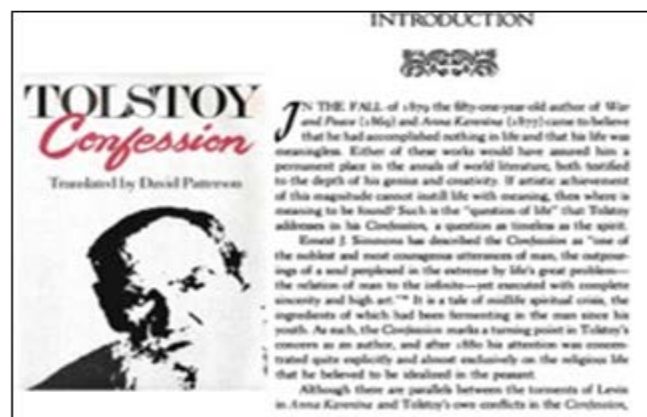
The leap in fuel technology was gas; which we started using indiscriminately. A TV drama on Pakistan Television, Koalaywala (the coal vendor), became very popular. A coal vendor faces customers shifting to gas from coal and how it affects his livelihood. Going house to house most refuse his coal but at one household, of someone thoughtfulness, his coal is purchased despite the fact that they too had shifted to gas. This goes on until after a long time he revisits the house and tells the owner that he can no longer supply coal since he has finished his business and become a gas technician, but he would find him an alternate supplier. The owner tells him he does not require further coal. Upon further enquiry he is shown the house's courtyard which is covered with coal. In his surprise he asks what this is. To which the owner replies there is something called Consideration.

Fragmentation

It is a condition where parts of a file are stored in different locations on a disk. When a file is fragmented, the drive's read/write head has to jump from place to place to read the data; if many files are fragmented, it can slow the drive's performance.

Frame-grabbing System

It is a combination of hardware and software, designed to capture individual frames from video clips for further digital manipulation, or consecutive replay on computer platforms.





THEN CAME GAS - AND WE
STARTED WASTING IT AS IF IT
WILL STAY FOR EVER

پی ٹی وی کا مشہور ڈرامہ
کوئلے والا
وضع داری اسکو کہتے ہیں

Topic 170

Studying Abroad

Earlier we were discussing how technology has evolved in context with my younger days in Multan. Those were days when in sizzling summers of Multan, we would study underneath the Neem Tree (*Azadirachta indica*). There were no air- conditioners or coolers. We had looked at technology in terms of transport, food storage, cooking and other aspects. Now we look at it in terms of communications. During the 60s in Multan the population was around six to seven million and there was only one telephone exchange. These exchanges were mostly manual, i.e. mechanical. We were lucky that we had a phone. Our number was 2427, which implied that there were approximately 10,000 phones in totality. For intercity call, call had to be booked. Direct dialing was introduced much later and considered a luxury. Telegraph was the other medium of communication. Late night telegraphs were a time of anxiety and happiness depending upon what news. Letters were regularly written, and mail did not imply email. Mail travelled through the Khyber Mail in those days, consequently the name, i.e. mail train.

My maternal grandfather Dr. Mohammad Wahid Mirza travelled to London for his PhD, during the early 30's. He traveled from Lahore to London via ship and it took him a couple of months. While in London his sole means of communication was by letter or telegraph. In 1962 my maternal uncle, Dr. Arslan, who went to become vice-chancellor at Quaid-e-Azam University, went for his PhD in biology to Wisconsin, US. We saw him off in a Boeing 707 and took him a few days to get to his destination. His means of communications was still letters and telegraph. During his seven year stay he only called, using the telephone, twice. It took him most of the night to place the call. One instance was when he had acquired his degree and the other at the demise of his sister my aunt. Sarah Fatmi was an English professor at Lahore College for

Women. Perhaps the mothers of some of you watching might have taught by her. Anyway, my son went for his doctorate to University of Central Florida, US in 2013. He travelled in an Airbus 780 and made his destination in a day. Through Skype and regular phone calls we were in touch with him continuously and knew what he was doing at most of the day and see his regularly. In those approximately 80 years, at intervals of 30 or so years, you can see how technology changed communications – from my grandfather to son.

Another grandfather of mine Dr.Abid Ali, also went for his PhD to Oxford, UK. He was the principal for Sargodha College during the 40s. In this picture you see his granddaughter, at VU, where we are figuring out how to teach graph algorithms. Komal completed her degree in TIE (technology innovation education) from Harvard. This program was replicated at NUST in the form of ITE (Innovative Technology in Education). (Picutres).

FTP

It is abbreviation of File Transfer Protocol. The Internet standard high-level protocol for transferring files from one computer to another across the network.

FTP site

It is a computer which stores files that can be retrieved using FTP. FTP sites which allow anyone to retrieve files (without having an account on that computer) are known as Anonymous FTP sites.

Effective utilization of ICTs for professional development

The need for trained teachers is great throughout most LDCs, and ICTs are increasingly seen as important tools in reaching and engaging teachers on an on-going manner.

1930



1969



2013



1935



2014



Topic 171

Evolving Calculations

In 1969, we started university, University of Engineering and Technology (UET). At the time there were only two engineering universities in the country; one in the West wing and one in the East. In those days' personal calculating devices, calculators and computers were nonexistent. In those days the only available option was scientific calculators. These used to consists of two fixed edges with a sliding middle portion. A short orientation course was available and required. Once understood, it was capable of performing quite complex calculations. I distinctly remember my first slide rule, manufactured by Faber-Castell, a German company. It was purchased from the Anarkali Bazaar and it came with a beautiful protective leather case: its use and memories cherished still. There was a certain pedagogy associated with the slide rule. The ruler was operated manually and used a moving/sliding cursor. Some of the calculations carried out were quite involved and involved multiple steps. The step by step process and sequential manual operation allowed lead to numerous chances of error. To counter this we were taught the use of paper and pencil, in arriving at an approximate answer or estimate. This is a good example of warning signals, at least flags the fact that, if your answer even in the ball park. Allowing you to revisit you answer or rely upon your intuition. With calculators this is no longer possible. As in the displacement problems discussed earlier, it was quite apparent, in the post mortem of results, that an immediate resort to a formulaic approach was taken; formulas relied upon even when not required. We need to revert back to an approach where a student thought about a problem prior to attempting to solve it.

A multiple approach allowing a student to countercheck their answers needs to be reintroduced. If the slide rule does not yield anything within the anticipated range, you would

probably anticipate user error. This pedagogy needs to be reintroduced through the use of technology. While the slide rule has been replaced by powerful handheld calculators and computers, it reminds us that a powerful affordance of an educational technology is to get one to think about the problem one is attempting to understand. The educational principle here is that reflecting on the nature of the problem being solved is often more effective than providing learning and understanding – a principle well worth remembering. This was the story of the slide rules.

It is a strange fact that I have never purchased a calculator in my life. As a professor during the late 70's and early 80's, at UET, my pay was not sufficient to acquire one. In those days calculators were rather expensive. A good friend Zafar Mahmood, my friend since FSc days in Multan, who later became my colleague at UET, went to Japan for MSc. On his return he brought with him a precious gift, a gift if given today you would probably throw away, it was a Casio scientific calculator.

Our university department purchased our first personal computer in 1980. All our department research funds went towards its purchase. It was the first computer which I touched with my own hands. You have heard its story before in detail. It was a RadioShack TRS 80 Model 2. At the time its memory consisted of 64k ram and contained no hard disk. In comparison the Apple watch I am wearing contains much more memory than the computer. From this you can realize how technology has evolved. These days all my children and even grandchildren have their own computing device; whether a laptop, tablet or android devices. All these are enormously powerful than our university microcomputer, which at that time was priced around Rs.100,000. This aside my driver, has an Apple Macbook Pro and an I Phone. He can utilize this to take any university lectures from top class international universities, if he wants.

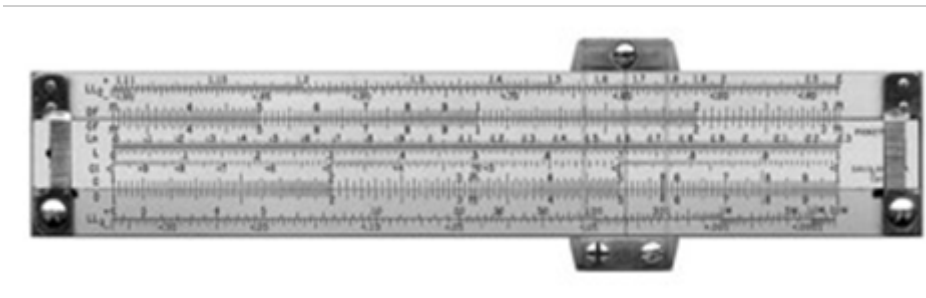
We next turn to new definitions of technology and how it can be deployed in education more effectively and efficiently.

Gamma

It is the curve that describes how the middle tones of your images appear. Often incorrectly referred to as "brightness" and/or "contrast", gamma is a non-linear function. Changing the value of the gamma affects middle tones while leaving the white and black of the image alone. It is used to compensate for differences between Macintosh and PC monitors.

Gamma Correction

It is the correction of tonal ranges in an image, normally by the adjustment of tone curves.





Topic 172

Technology Changing Society

Now that you have heard my story of three generations: from my grandfather to my grandchildren - approximately, a period of a hundred years. The changes in technology, during that period, and its speed is mind boggling. A change is only construed as such if it is for the betterment of society or if it creates a change in society.

Change is integral, nothing remains constant but the divine. You could possibly question the betterment part of the definition, in terms of technology. Have more benefits accrued in comparison to disadvantages? Yet, you cannot question the fact that there have been tremendous changes. Some critics would point out that some groups have been disproportionately favored, while others left behind. While this may be true, it is only through technology that we will be able to create equilibrium.

Technology involves the practical application of knowledge, for a purposeful change; again a point which can be debated. Knowledge is progressing and so are the goals and intentions of people. People's demands and requirements are progressively increasing. In the old days travelling from one city to another was a chore. Now it is easy to travel globally even to the earth to space - the destination of dreams one which will never be arrived at. Technology is all pervasive and changing every aspect of our lives. How you communicate, live, educate and relax. "Technology changes people and people change technology," a famous quote. We are now transferring a treadmill of technology, which perhaps is out of our hands. Who affects who, no one quite knows. A thousand years ago or for that matter a hundred years ago, the pattern of living around the globe was quite similar. Starting with the advent of the steam engine the fragmentation caused by technology is quite evident: the fragmentation of empires and the rise of

new entities and territories. Change is the imperative word; it has now assumed a life of its own and beyond anyone's control. We need to harness this Frankenstein to play a useful role in education. Technology makes things possible which were not possible in the past. The earlier stories all contain elements which point towards this trend. Similarly, what is impossible today will be tomorrow's possibility.

We now turn towards education and see what new definitions we can derive. Our course, the use of technology in education, raises certain questions. Why is that with all the changes in technology our classrooms, course books and other educational tools are pretty much the same as they were a hundred years ago? This is the main purpose behind this course to tackle this issue of education.

Gamut

It is the limited range of colors provided by a specific input device, output device or pigment set.

Gang Scanning

It is sequential scanning of multiple originals using the same previously defined exposure setting for each.



LOOK AT THIS LATEST
BOOK WITH A NEW
DEFINITION OF
EDUCATION AND
TECHNOLOGY
PUBLISHED IN 2016



Topic 173

Purpose of Education

Education is defined now, as always, involving a purpose, or goal and a process to achieve this. Knowledge and process of learning are essential for any educational experience. Learning is a change in what a person can do or believe. There should be a marked change, whether in thinking; a change from a geocentric belief to a heliocentric belief of the universe. 35:35 this change in thinking took place over a period of hundreds of years, as did the ideas of Newton and Einstein. There are many things which have been made possible by their discoveries. Using Newton's laws we could predict planetary motion, on which we consequently based our satellite orbits. So learning implies a change in what a person believes or can do. Similarly, new knowledge is a change in what a person can think, do or achieve which was not possible without this knowledge. This sounds very familiar to how we had defined technology earlier. It seems like we are talking about education yet it appears as if we are discussing technology. So learning through new knowledge is a change in what a person can do, to achieve a purpose. Similarly, technology involves the practical application of knowledge for purposeful change. A point which the author is trying to reinforce, that technology and education are similar in terms of objectives. A concept which has enormous implications for our course since based fundamentally on technology and education. The commonalities are shown in the diagram which follows. On one side you find Learning/Education and the other Technology. The underlying elements under the two are as follows: Learning – Knowledge, Change, Process and Purpose

Technology – Application of Knowledge, Process and Purposeful Change From this diagram distinguishing the two is hard. Now we turn towards aggregating the two, education and technology and defining them. Educational technologies involve the disciplined application of

knowledge for the purpose of learning, instruction or performance. The notion of disciplined application of knowledge is included to reflect the view that educational technology is an engineering discipline. The major difference between a science and engineering disciplines is that; science studies existing phenomena. Engineering meanwhile creates new systems and structures. Thus the classification of educational technology as an engineering/design discipline, through which creates something new. All new design solutions based on scientific theories; educational technology based on learning theories. Past experience and empirical evidence guides what professional educational technologists do. This is the reason that in the book teaching is described as a design discipline/science. Its principles derived from basic sciences and empirical research (students teaching methodologies) and rooted in constructivist and behavioral theories and beyond these in; cybernetics, communications, cognitive, design, organizational theory, psychology among other. A truly multidisciplinary based approach. Educational technology is inherently an interdisciplinary enterprise, which makes it rather complex. This is the reason that problems in education in Pakistan. While various departments in isolation are strong, since there is interdepartmental communications education suffers. Very few educationalists are available in comparison to teachers for other disciplines.

The principle of encouraging problem solvers to reflect upon the nature of the problem, can be traced back to cognitive psychology, perhaps it goes back much further. This reliance upon a multidisciplinary approach is based upon the fact that problems in educational technology are complex (wicked problems) and challenging. For these problems, it is especially important one think about what one does and naturally, why and how, in a disciplined and systematic manner. It is of great import that this research commences in Pakistan, for instance the MSc program we had introduced at NUST. In other educational programmes there are separate

department for pedagogy, technology, social sciences and basis sciences (Pedagogy-Content – Technology) yet no department talks to another. Till the time they communicate educational technology will not be developed.

Gateway

It is a special-purpose dedicated computer that attaches to two or more disparate networks and converts data packets from one form to another.

GB

It is called Gigabyte. It is a unit of data storage size which represents 10^9 (one billion) characters of information.



Topic 174

Framing Learning Changes

Learning is a holistic concept that involves cognition, cognitive memory, mental constructs, language associations, non- cognitive emotional states and physical conditions, aspects and constraints. All these variables have an effect upon learning. Suppose a student is not motivated to attend class? We have extensive knowledge of human physical development but more limited knowledge with regards to other aspects of human development – cognitive, emotional and social development. My daughter, who is going to for her MSc, in her college Colombia they do not distinguish between the three aspects of development. What students understand in class is not solely a cognitive problem but also of one of emotional and social development and cognition. Much learning is unplanned and incidentally associated with a variety of actions, while much of the learning which occurs in educational programs is planned. This is where the problem lies, why can little children learn on their own but as soon as put into school this ability deserts them. Planned learning activities typically occur in complex environments; classrooms, online education, blended learning and workplace locations, with many things which can enhance or otherwise inhibit learning. Strangely, planned learning activities instead of enhancing often prove to inhibit learning. Our school environments distort the thinking processes of our students. Solely focused on getting good grades and able to move in to chosen fields of vocation. Determining the extent to which learning has occurred involves the analysis of measure or indication of change; before, during, after and long after. In terms of lecture this would mean examination of student prior knowledge before a lecture, the understanding taking place during the lecture and what retained after it. Also a test of long terms retention is also required. Why learning occurred to the extent measured or observed is even

more challenging. Here the scientific cognitive theory helps in gauging learning; how much, why, and how. It also deals with how student attention and retention can be enhanced; an area in which technology can play a crucial role. Performance is something that can be observed, assessed, measured and rated against a standard or other point of reference. This can be greatly enhanced by technology, a case of which we will discuss in detail later. Change in performance, especially improvement in performance is particularly important to an instructor. Providing feedback on performance very soon, even instantaneous, after the actual performance, is very effective in improving performance; if feedback is specific and constructive.

As a dean in university, our institution has a system of quizzes, leading to midterms, leading to Finals. We have discovered that if quizzes are graded in time, positive and constructive feedback provided the midterm results are positive and vice versa. The role of administration, along with technology and education necessarily implies the human factor. This is the hardest to change and requires a long time. In top universities of Pakistan, quizzes are still not graded on time. To counter this technology is greatly utilized in mitigating the human factor and providing instant feedback. The approach of preaching and teachers training which has been going on for 50 to 60 years has proven ineffectual.

GB

It is called Gigabit. 10^9 bits of information (usually used to express a data transfer rate; as in, 1 Gigabit/second = 1Gbps).

GCG (Grey Component Replacement)

It is a technique for replacing all the neutral tones of an image with an appropriate amount of black.

Educational technology involves the disciplined application of knowledge for the purpose of improving learning, instruction, and/or performance. The notion of disciplined application of knowledge is included here to reflect the view that educational technology is an engineering discipline in the sense that principles based on theory, past experience, and empirical evidence guide what professional educational technologists do.

These principles are derived from basic science and empirical research in such areas as cognition, cybernetics, information science, human factors, learning theory, mass communications, message design, organizational theory, and psychology. Educational technology is inherently an interdisciplinary enterprise. The principle of encouraging problem solvers to reflect on the nature of the problem first can be traced to research in cognitive psychology (perhaps it goes back much further).

LEARNING - PERFORMING - ASSESSING -
DOING - EXPERIMENTING - EVALUATING
-CONNECTING - REFLECTING ARE ALL
ARE HAPPENING AT THE SAME TIME? WE
SHOULD NOT SEPARATE THEM?

WE HAVE SEPARATED THEM INTO
LECTURING - EXPERIMENTING AND
EXAMINING?

WE HAVE SEPARATED THEM INTO
LECTURING – EXPERIMENTING AND
EXAMINING?

THE FACTORY PIPELINED
MODEL WHERE THERE IS
NO TIMELY FEEDBACK?



Topic 175

Instant Feedback

What happens if there is no feedback? When coming over here one was not feeling well. Seeing my rather depressing posts on Facebook, even my son abroad called and enquired as to why I was so depressed. The matter was that at our university the final exams were to take place in a week or so. In most subjects almost half of the class has dropped out. The reason, I believe was that we had not provided them with appropriate feedback in their quizzes, labs, assignments and midterms. Due to this the final year came as a surprise, full of uncertainties. Now each of these students had paid Rs.25,000 for each course. For a middle class family, this implies a serious amount of money. Unfortunately, the onus of this failure resided with us, in terms of not providing appropriate feedback, and consequently my anxiety. It was not a matter of replacing teachers, since a significant majority of new teachers would also repeat the same mistake. Change is the most common denominator which can be measured quantitatively through technology. If we had gotten the appropriate warning signals in time we could have attempted to tackle the situation. If learning has occurred a person can do something which he or she could not have done earlier, showing change has taken place. New technologies and knowledge have made this possible.

Developing an individual's ability to monitor and assess his or her own performance is often a desired and measurable goal for advanced learners. For example in English we can break down the subject in various smaller components, i.e. comprehension, reading, expression, etc., and assess where he stands and then focus on that aspect to make sure everybody is on par. The one size fits all approach does not work and technology is the answer. Also, this is a cheaper

solution. Technology does not scold or indulge in sarcasm and is also available anytime and anywhere.

Performance is a holistic concept that typically involves cognitive as well as non-cognitive abilities. Performance may vary with an individual's mood or other factors affecting the individual's performance at that particular time. Our understanding of human performance is reasonably well developed but far from complete. Any variations in performance across different individuals and tasks are not predictable based on current evidence, knowledge and theory. Performance and learning are closely coupled concepts. Performing tasks and activities can result in learning; as learning develops in a particular domain performance in tasks in that domain is likely to improve. Learning, performing, assessing, doing, experimenting, connecting, thinking, reflecting are all happening at the same time when learning is taking place. We should not separate this, something which we have managed to do. This happened due to the absence of technology which could have consolidated disparate facets. The straight pipeline model (manufacturing), where there is no feedback, cannot work. A feedback pipeline model will need to be developed, where learning, experimenting, teaching, doing, reflecting, connecting, evaluating, examining and assessing will need to be carried out in conjunction.

GIF

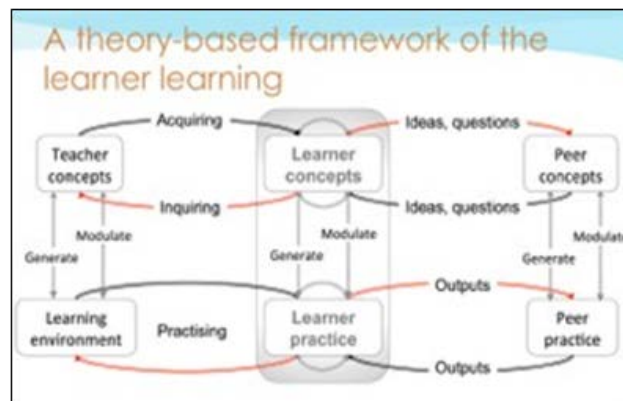
It is abbreviation of Graphic Interchange Format (pronounced jiff). It is a file compression format developed by CompuServe for transferring graphic files to and from on-line services.

Define Webmaster

It may be defined as a person or group of people who maintain and administer a web server. Webmaster also refers to a standard Email address at most web hosts where comments and questions can be sent.

Impact of Computers on learning

A comparative study of the benefits and costs of placing computers in computer laboratories, individual classrooms, teacher offices, libraries, and/or community centers would be quite useful.



Topic 176

Turning of Days

There are many perspectives from which we can look at our course:

Kyon gardish-e-ayam say ghabra na jai dil

Insan ho, pyala-o-sagar nahi ho mai

Why should the heart not troubled by the turning of days

A human after all, not a cup or wineglass am I

The significance of this famous Ghalib couplet will be highlighted later. Prior to that, since we have covered a significant portion of our course, a summary might be in order, along with what we plan to cover later. Earlier we had discussed the design of intelligent learning applications. The critical points were its adaptive nature: that it is able to adapt itself to cater to individual student learning styles; Interactive and provision of instant feedback; record keeping, keeping track of individual learning styles. We have also learned some golden rules. Whether the form of teaching; classroom setting, interactive or blended learning, the problem should be presented in the form of a story and not in symbols. Mathematical if mathematics, equations in chemistry and so on. This is now a universally accepted norm which unfortunately here we do not follow. We also assume that before commencing 'prior knowledge' is tested for. This can only be achieved through technology, its methodology discussed earlier. No learner would be allowed to progress, until his prior knowledge is certified.

In the university I am currently associated with approximately 700 students apply for computer sciences. English is a key issue. Reading with comprehension and writing with understanding, listening and speaking. If you are not proficient in these facets, learning computer science is extremely difficult. If you cannot express what is in your mind properly you cannot

clear an exam. Here I am not saying that all students should be able to speak with American accents, rather you should be able to express yourself with clarity and simplicity. Apart from any learning institution, these conditions also apply for any organization, where you may choose to work. The students which apply to our university comprise of great diversity. This is especially true for English, in other subjects there is more correlation. The main issue is with speaking and comprehension/listening. While this diversity may not exist for LUMS, NUST or for that matter UET, it is a serious problem in our university. This is a great challenge for us, i.e. the use of information technology for English. Yesterday an applicant for a teaching position visited an MPhil and a position holder from Kinnaird College. She was expecting an interview, instead I told her; this is our problem and if she could solve it. I told her she would require various resources and deliberately omitted technology, to bring all the students to a standardized, acceptable minimum standard. We have three English courses and the lecture is standard for all and does not cater to the diversity. I asked her if this was impossibility and if not how do we solve it? I told her I was not expecting answers to the effect that one would try their best or that the children are not intelligent enough. We had discussed this point earlier, whether pedagogy, technology or its mix, in universal learning design, no student is unable to learn. The fault lies with the teacher. When I joined and reiterated this point there was a great hue and cry. Strict administrative measures were required to get this implemented.

Gopher

It is a consistent user interface and gateway into many on-line white pages and other address databases.

Graphical User Interface (GUI)

It is the graphical visual representation of the working environment that presents the elements of your computer as objects on a desktop.

Topic 177

English Vinglish

This problem with English is not only confined here, rather an international one. There are some vested interests, for example students enrolling in MIT, Stanford and other top universities, and at forefront of increasing American knowledge/research mostly foreigners. These graduate students go for their MSc and Phd degree with English as prerequisite. So it is imperative for these universities to provide quality English education: a matter of survival if you will. Now this teaching is not possible through conventional lecturing. A lot of financial resources were directed towards fixing this problem and various packages designed, which we attempted to explore. The best automated software we were able to locate, one used by the earlier mentioned top universities, was provided to us along with a license for ten students and three teachers. The first thing, which the interactive package, does is to analyze where each student currently stands; without embarrassing them. After assessing their placement it starts bringing them to par. It also allows for teachers to localize or adapt the software. The software also allows for flexibility in terms of time required for completion of certain proficiency. Now if you compare this to what we are doing currently you will see how our standardized system compares with this variable one a complete paradigm shift.

Returning to our summary, a learner is also encouraged to look at the bigger picture. Instead of attempting to solve a part of the problem students should look and think about it holistically. Feedback too is provided continuously, instant feedback, so periodic exams do not come as a complete surprise. We had also discussed cognitive disequilibrium and the importance of danger signals: how to create, generate and exploit them. We stressed the need for approaching a problem from multiple perspectives/methodology and the importance of

artificially introducing cognitive disequilibrium in a problem. We had also addressed conflict resolution. We also looked at concept maps, a pedagogical tool, which helps in charting your thinking processes and another aspect of technology, i.e. unlike humans its capacity for infinite patience. Now we turn to the remaining course and some truly exciting and ambitious problems. Here we will only describe these problems to pique your interest. Most of these are based upon real challenges which I am facing as a dean at a prominent university.

Grey Balance

It is the balance between CMY colorants required to produce neutral greys without a color cast.

Grey Levels

It is called discrete tonal steps in a continuous tone image, inherent to digital data. Most CT images will contain 256 grey levels per color.

Topic 178

Of Universities

Despite the fact that it is not customary to talk about universities in a public forum, probably extension of the fact that we do not like to express our emotion or thoughts, especially publically - a fact of which I am very much guilty. I have been asked by numerous universities, with which I have been associated, not to do so. But how can one not express himself? Ghalib captured this nicely in his couplet, with which we started this section. At my current university I have been given three year tenure, as dean of computer sciences, in which I have to rectify the situation. Of the 700 students enrolled every year, fifty students each are placed in 14 sections. When I joined the university, I was not told why the earlier dean had been dismissed on days' notice. Also my challenge was not highlighted. Sounds interesting, to be given an important post and not told what you are supposed to be doing? Rectifying a problem without its identification can be problematic. So as I mentioned the division of students in sections is not a problem.

We have well stocked labs. Computers are available. The fee structure I had told you about earlier: a student pays Rs.25, 000 for a course and in a semester the typical workload is four courses or Rs.100, 000. So if all this is available, what is the university's problem? Very few universities have this kind of infrastructure and a custom built facility. Teaching Assistants and graduate students, who also serve as laboratory engineers, are there to assist faculty and students if they choose to do so.

Library facilities are world class, with no financial constraints. Since it is a private school, there is no bureaucracy or red tape; decisions are immediate. Multimedia is available in each class. In our days in UET there were only overhead projectors, most of which were dysfunctional. I remember that for my PhD symposium I had to borrow one. There is student

unrest in terms of unionization and students receptive to faculty and administration. All classrooms are air-conditioned. The university buildings are surrounded by beautiful landscaping and contain appealing artwork. The university higher fee structures should point to intrinsic motivation on part of the students. In comparison the fees at UET were so low that people said that this was the reason students were not motivated. Infact the salaries of accountants for monitoring fees were higher than the actual fees. The converse excuse given at our university is that since rich kids not really interested in studying. The university curriculum is the best I have seen; despite being associated with many universities. Yet, a sense of inferiority complex remain, upon enquiry internally I was told that there is no comparison to LUMS. A perception that is factually wrong. In large universities the scale is so extended that it is difficult to alter subjects within a curriculum. At our university this can be done with a short notice. A student evaluation mechanism exists for instructors. Some of the best faculty is available, a significant number pioneers in their fields and involved in setting up these disciplines in earlier varsities. Based on these attributes, what should be the performance of such a university?

Grey Scale

It is a continuous tone image comprising black, white, and grey data only.

Hacker

It is a slang term which is used for a technically sophisticated computer user who enjoys exploring computer systems and programs, sometimes to the point of obsession.



Topic 179

Communicating and Feedback

Upon joining the university, meeting with faculty and students, looking at the syllabi and visiting the facilities, there seemed to be no apparent problem. On further investigation, the underlying problem turned out to be that the students were completely fearful of talking. There was no interaction between them and faculty and administration. The only medium of communication open to me seemed to be Facebook. Despite it being misused often, sometimes people do actually put down their actual thoughts. When I started using this medium I was told it was beneath the dignity of a dean. Also they had issues with the fact that all university matters were posted on an open forum. There are many detractors to this medium, for instance a head of computer sciences at another university wrote a Ramadan greeting and then posted that he would not be using this medium for the rest of the month. The implication being that somehow it interferes with ones religious obligations. The story of how Facebook started is quite interesting.

Started by Mark Zuckerberg at Harvard; the Harvard Crimson states that it started off as a social dating site called facemash. It consisted of photos of female students from facebook of nine houses at Harvard. Two pictures of girls were paired and posted for comparison as to who was the 'hottest'. These pictures were often hacked and taken without permission. The initial site mirrored and paired physical community with real identities; this later went on to become Facebook. The initial reaction from Harvard was one of great opposition and even legal ramifications. Naturally later on this was modified due to its success.

At the time Mark Zukerberg wrote in his blog; "I am jerk for making this site, oh well, somebody was going to do it eventually." The site was sent to several servers at other universities but was shut down by the administration. Mark faced charges of copyright

infringement, breach of security, invasion of privacy, theft among others. Eventually all charges were dropped. Many stories circulate as to what transpired: after having been called in front of committee, Mark appeared totally unrepentant and said he would do so again. While the site was up Harvard's computer was overwhelmed in 25 minutes to which Mark replied it took only two. Realizing what had happened, the Harvard people realized they were onto something. A teenager had created something innovative which they could never think of. In terms of ethics was he very much different from how large corporations operate? Its current popularity attests to the founder vision. Yet, some negative connotations associated with Facebook still linger. People, including my wife, object that why are you always on Facebook. For me it's a medium of expression. In our MSIT program everyone was connected via Facebook and any problems and issues were immediately posted to be tackled. Also, I revert to it when despondent. Facing resistance at work I often express my feelings on Facebook, something which is not looked upon favorably by the university's administration. To which I replied with Ghalib's couplet mentioned earlier: Why should the heart not be troubled by the turning of days, a human after all, not a cup or wineglass am I.

Let us look at who else was opposed to technology. Cathy Davidson, in her interesting book, "Now you see it", tells the story of iPods at her university Duke. The news at that time appeared on the cover of Newsweek; Steve Jobs quoting – iPod there for I am. Duke University was distributing iPods to all its undergraduate students. This was frowned upon. Why a world class university with plush endowment, stoop to such a cheap publicity stunt. The Chronicle of Higher Education, one of the popular research magazines, for which professor vie to be published, had this to say: "Transforming the iPod to an academic device, when the simple fact of the matter is that iPods are meant to listen to music. It is an unnecessarily expensive toy that

does not become an academic tool simply because it is thrown into a classroom.” Most other senior members of the higher education community agreed with the assessment. If this was the opinion in the US, think what opinions about technology exist in our context. Whatever the pundits thought Duke’s efforts were based upon an Apple initiative in which it would partner with six campuses in sharing its technologies for proposed campus wide usage – a business and education partnership. The partnership was entirely exploratory in nature and no concrete use, perhaps the reason for the intense opposition. One university selected Apple’s PowerBook loaded with iLife video and audio software. Another chose e-portfolio online workspace, which would develop multimedia projects (quite understandable from an academic viewpoint); while another Apple’s audio software for creating audio archives. Duke’s choice of flashy music listening gadgets, which young people loved but baffled adults, created a strong reaction amongst the academia. At this time the iPod had no single educational application, since it was not designed as such. Nor did it seem to fall in the overpriced category of information technology (IT). In the backdrop meanwhile huge billboards had sprung up everywhere showing young people dancing against vivid colors. What could all this have to do with education? At the time no one was thinking in terms of their learning potential, since they were clearly about young people and not IT administrators. This was the reason that that intrigued Duke. Their thinking was that educators had to start thinking about the incoming students, who were born after the information age was in full swing. They were the last entering class who as a group would remember the before and after of the internet. If born roughly in 1985, they would have entered grade school when Tim Berners-Lee was writing his first protocols for the internet. These kids would have grown up searching for information online, socializing online, playing game and sharing music. Categories and distinctions that an earlier generation of students would have

observed in school and at home, between knowledge making and play, which came bundled in a new way for this first generation of students who were learning in an informal way the boundaries between learning and play were eroding. Their schools had not changed much but at home online they were already information searchers – they had learnt by goggling. What if instead of telling them what they should know, we asked them to continue the lesson on the internet itself? And let them lead us to a new exploratory way of learning to see if this self-directed way might mean something when it came to education. What if we assume that their experience online patterned their brains to a different kind of intellectual experimentation and what if let them show us where the pedagogical results of such an experiment might lead to?

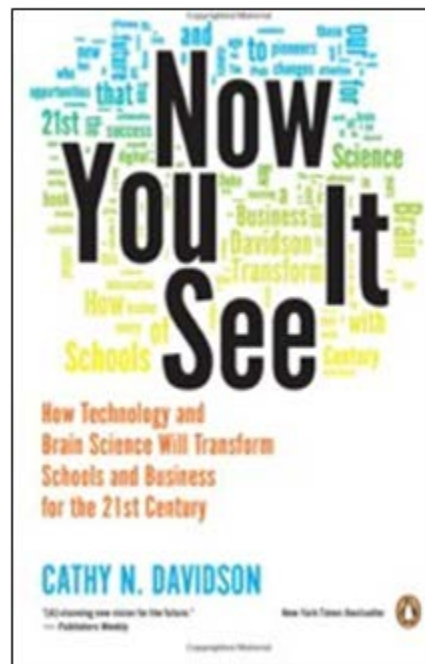
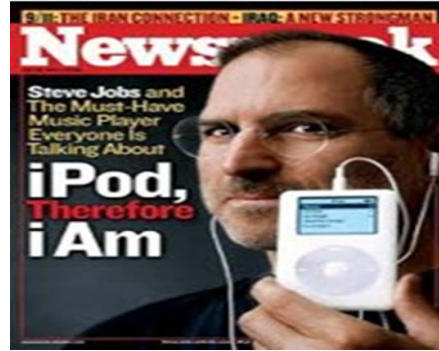
Halftone

It is a simulation of continuous tones by the use of black or overlapping process color dots of varying size or position.

Halftoning Factor

It is a multiplication factor (between 1 and 2) applied to output screen ruling to calculate scanning resolution for optimum output quality. This is also known as the halftoning factor.





Topic 180

The Future of Thinking 1

Another book by the same author “the future of thinking,” narrate the story of all the professors who might be found my college. Like all god teaching our lesson begins with a story of a typical classroom. The instructor is reading aloud from a passage in the assigned work for the week. He looks up to find all students deeply engrossed. Their rapture is alas is not for him but their laptop screens. Their attention on messaging on Facebook. These children do not want to learn. What needs to be done, the professor did what any professor would do. Later that day the professor sent off an email to his colleagues stating that laptops during class should be banned or at least at the minimum Facebook should be banned. What could you comment on this? An outrage! More familiar that students are distracted by their laptops, their attention too readily drawn away from class activities and lessons.

Perhaps another response might be that I am the most hated teacher, few would go down this route. Perhaps the professor should not be sitting at his desk and reading aloud from a book – something quite basic. Incidentally an interesting story: a student came to me in college and told me that he attended every class and lab, a fact which can be verified and consequently I should let him pass. I said without an exam and he replied yes. Upon asking why he said that each teacher goes to his class, you too as a dean come to your office, yet we do not what you do since there is no exam and you get your salaries. Consequently I should be allowed a passing grade for just attending. Technically he was right, if the teacher’s sole function is to read aloud from a book. Each of the responses deserves some consideration. The latter/last raises some questions. Our teaching methodology needs to be reanalyzed.

Halo

It is a light line around object edges in an image, produced by the USM (sharpening) technique.

Handle

It is unique character string identifier assigned to each entry in the NIC WHOIS database.

Use of ICTs in Classroom

The use of television and LCD projectors are in wide use in many LDCs as information presentations.



I Introduction and Overview: The Future of Learning Institutions in a Digital Age

A university classroom. The instructor is reading aloud from a passage in the assigned work for the week. He looks up to find all his students all deeply engrossed. Their rapture, alas, is not with him but with their laptop screens, their attention worlds away. Later that day, the professor res off an email to his colleagues suggesting that laptops be banned from the classroom because of incidents like this. At least FB should be banned?

Why is a professor at the front of the classroom at all? Why is he reading out loud? What are the forms of learning implicit in such an act, and how do those forms edify our concept of learning, education, and the whole process of communicating ideas? How do laptops change the way we learn? And how *should* they change the way we teach?

Topic 181

The Future of Thinking 2

The book we are discussing is a very recent one. The question raised in it critical and much more basic. Information technology and other considerations come later. Why is that classroom architecture designed to give prominence to the professor? While the students are ignorant the professor's job is impart /transfer knowledge; known as the banking model in pedagogy. What are the forms of implicit learning in this act of reading aloud? Is there any learning and what type? How does this edify education, learning and the whole process of communications? The question should be how laptops can be used to teach. Instead of banning them we should utilize them. But prior to that we need to understand what learning is. In our college we face the same problem; here teachers can even fine students. The wish most universities to become highly ranked through strict academic discipline and coercive measures, is paramount.

This was the crux of the problem at our university. Faculty point out that multimedia and technology are the culprits. As a student sits in front of the dean in fear, once Mark Zuckerberg once sat in front of the committee at Harvard. One day I enquired of my senior instructors as to the strengths and weaknesses of our students. The strengths were stated in writing and included: reasonable intelligence; posses' rudimentary knowledge; no learning disabilities; digitally smart; energetic; will to learn; optimistic; and healthy. What then is the issue? Out of the 700 students enrolled, 550 dropout every year. We will discuss this in detail later. The stories behind this started coming to light through face book. In terms of weaknesses: there failure rate was very high; did not possess academic habits (what are these); no fundamental knowledge of morality; short attention span (but this common according to pedagogy); zero expression; weak analytical

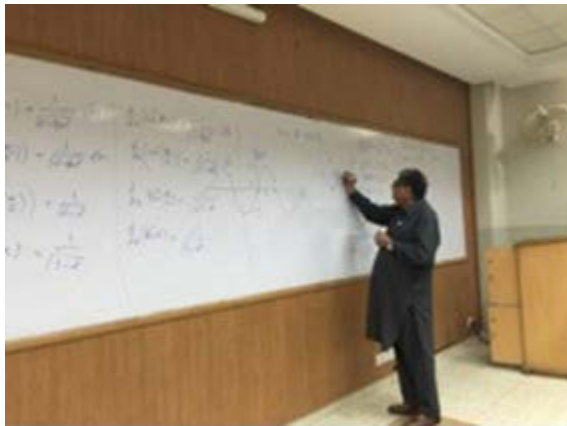
skills (these need to be de-learned from earlier teaching); no contemplation; cultural change; distraction due to girls. Now from this book, “disrupting class,” we will try to look at what constitutes a good university.

Handshaking

It is the process computers and modems go through in order to establish a connection and agree on the speed and protocols for data transmission.

Hayes Compatible

It is a modem is considered Hayes Compatible if it supports the Hayes command set (a language used to communicate with and control a modem).



Topic 182

Troubleshooting at the University

Earlier we were discussing that upon joining my new university one came across multiples issues in terms of: education; pedagogy; systems; classroom; examinations; lab conduct; etc. Now this has two aspects; the diagnosis of the problem and its treatment. As to the diagnosis part this is done through an ‘educational audit,’ something which is not available in Pakistan. Unlike a financial audit this is not a purely a quantitative exercise. Once the correct diagnosis is made a remedy can be sought. Again two areas need to be analyzed: systems; mostly procedural including, books prescribed, how exams conducted, what pedagogy utilized, and other similar aspects.

The other consideration is, a system’s implementation, how efficiently, with what intentions, biases is this implemented. Problems can exists in each individual aspect, systems and Implementation, but when compounded a serious issue. There are no ideal systems, every system has some flaw. Yet, some systems are naturally more conducive; greater output with little effort. Leaving human inefficiency aside, constant for our purposes, what is the better system? We look at this in context of an actual case study at my new university. We will analyze all aspects in relation to education; administrative measures, assessments, classroom and lab pedagogies. We would discuss these based upon the prior knowledge we have acquired earlier in the course. But before opening this Pandora’s Box, I would like to narrate some of my own stories in relation to our discussion. These personal stories are closely intertwined with tales of technology, pedagogy and a blending of the two. With this let us look at how we can arrive at a correct system assuming that the human element remains the same. What changes and system is required?

Header

It is the portion of a packet, preceding the actual data, containing source and destination addresses, error checking and other fields. A header is also the part of an electronic mail message that precedes the body of a message and contains, among other things, the message originator, date and time.

High Key

It is a light image that is intentionally lacking in shadow detail.

Topic 183

Early Learning

When I was around nine or ten my mother was quite taken by the desire for me to formally learn recitation of the Quran (Qirat). My maternal grandfather, who was proficient in English, was also able to read and write Arabic. The task of teaching me fell on him but after a while a formal teacher was appointed. From a pedagogical perspective there were issue of understanding, making understand, remembering, goals and my reaction. I was attempting recitation in a language of which I had no inkling. Since no prior knowledge, mentally I could make no connections. The daily lessons of around an hour were a virtual impossibility for me. There was a rebel within me, who questioned the teaching being imparted to me and its requirement. My mind's pedagogical setup was not designed to understand and this lead to frustration, confusion and impatience on my part. During this time, the teacher, who was rather old, passed away. His son took over and since he was closer to me in age we formed a friendship. This allowed me take frequent breaks, during which we chatted. Those were painful days and time dragged on. My mother seeing my distress stopped the lessons, but handed me over to her father my grandfather. My grandfather apart from teaching me recitation also taught me English. The textbook used was called 'Radiant Reading' a very popular book with lots of illustrations. In those days little was known of pedagogy and now it is know that there should be entirely separate textbooks for foreign learners and native speakers. I became deeply troubled and tended to avoid my grandfather at other times. My aunts used to tell me how lucky I was to be taught by him – my luckiness I failed to understand. Now in hindsight I realize why my brain was resistant to this teaching. Our brains are designed that if you try to teach something which he cannot relate to (pedagogical theories) any prior experiences and will face great difficulties in learning.

I had started school rather late, apparently around the 4th or 5th standard. Before that I spent most of my time with grandmother, since my mother and aunt's away teaching. My grandmother would sit out in the verandah. Our house contained many quarters and also temporary low income housing. In the winters I remember gypsies too would join this community. Young girls from these families would come to my grandmother for lessons in the Quran and basic Urdu: which she did without charging anything. The presence of these girls at time would annoy me since after a couple of month they became quite proficient in their studies. Naturally, I considered them to be less intelligent than I was. There was no fixed time for their teaching, since most also worked in other houses; they would come to my grandmother when free. She would cater to this diverse student body by understating and catering to their unique individual styles of learning. At this juncture you must be questioning the reason behind my narration of my grandmother's story. Was she a great pedagogist –no! But what she managed to do was to treat every girl at her level. For girls whose basic knowledge was lacking she used to focus on that aspect. For more advanced students she focused on their particular requirements. She also took into consideration at what speed, depending upon her ability, a girl was able to learn. Due to their humble background most of these girls were quite submissive and used to obey their teachers instructions, unlike me. What was interesting that they were all pretty much able to become well versed in the subjects? Something in which I had failed to do! The reason I believe was that my grandmother taught each girl, according to her ability to learn. She did not treat the entire group, in which there was great diversity, in terms of age and ability – not a standardized lecture. In today nomenclature and in terms of science of pedagogy this is called 'formative assessment.' This entails making a student aware of what his mistake is immediately and only giving him a workload which they can handle. Now I realize that despite her times she

was using a very modern methodology, which was quite prevalent at that time, especially in Madressas. In the next module I take you to another personal story which has great linkage with ‘formative learning’ and ‘summative learning’.

Highlight

It is the lightest tones in an image. It is a spectral highlight is a bright, reflected light source.

Histogram

This is a chart displaying the tonal ranges present in an image as a series of vertical bars.

Topic 184

Board Examinations

When we came to school, naturally the annual grade system was in place. The school was part of the Punjab Board system and we fell under the Multan Board. Board examinations were held after the eight standards, which very few students opted for, or after the tenth standard. Our schools conducted its own exams, attempting to make the board's summative system into a formative one. The purpose, collective progress for all students: the more the more successful their attempt the better there exam results. My results were quite good for FSc exams; securing 10th place in my board exams for Punjab. Applying for my undergraduate degree, I chose UET, which at that time was known as PUET and had one wing in West Pakistan and one in East Pakistan. For admission first priority was given to those who had graduated from Punjab University and the left over seats allocated to other colleges. It was big achievement for me, since at that time the cream of undergraduates from all over Pakistan and even abroad used to constitute the college body. Upon entering the university we discovered a strange system, which by now is forgotten. It was entirely summative system based upon annual examinations. Classes continued for the entire year with little interruption for exams. Throughout the year a lecturing format was used and annual exams the sole tool for student assessment. Those who could not pass their annual exams were given supplementary exams. In the laboratory things were the same; open experiments in which discovered, were not allowed. The experiment, methodology and reading were agreed upon earlier and results recorded. Annual exams were given every year. The irritants which are a part of any system become necessities. Our summative system was no different. One of these necessities was the demand of students for preparatory leaves prior to exams. Having attended the entire years lectures, mostly idling away of time, nearing the

exams the students realize their predicament. Since no feedback was received during the entire year, we were quite unsure of our grasp and understanding; also probably looked at their textbooks for the first time. It is not that the students were not intelligent but intelligence does not imply cramming a year long course in one week.

Continual assessment is required which a summative system does not account for. The students, cream of the country, were there to study. They were motivated and wanted to get good results in their exams; consequently the demand for study leaves. These were usually granted, after student protest, implying about two months wasted in a year. In the exams 40% were considered pass marks. Out of a total of eight questions, five had to be attempted. This meant that focused and limited preparations were made to secure passing marks. If there is an inherent limitation in a system it is natural for it to be exploited – the blame does not lie with the students. For those who could not pass the exams, supplementary exams had to be taken. The logic behind these exams was strange. For those who had studied throughout the year there was no room in the summative system for repeating earlier learning. Due to this students and some faculty started protesting concerning the deficiencies of the system. This was happening with the backdrop of political changes – democracy was reintroduced.

These political changes were reaching to universities, where there was hope of change. Students sensing the changing atmosphere increase their demands: grace marks for passing an exam were demanded. To counter this university had two options; either to eliminate the existing flaws in the system; or comprise with the system, giving way to student pressure – this is what eventually happened with the university succumbing to the system. The system contained no margin for repetition of learning, without which they could not pass their exams. Consequently,

the need for supplementary exams and grace marks. Despite its limitation this system still managed to produce many highly qualified engineers, entrepreneurs and innovators.

I remember distinctly that a movement was incepted at the university around 1974, in which some faculty and students pointed out that this was an obsolete system, whose time was up. A new system was required in which teaching occurred in parallel. When these demands became widespread and with a changing political and social scenario in the background; a committee was formed comprising of students and faculty. The committee's mandate was to bring about an alternate system; in which testing and assessment would take place in conjunction.

Home Page

This is the document that is displayed when you first open a web client program. Also, commonly used to refer to the first document you come to in a collection of documents on a Web site.

Host

It is the main computer system to which users are connected.

Topic 185

Adopting the American Model

I too was part of this committee, along with two professors who played an important role in molding and defining me. I am where I because of them - Asif M. Hassan and Dr. Q. U. (Qamaruddin) A. Khan. At the commencement of the committee meeting we were given exciting new ideas. We were told of their experiences in America; semester system; quizzes, graded on the same day; midterm and finals with results available in a couple of days; no preparatory leaves; repetition of course for those who do not pass. All these were new ideas for us and underlining this was the fact that teaching and assessment should be done in parallel. At that time we did not know what the pedagogical difference between an annual and semester system. While was entirely summative in nature the American system was designed to be formative. In the committee we proposed a semester system, consisting of four month and an evaluation/exam for every month. No midterms or finals and weekly quizzes would be introduced whose results would be provided immediately.

Theoretically this constitutes a move towards the formative system; in an ideal system there would be instant feedback. There was an almost palpable quality to our anticipation of changed system. The change was later made but unfortunately due to lack of proper implementation the annual system was reverted to. It was not a failure of the system rather the associated people, whose work load increased. Apart from the extra exams, quizzes and assignment needed to be marked and returned to the students. Even till date the most answer papers are not shared, let alone discussed. If you ask for rechecking of your paper today at the Punjab University, it implies that the paper is checked to see that all the answers are marked and total added properly. If a student has another query or proposes another viewpoint contrary to the

examiner or enquires as where we went wrong there are no answers - a major deficiency of that system. After these stories we turn back to the problems facing our existing university and see how the two are related.

Hostname

It is a name which officially identifies each computer attached to the Internet.

Hotlist

This is a file containing lists of Internet Web pages, enabling you to easily access your favorite Web pages.

Emerging Practices of using ICT in Education

Handheld devices have one great advantage over many other types of ICTs – they are mobile – and anecdotal evidence suggests that this mobility could provide help meet certain needs of schools and educational systems.

Topic 186

Flawed Logic

In the committee discussed earlier, the point was often made that there was no need for a new system. If all the students start studying and teacher's teaching properly there would be no issues. An argument much like: if all human beings became good there would be no problems. Dr. Asif's retort to this, I still remember after a gap of around fifty years, it is people that affect the system and the system effects people. There is a system which needs to be implemented and people that need to implement this system. How these people tackle various variables during implementation would discover its effectiveness.

Similarly, once the system is place it determines what output would take place, i.e. summative system leading to annual exam based system. So according the professor it is the right combination of people and system which effects society as a whole: invaluable advice which remains with me. You will see how this applies in our case. In our developing country there are major issues of; poverty, lack of resources, little education, issues of economic and social justice, intolerance and violence. Is it because we as a people are terribly flawed? There are other nations where things are entirely the other way around. Are they better human being than we are? No rather it the system which affects the people and people the system; the right combination of the two effecting society.

HP

It is abbreviation of Hewlett Packard.

HTML

It is called Hyper Text Markup Language. A system for tagging various parts of a Web document that tells the Web client programs how to display the document's text, links, graphics and attached media.

Topic 187

Problems despite Facilities

But before going onto the actual problems we also need to look at all the salient points of the existing system. Our task then would be to take a look at the entire scenario and determine if it is a problem of the people, system or of combination of the two. We have qualified instructors, teaching assistants (TA) support and assistance from laboratory engineers (graduates), again rare in our context. Customized university infrastructure with a well-equipped library is also situated. Since a private college, things are expedited and there is no red tape. No student unrest or demands; a semester system in place; air-conditioned classroom; most modern and excellent curriculum; teacher evaluation by students in place. Fees are relatively high which should create extrinsic motivation for students to study. Exam and quizzes answers shared with students or supposed to be?

We now need to look at the symptoms of the prevailing disease/problem. Having described the inputs we need to see what output. Student dropout rate within two years is more than 70%. I equate this to a 100 customers walking into a giant retail shop and purchasing goods, for which they have already paid, yet receiving nothing. Can this shop remain viable? This dropout rate is truly surprising and the fee money involved more than significant for a majority of these students. Once they dropout they go to places which are comparable in terms of fees and facilities but lacking: qualified and celebrated faculty; no quality control function as in our university; no TA support afforded or provided. The students who choose to stay are in a constant conflict and disgruntled, despite all the facilities. They complain that they have lost their self-confidence. They are extremely dissatisfied and not optimistic about their future. They are disturbed and extremely fearful.

Hue

This is the color of an object perceived by the eye due to the fact that a single or pair of RGB primary colors predominates.

Hypermedia

This describes hypertext in which various types of data can be stored - sound, images, video and so on - as regular text.

Topic 188

Educational Audit

Now we turn towards the underlying reasons behind the prevailing situation. The administrative challenge of a 70% dropout in two years is daunting and the main reason for my induction into the university. To come to grip with this challenge I approached the qualified faculty members. A well know faculty member and celebrated professor in Lahore, upon enquiry, had this to say. First the strengths were highlighted: reasonably intelligent; rudimentary proficiency in subjects (prior Knowledge); no learning disabilities; digitally smart (proficient in terms of information technology); optimistic, or at least were prior to joining; fundamental will to learn; energetic and physically healthy. What then are the weaknesses?

The main ones lighted include: no academic habits, implying lack of learning habits, not their faults since we taught them (wrong pedagogy) and we have to make them unlearn; weak moral sense (not sure how that applies); short attention spans, this is how we are designed; weak comprehension, issues with speaking and understanding since immense diversity; non-reflective capabilities, no thought on repercussion or reflection; weak analytical skills; cultural shock via gender issues, not really understood. As you know there are three presences in a classroom; social, teaching and cognitive. The absence of any one of these elements can result in failure in learning; the most important of these elements is social. A classroom atmosphere needs to be nurtured which is open and where students co-mingle and discuss (peer learning). Reasons like these only seemed to me an effort in glossing over their own incompetency. At this point we felt the need for a comprehensive survey which was carried out. The results not only informative but critical to our diagnosis of what was ailing the university.

Hypertext

It is a text-linking strategy that lets you jump between related information in a document by clicking on a button or highlighted word. On-line help systems often use hypertext links, as do some programs designed for the electronic distribution of documents.

Define Scripts

This is a type of program that consists of a set of instructions for another application or utility to use.

Topic 189

Balanced Results

This course varies in comparison to other courses. The modifications and advances we face in terms of technology have to be considered for the course to remain relevant. Today, again we commence with a line from a couplet which I had recently received from quite a few people:

Bohat dair kar di meherban attay attay

Too late your arrival o benevolent one

As you know I am currently dean at a university. We have a semester system there and recently our final exams were held. These have been marked and the results in process of compilation. According the existing process, before finalization these papers come to me to be signed off. The first thing I see is that if the results are balanced. Recently the aggregate result for calculus papers came to me. The subject is being taught by a seasoned and senior female instructor. She is considered the most qualified mathematics teacher in our department. When her results came in, I saw that half of the class had failed. Upon, enquiry she told me that she had not failed them rather they had failed themselves. If they were not going to study properly or prepare what other results were expected? At this juncture I showed her an email by one of her students, describing her class: what transpires in her class; what pedagogy used; interactivity; her non-receptiveness to student queries; and lack of support to students. To this she rightly said that it was not appropriate that I should blame her based upon one student's input. I told her that I too had audited one of her classes and prevailing atmosphere was at the least not friendly. Her response was that education was not designed for friendliness and is not conducive to learning. She kept referring back to the fact that I had blamed her for failing 50% of the class and broke down and started crying. She kept insisting that I had questioned her dedication and integrity.

Mathematics is one of the fundamental subjects and much work had been done on it in context of education and its learning globally; especially in its branch of calculus. I told the lady teacher that she must be around 45 and associated with teaching mathematics for the past 15 to 20 years; there was one question I would like her to answer; a question which I would like to put across through the assistance of this program and virtual university; to the million of mathematics teachers in Pakistan and globally. You must have studied hundreds of books on mathematics, yet have you come across any books; one book, on pedagogy, learning, problems of learning of mathematics or teaching mathematics? A fundamental question the answer to which I knew. She replied in the negative and despite the fact that she did not say so; I felt that she did not feel the need for it. Having completed her higher studies in mathematics, she felt that this was enough: a fundamental flaw or oversight which extends across most of our faculty. As soon as she left my office, I thought about all the books and resources available and the tremendous amount of work that is taking place in this field. I have been involved in this field for the past fifteen years, yet every time I go to a website like Amazon or search for new pedagogical resources, so many new things come up that one is amazed and joyous at that same time. On her departure I again searched for these resources and found a book, which I had not come across earlier. The book was entitled, 'Mathematical Mindsets'. I greatly felt the desire to order this book and others related books; the website provides you with similar books being purchased. I could not wait for it to come in time for the workshop on mathematics, which I am trying to organize. For the workshop I would be distributing books and relevant literature. Unfortunately, I did not have my credit card at the moment, but Amazon allows you some preview of the content, which I thought could be posted on Facebook. While reading this content

had a strange sense of déjà vu, the author was reaffirming my viewpoints, intuitive, but with neurological and psychological justifications. The topic was conflict resolution.

Having posted it on Facebook and requesting my friends that if they were coming to Pakistan, they should pick up a copy for me, I went home. Opening my computer and lo and behold on my messenger the book was available – technology. I immediately started reading and found it to be very enjoyable. I had received another message from a former student at LUMS who was traveling to Pakistan, over the weekend, saying that he would bring the hard copy of the book for me. Initially, I refused since I already had a copy, but requested him for a hardcopy. What was so interesting about the book?

In the first page of the book, it refers to a Stanford professor who has this to say: “Every time a student makes a mistake in mathematics: they grow in knowledge; they move forward in learning something important; they move a step higher towards knowledge.” While what we do is to fail them. When he made this statement at Stanford, a hush fell over the entire room. Most thought this was entirely absurd. “One reason it is so significant that it speaks to the huge power and value of mistakes, although students everywhere think when they make a mistake they are not a math’s person or worse that they are not smart in mathematics. Many good teachers have told students for years, that mistakes are useful and that they show we are learning. New evidence on the brain and mistakes says something much more significant.”

I-beam

This is the blinking vertical line that shows the point at which text or graphics will be inserted.

IBM

It is the abbreviation of International Business Machines Corporation.

JO BOALER

FOREWORD BY CAROL DWECK

MATHEMATICAL MINDSETS



Unleashing Students' POTENTIAL Through
Creative Math, Inspiring Messages and
INNOVATIVE TEACHING



JOHN F. BAKER
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The Power of Mistakes and Struggle

I started teaching workshops on how to teach mathematics for a growth mindset with my graduate students from Stanford (Sarah Kate Selling, Kathy Sun, and Holly Pope) after principals of schools in California told me that their teachers had read Dweck's books and were "totally on board" with the ideas but didn't know what it meant for their mathematics teaching. The first workshop took place on Stanford's campus, in the light and airy Li Ka Shing center. For me, one of the highlights of that first workshop was when Carol Dweck met with the teachers and said something that amazed them: "Every time a student makes a mistake in math, they grow a synapse." There was an audible gasp in the room as teachers realized the significance of this statement. One reason it is so significant is that it speaks to the huge power and value of mistakes, although students everywhere think that when they make a mistake it means that they are not a "math person" or worse, that they are not smart. Many good teachers have told students for years that mistakes are useful and they show that we are learning, but the new evidence on the brain and mistakes says something much more significant.

Topic 190

Reasons for Failure

In our department at the university, out of 700 total admissions, the failure of 550 or so students was the topic we were discussing earlier. The major reasons highlighted by our highly qualified faculty: : no academic habits, implying lack of learning habits, not their faults since we taught them (wrong pedagogy) and we have to make them unlearn; weak moral sense (not sure how that applies); short attention spans, this is how we are designed; weak comprehension, issues with speaking and understanding since immense diversity; non-reflective capabilities, no thought on repercussion or reflection; weak analytical skills; cultural shock via gender issues; were not really acceptable to me. We should then look at what the experts (technology, assessment, learning and pedagogy), in this field, have to say: facts which should help us in identifying the critical weaknesses in our case study.

Before we turn to them we should also look towards you, who by now should have acquired some expertise. What if this problem at my university was put to you? Why the high dropout rate despite the availability of all facilities; how would you go about an educational audit? This forms a critical component of our course. Information technology and pedagogy, there right blend, can only play a role in education, once the underlying problem has been identified. This is what happened in the West, statistical data on these problems was gathered and studied based on the rules of science of learning. The findings thus are not solely based upon teaching experience, rather on actual research. The book ‘Academically Adrift’, contains the following quotation from Harvard University’s president: “Colleges and universities, for all the benefits they bring, accomplish far less for their students than they should.” Many students graduate college today, according to Bok, “without being able to write well enough to satisfy

their employers ... reason clearly or perform competently in analyzing complex, nontechnical problems.” While concern over undergraduate learning in this country has longstanding roots, in recent years increased attention has been focused on this issue not only by former Ivy League presidents, but also by policy makers, practitioners, and the public.

Icon

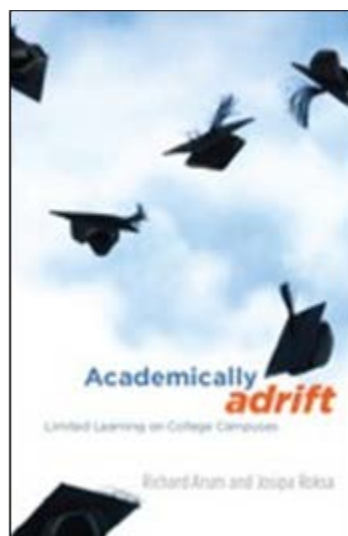
This is a graphic symbol, usually representing a file, folder, disk or tool.

Image Map

It is a graphic which is divided into regions or "hotspots". When a particular region is clicked, it calls up a web page that has been associated with that particular region.

Existing models of ICTs

The community telecentres movement has grown enormously in the past decade. However, sustainable models for such centres have not emerged. Increasingly, multi-purpose centres based in schools are seen as having a greater likelihood of achieving sustainability, but little data supports such beliefs.



“Colleges and universities, for all the benefits they bring, accomplish far less for their students than they should,” the former president of Harvard University, Derek Bok, recently lamented. Many students graduate college today, according to Bok, “without being able to write well enough to satisfy their employers ... reason clearly or perform competently in analyzing complex, nontechnical problems.” While concern over undergraduate learning in this country has longstanding roots, in recent years increased attention has been focused on this issue not only by former Ivy League presidents, but also by policy makers, practitioners, and the public.

Topic 191

Critical Thinking

What was happening was that at top American universities only the content knowledge was tested. Critical thinking, analytical and problem solving abilities were not tested and compared to what abilities they possessed upon graduation. Upon testing the results, which we had mentioned earlier, were shocking. In the top university the difference was not apparent, since the output was of high quality so was the input. While in the less academically sound universities the difference was apparent; somewhat similar to our situation here.

Another Harvard professor, from his book 'Disrupting Class' initially attempts a definition of education: I continually reiterate this point since an innate understanding of education is prerequisite. The main points are: maximize human potential; facilitate a vibrant participatory democracy, in which we have an informed electorate that is capable of not being spun by self-interested leaders; capabilities, skills and attitudes that help our economy remain prosperous and economically competitive; nurture the understanding that people can see things differently, and that those differences merit respect rather than persecution. In the book he goes on to further state that: "managing innovation successfully has been the primary focus of my research and writing at Harvard. I am teacher, the husband and son of teachers, but I am not an expert in education. I have practiced it for sure but until we began writing this book I had not studied education." Strange thing about most people involved with technology, pedagogy and education come from other backgrounds. For instance by training I am an electrical engineer. "Nearly a decade ago however, representatives of a national network of school reformers called education evolving men such? who had played a pioneering role in the chartered school movement, visited me with a proposal." Their main query concerned the amount of innovation in

what was being taught. This is the point where he started thinking about education and how content knowledge is not a sole academic criteria/goal. “We simply cannot get all the schools; we need, by trying to fix the ones that have compelled me to accept this invitation. I thank these pioneers who have dedicated their lives to the improvement of our schools in persuading me to join the movement.” The movements main thrust was on reduced costs in providing world class education to the poor and the oppressed while remaining sustainable – anywhere, anytime, anyone. This naturally hinges upon effective use of information technology.

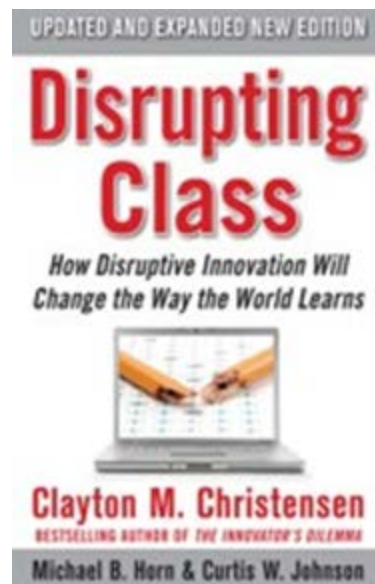
In terms of American schools, the author feels they are a failure. He accepts that they meet none of the objectives of education described earlier and are in a rather sad state. As we look at what deficiencies, he thinks exists, we can perhaps compares to our own case here. “We are not doing very well in the journey towards these aspirations, weakening churches and families must shoulder their share of the blame for our backsliding and wheel spinning, but most of us wish schools were playing a much more effective role in moving our society towards goals like these.” Our university here too falls within this category. “Why do school struggle to improve, everyone has a favorite theory; one is that schools are underfunded.” To the contrary our university is overfunded. There are no limits on the capital expenditure which I can approve on improving facilities. In fact we have now started to use classroom response systems, which provide instant feedback. So this issue of funding does not seem to apply.

Image setter

It is a device used to record digital data (images and text) onto monochrome film or offset litho printing plates by means of a single or multiple intermittent light beams. Color separated data is recorded as a series of slightly overlapping spots to produce either solid areas of line-art or halftone dots for printing continuous tones.

Import

It is the process to bring data into a document from another document, often generated by a different application.



We're not doing very well in the journey toward these aspirations. Weakening churches and families must shoulder their share of the blame for our backsliding and wheel- spinning. But most of us wish schools were playing a much more effective role in our efforts to move society toward goals like these.

**OUR UNIVERSITY IS
DOING A PATHETIC JOB?**

Why do schools struggle to improve? Everyone has a theory. One is that *schools are underfunded*.

**BUT OUR UNIVERSITY IS
OVER FUNDED?**

Topic 192

The Blame Game

Another reason put across by people is; not enough computers in the classroom: another myth which we seem to have accepted as reality. Again in the case of our university, it is flooded with computers and there are no restrictions in purchasing any further amounts. Another camp blames students and their parents. Teachers often resort to this point. “Educators complain about students who are uninterested and not ready to learn and parents who do not monitor homework or show up for conferences this argument resonates with the public. They say that kids on street corners with their hats in backward pose and their trousers dragging and droopy.” Do these look like kids who want to study? “Indeed just to explain the problem facing schools, a number of students from minority backgrounds who have historically performed least well in the US, have skyrocketed in the recent years: from just over 20% in the 70s to around 35%. This entirely refutes the theory of students not wanting to study. “The population of those who do not speak English at their homes, a population that has also underperformed historically in the United States, has also climbed from just under 10% before 1980 to around 20% today.

Another camp blames the students that they do not want to study. I have taught in five universities and am a witness to the fact that students want to study. The complaint that they are discards is not valid and the students are equally intelligent on the average as in other universities. I have measured their comparative learning ability and they turn out to be similar. Other people point out another flaw, but which is only applicable in the States: teacher unions must be the problem. Here we do not have any teacher unions, particularly in our university. As the dean I have a complete mandate to hire and fire teachers. “Whatever has been described here shows that whatever is needed in a good university is already available, here in my university.

This is rare in a context of a developing economy like ours, where a small minority can afford such education. What then is wrong? Currently my opponents in context of technology and admirers of conventional learning feel that there is no need for distance, virtual or blended learning. The classroom provides the right teaching atmosphere and the physical presence of the teacher is critical to student-teacher interactivity. In response this is what is currently happening, high classroom facilities and qualified faculty is available. How do they explain this anomaly? The example of our university challenges the very basic assumptions about learning.

If we look at the dropout rates in America. There are significant dropouts during transition from high school to college. Why in our context? This is a dangerous exercise we are attempting and can lead to significant redundancies. We will have to be honest and transparent; otherwise we cannot get to the actual root cause. Thus the necessity of an online survey to find out what the students thought about us: our primary customers and hence their importance. Why the alarming dropout rate, enormous prepaid fees and their consequent decision to opt for lesser educational institutions? One thing is quite clear that the major failure of our system is that there are no inbuilt warning signals to flag the problem. We only find out once they dropout; something prevalent in all universities. We only find out about their failure once they fail their midterms. The heads of department do their best by running around to find out what is happening in the classrooms; even if it is taking place - a fact which can be easily ascertained through technology. Automatic and instant feedback is missing. An alarm should continue to beep in all the concerned offices if no learning is transpiring in the classroom. This feedback should be openly available to all concerned. Here we are talking about a paradigm shift in which only technology can solve these problems.

Inactive Window

It is a window that is open but is not the top window.

Info Path

It is also called as Internet gopher for the University of California at San Diego.

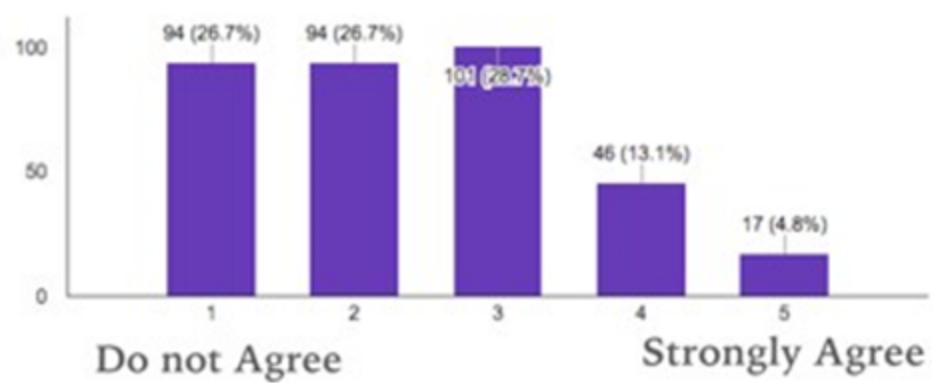
Perhaps there's a problem
because there *aren't enough*
computers in the classroom.

**BUT OUR UNIVERSITY IS
FLOODED AND MORE CAN BE
PURCHASED?**

**THE BASIC ASSUMPTION THAT WE SHOULD HAVE
NICE INFRASTRUCTURE - GOOD INSTRUCTORS -
LABORATORY SUPPORT AND YOU WILL END UP
WITH A FIRST CLASS UNIVERSITY?**

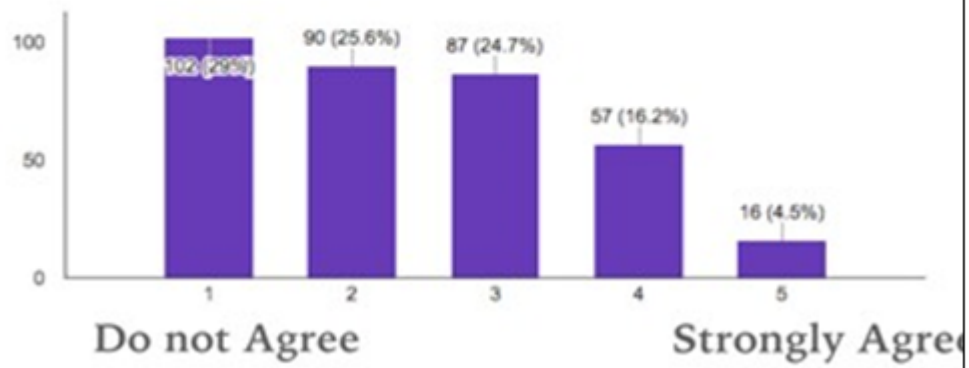
**THIS CHALLENGES SOME OF THE
VERY BASIC ASSUMPTIONS?**

Instructors make the subject interesting (352 responses)



Feedback regarding how to improve my work is provided

(352 responses)



Topic 193

Assessing Instruction 1

What did we discover through our survey? The survey comprised of 352 students. Do the instructors make the subject was the first question; an important one. The unanimous consensus by the students was that this was not happening. Is there feedback concerning improving my work? The majority replied in the negative. If their mistakes are not highlighted what chance is there for improving their work. What is the point of a singular lecture? You could just provide a recording of it. The question was; that do the instructors test what I have learned or memorized. The third consensus was that memorization took precedence over understanding. What is the point of rote learning instead of exciting problem which they can grapple with? A deliberate question inserted was that does the presence of other gender effect you studies negatively. Overwhelming majority responded that it was not an issue. Some even responded that not our issue but perhaps for teachers it was. The question was reinstated; do you feel comfortable working with the opposite gender for your projects. Again, the answer was no; they had no issues. As it is once in the workforce this is what the situation would be. Have you thought about leaving this university program? Most answered in the affirmative since no actual opportunities of learning. As mentioned earlier mistakes are way to advance knowledge; in our environment the opportunity to leverage their mistakes to learning is not provided: only used to cut marks and their self-esteem. In conjunction these are not small errors but a huge mistake that we are making.

Apart from these questions we also asked for individual comments in the survey. The majority believed that the classrooms were not friendly. The social presence in terms of a pedagogical perspective was missing. An environment of fear prevailed. Discussions,

interactions, questions were discouraged. Some of the teachers were more interested in exercising their authority rather than focusing on learning. A quotation by a Harvard dean seems applicable: “A university is composed of senior scholars who are faculty members or instructors and junior scholar who are students.” There is no differentiation per se between the two.

“Another third entity known as administration, functions to facilitate senior and junior scholars in advancing knowledge.” This learning is a mutual exercise with the students sometimes teaching us and us teaching them. In our case most of the instructors are not teaching instead covering the course outline. Another common comment was that the teachers did not understand the students. They are not aware of what the students are thinking. We had earlier spoken of the Harvard gazette advertising a workshop which was needed for all universities on what the students were thinking. This has implications for prior knowledge, if the teachers do not know this how they can hope to advance learning. The interesting thing here is that the students are diagnosing their own ailments. An often recurring comment and a strong one at that was: the teachers are only interested in covering the course outline on time and if someone did not understand, to hell with them.

Information Technology

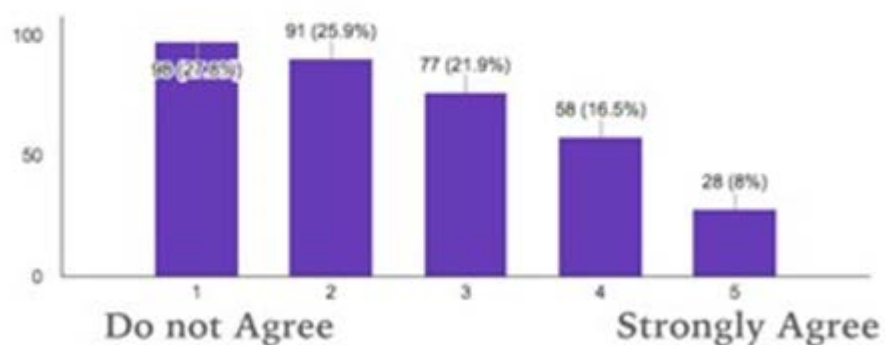
It includes matters concerned with the furtherance of computer science and technology, design, development, installation and implementation of information systems and applications.

Initializing (formatting)

This is setting up a disk (any kind) to receive information. When a disk is initialized (formatted), its magnetic media is divided into tracks and sectors, and structure files that your computer uses to keep track of data are created.

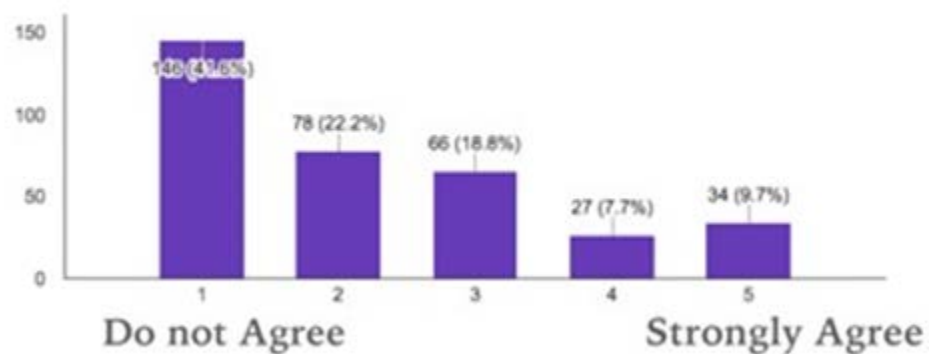
Instructors test what I have learnt rather than what I have memorized.

(352 responses)

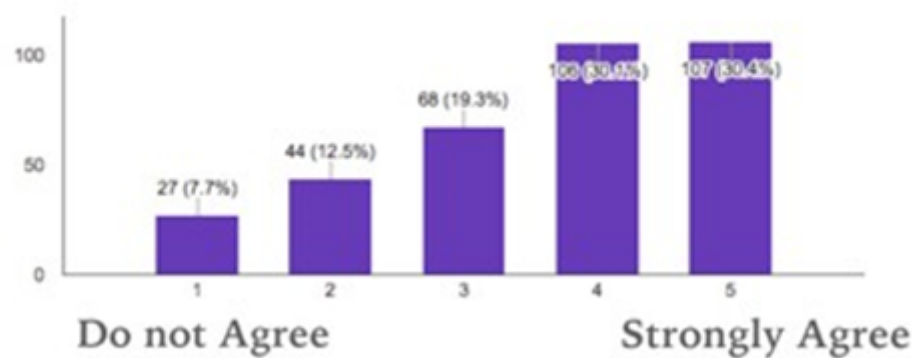


You think opposite gender negatively affects your academic performance

(351 responses)



You feel comfortable to work with opposite gender for your academic project
(352 responses)



Topic 194

Assessing Instruction 2

These comments seem like they are coming from pedagogical experts. They say a friendly atmosphere should prevail in the class and teachers should not ridicule our questions rather try to make us understand our mistakes through example which have application in real life: leading to us being motivated rather than depressed. The linkage of prior knowledge to new knowledge is the first principle of pedagogy. There has to be conflict in terms of prior knowledge for the requisite excitement and friction to generate new knowledge. We should be learning from the students. I realize these views do not add to my popularity, which is rather low as it is. In other comments the students address the second principle of pedagogy.

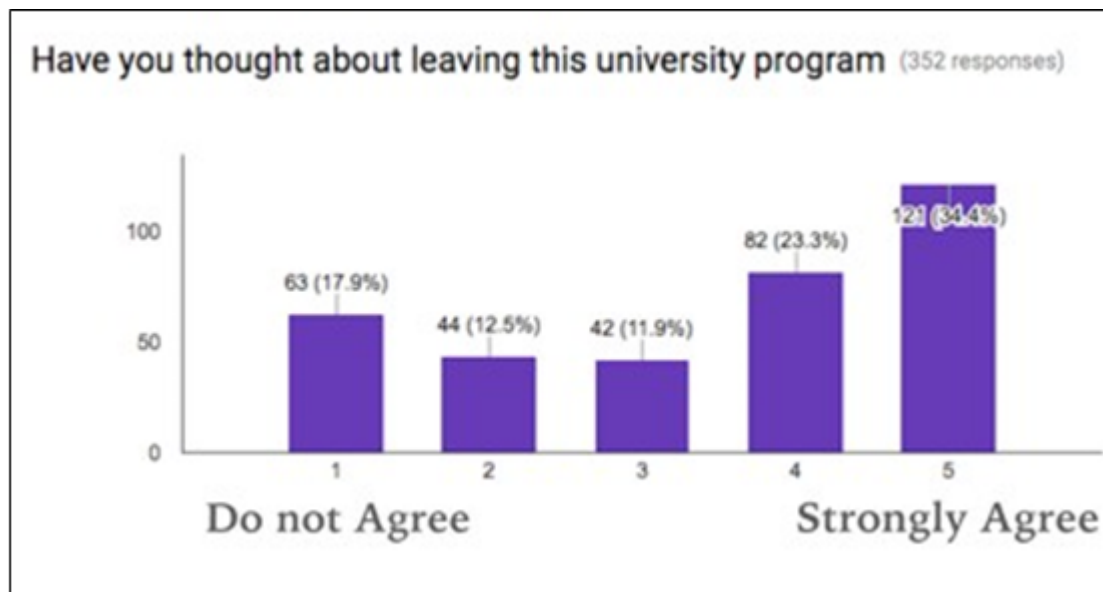
Feedback concerning our assignments and quizzes are not provided. Through this comment the entire problem lays bare. The patient himself has diagnosed the ailment. This is truly an astonishing insight from so young and those considered not quite intelligent. That we have to learn from them is a sad state of affairs. They go on further to add that without feedback they have to appear in exams, where they are doomed to repeat their earlier mistakes. This leads to disagreements between the faculty and students, since they want marks for mistakes which they do not consider as mistakes: a vicious cycle which leads them to eventually dropout from the university. In the context of this extensive educational audit the next step was verification of these insights. This I carried out through attending classes and labs.

Inline images

These are graphics that are contained within a document's textual information. In a Web document, these graphics can either be loaded automatically when the page is accessed or loaded manually by clicking on the image's icon.

Installer

It is a utility that copies system software or an application from floppy disks or a CD-ROM to your hard disk. An Installer may also decompress the new files, remove obsolete files, place extensions and control panels in their proper folders, and/or create new folders.



Topic 195

Enhancing Student Learning

The nations that focus on education, especially its equitable distribution and prevalence, will only be able to advance. The extremist strains and increasing incidence of terrorism, that you observe around us is the other alternative. We now look at “ “, also seen earlier and revisit some quotations. In the introduction there is one from George Bernard Shaw: “The reasonable man adapts himself to the world.

The unreasonable one persists in trying to adapt the world to himself, therefore all progress and change depends upon the unreasonable man.” Some of you upon listening to this must have felt unease or are unsure about the statement. This is good and one of the purposes of our course is to create this friction/conflict for you to learn. The second quotation is, directly related to education, teaching and the classroom by Bertrand Russell (philosopher, mathematician, social scientist) from his book, ‘the unpopular essays’ concerns the flaws within our society, especially its double standards. “Passive acceptance of the teacher’s wisdom is easy for most boys and girls. It involves no effort of independent thought and seems rational because the teacher knows more than his pupils. It is moreover the way to win the favor of the teachers unless he is a very exceptional man. Yet the habit of passive acceptance is a disastrous one later in life. It causes men to seek, to accept a leader and accept as a leader whoever is established in that position. ” We had discussed this earlier in the ‘Pedagogy of the Oppressed’, and is known as the banking model. As you can see from the quotation an entirely false premise and one which require a complete paradigm change? Existential threat

The final quotation from: “Thus an educational experience has dual purposes: the first is to construct meaning; reconstruction of experience from a personal perspective; the second is to

refine and confirm this finding collaboratively within a community of learners. At first glance this dual purpose would seem to reflect respectively the distinct perspectives of the teacher and student, however close consideration of the transaction reveals the inseparability of the teacher and learning roles and the importance of viewing the educational process as a unified transaction. We are simply viewing the same process from two different perspectives; teachers are learners and learners are teachers. These two perspectives raise fundamental questions concerning responsibility of learning and control of the process.” These are radical thoughts and ones which need careful contemplation by you.

In our department at the university, out of 700 total admissions, the failure of 550 or so students was the topic we were discussing earlier; this is the dropout rate in the first year. This implies a financial loss of Rs. 55 to 60 crores to the university - a bad business proposition. The prevailing state of affairs, that can only be addressed through the right combination of technology and pedagogy. The amount of financial loss can be utilized for education and a quantum for augmenting learning in the classroom through automation of assessment. In all respects a successful business. Last time we had looked at some comments from students, most of whom were seeking a friendly environment; instead of one where they were belittled upon their queries and their misconceptions not addressed. Also feedback on assessment is demanded, which is not presently occurring. Once the backdrop and a context to the prevailing situation were acquired, the next step was to actually visit the classrooms and witness what was transpiring. With my extensive educational experience the least I can discern, from the students expressions and their body language, is their state and the amount of learning taking place.

Interface

It is the way a computer interacts with a user or a peripheral.

Internet

The Internet (note the capital **I**) is the largest internet in the world. It is a three level hierarchy composed of backbone networks (e.g., NSFNET, MILNET), mid-level networks, and stub networks. The Internet is a multiprotocol internet.

Use of Different Network to enhance education

Potential target countries for this study include those with established experience and expertise in using radio for developmental purposes outside the formal education sector.



Re: 70 percent dropouts

24 June 2016 at 2:16 PM

Fee per semester = 125,000

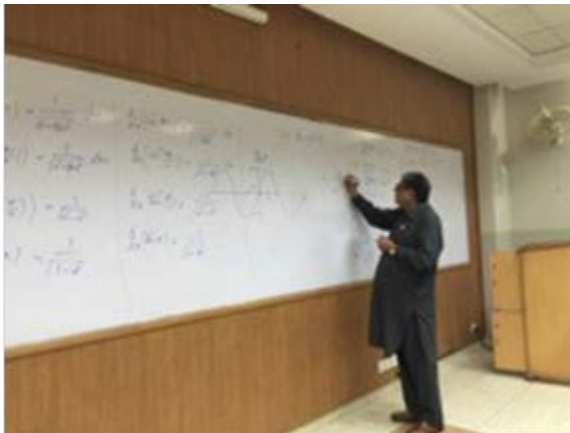
Number of dropouts (estimated) = 500

Total loss per semester = $125,000 \times 500$
= 62,500,000

Total loss per year = $62,500,000 \times 2$ =
125,000,000 (not including Summers)

Tentative Degree duration = 4 years

Loss incurred in 4 years = $125,000,000$
 $\times 4 = 500,000,000/-$



Topic 196

Teaching Class

At the crime scene, where a crime was occurring on a daily basis, this is what was happening. The teacher was writing equations on the board and lecturing. Upon discreet enquiry from students, I understood that nothing was being absorbed. Not being able to contain myself, I enquired of the teacher about functions. The teacher immediately drew a diagram of what functions he was trying to integrate. Yet, all this teaching had no connection in the students mind and it was just a case of the teacher imparting his wisdom to the students, which they were passively receiving. This harks back to the traditional class structure, where a teacher on a high pedestal, in front of the class, with students looking up to him. In the book, “Educational Innovation beyond Technology’ by Nancy Law, Robert Yung and Robert Fox, they address the question of” nurturing leadership and establishing learning organizations.” “Many commentators and researchers consider the classroom environment in which the teacher teaches in front of a class of students seated in rows as an educational solution suited only to the industrial age, where mastery, conformity and obedience were seen as the prime goals, rather than creativity, realization of individual talents and empowerment. Many of today’s classrooms are still typically organized in ways that allow teachers to teach students so that they can master the knowledge and skills in the prescribed curriculum and have the quality of their learning assessed through standardized tests.” Here we are talking about summative assessment.

Now that you have glimpsed a typical classroom we look at the laboratory settings. Why go to a laboratory, many students and teachers enquired of me. The dean’s job is academic administration. Earlier the same people had come to me, upon joining and my announcement of an open door policy, to advise me to not to do this – a view based on a conformist perspective.

They told me that once I open this door, it would imply opening a Pandora's Box. This only added to my anxiety. Only a month had passed and I remember the day distinctly. They told me that in one of their programming labs 'programming fundamentals', I should consult the register. It has been four to five weeks since the lab has started; it begins with a theory class. After which the students are told to translate the problem provided in theory to actual program and run this. In the lab there are qualified laboratory assistants, from FAST, who are also highly motivated. These enthusiastic engineers showed me the register which surprisingly contained zero marks in front of the each student's name. The first zero was for the ungraded portion of the task assigned soon after their first lecture; which was derived from it. Apparently this was their first or second exposure to programming. They were allocated a fixed time and the task assigned relatively simple. Yet, no student managed to complete his program and consequently was given a zero. Upon my enquiry that if any feedback was given concerning their mistakes?

The reply was none, since only asked to assess their programs – a perfect example of summative assessment. Next the students were assigned another small programming task, with some linkages to the earlier task. Again the results were the same. No student managed to complete the assigned task in the required time. Consequently they were awarded zero marks. There was also a third ungraded task. I was sure that again the students were given a zero and this proved to be correct. Out of a three hour lab, in the last hour their graded task began. I told them they did not need to tell me what had happened and what marks were given. In the initial ungraded part, which they could not finish, due to a small syntax error, their mistakes were not pointed out. How could they hope to accomplish the remainder of the tasks? But that was the system and one which they considered very efficient, transparent and strict. It would only allow the students capable of advancing to go forward while those who could not discarded. I told what

was the need of formality and wasting of time, since you could have directly given them a zero. They disagreed and since it was their duty. When all students continually got zeros for five or six weeks, the students that since a new dean has joined and has declared an open door policy, perhaps we can approach him.

They came collectively and hesitatingly (passive acceptance, obedience, fear and respect) to my office - a victim of our society's conformist norms. I told them to sit; which they again did with great reluctance. They explained to me their entire dilemma, which led me to visit their labs and the crime scene. When I got there they were attempting the first ungraded part; most could not complete the compilation and those that could not get the appropriate result. I enquired why did they did not consult the lab assistants or among themselves, they said they were not allowed. I sat there for an hour and half and saw that the students wanted to learn, they had paid serious fees to attend, and teachers wanted to teach and paid got rather well to do so in addition to the lab assistants. Yet, the learning outcome in labs was zero. I called a teacher and showed him the students and their frustration. No one seemed to be enjoying the experience. If things continued in this manner all students would be failed. He agreed and told me this is what happens and after the first year they dropout.

They are not capable of learning programming. How did you come to this conclusion I enquired and he told me he had explained all this in his lecture, prior to the lab. He also told them that I had two hours of office timings in the evening, where they could sort out their queries. I knew that the teachers were honest and there were no questions of implementation. He also said that not even a single student came to him during his office hours: a clear reflection of the student's unwillingness to learn. I told him that this could have another meaning from a pedagogical and learning perspective, if he had bothered to consult it. You would then know why

they do not come to you. Since they have no inkling in class, what would they consult you for? New knowledge requires connection with prior knowledge, where conflict is created, and friction happens and a student feels uncomfortable, errors take place along with neurological changes take place. On the other hand if teaching is entirely receptive (passive); no questions arise. To this he told me that if they wanted to study, they would have made an effort. I told him perhaps this is the treatment which should have been meted out to him – perhaps that is why I am the most hated dean in the university. Teachers should realize that if the students are not learning anything in class, it is entirely their responsibility.

If you remember what Harvard University's physics professor Novak had to say about this: stage occurs where the teacher becomes unsuccessful and is not able to put across learning to students. He just falls into a rut, where he repeats himself. Some intervention is required. Students would not be embarrassed in front of technology. In our English language center when we put students in front of technology, to learn proper pronunciation and speaking, some students asked us to leave the classroom. These were students with the worst pronunciations and who never spoke English in front of us. Yet, they had no hesitation in speaking front of the computer.

Interpolation

In the image manipulation context, this is the increase of image resolution by the addition of new pixels throughout the image, the colors of which are based on neighboring pixels.

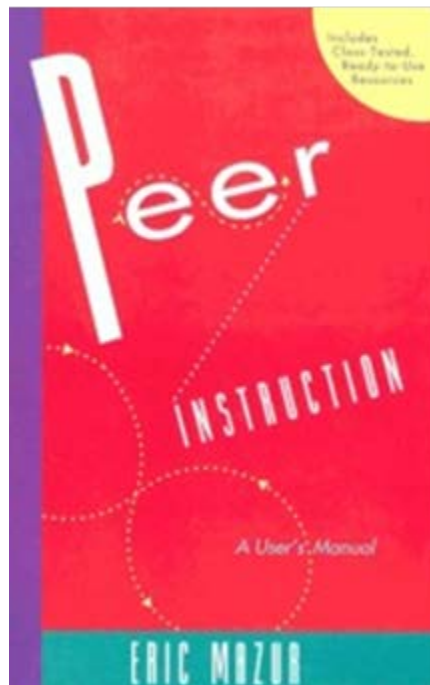
Interrupt

It is a brief interruption of the computer's activity so that an urgent task can be performed.

Many commentators and researchers consider the classroom environment in which the teacher teaches in front of a class of students seated in rows as an educational solution suited only to the industrial age, where mastery, conformity, and obedience were seen as the prime goals, rather than creativity, realization of individual talents, and empowerment.

Many of today's classrooms are still typically organized in ways that allow teachers to teach students so that they can master the knowledge and skills in the prescribed curriculum and have the quality of their learning assessed through standardized tests.





ACTUAL PRACTICE: THE FAILED
PHYSICAL CLASS ROOM? WHY?

IT IS ALSO THE FAILED
LABORATORY EXERCISE?
WHY?

Topic 197

Programming 1

Now we look at a holistic picture of how all these elements integrate. A failed classroom, a failed laboratory exercise, a seriously failed assessment system: a crime which is taking place all over Pakistan. Teachers who have taught for around thirty years are relics from the past. Their mantra is that; those who want to study do and those that do not do so. The students who still manage to transverse the system end up as the best in the world. I told them that I had interacted with the best students, since I had worked at the top universities in Pakistan and consequently fired from them. In my opinion the dropout rate was directly affected by the failed assessment system. The root cause of this failed system was its philosophy. There could be issues with implementation but these could be rectified.

Now let us turn towards assessments; what is happening and what is possible. First programming, since relevant in our context and also a basic understanding critical for all disciplines: a paper entitled, ‘problems in learning and teaching programming, a literature study for visualizations’. The paper states that: “learning to program is generally considered hard pedagogically and programming courses often have high dropout rates. It has even been said it takes about ten years for a novice to become an expert programmer. A student can learn to explain and understand a programming concept for example what does a pointer mean but still fails to use it appropriately in a program.” There is a vast difference in understanding a concept theoretically and its practical implementation. If the faculty does not realize this fundamental difference then a system like ours would be in place. For most of you some of the mystery should have cleared up by now. “Winslow noticed that students many now the syntax and semantics of individual statements but they do not know how to combine these features into a

valid program.” Prior to solving the existing problems through pedagogy and technology our primary task is to identify these problems. “Even when they know how to solve the problem by hand, they have trouble translating it into an equivalent computer program.” What was then the proposed solution? “To help students learn concept knowledge and strategies for their use in programming they developed a problem solving approach for a programming course. In their approach language features were introduced to students only in context to specific problems, little by little. This was shown to have positive effects on student score results and their programming confidence.” There are other solutions then failing all students or telling them that they are incapable of programming. Destroying their confidence and self-esteem is a crime.

IP

Internet Protocol is the standard that allows dissimilar hosts to connect to each other through the Internet. This protocol defines the IP datagram as the basic unit of information sent over the Internet. The IP datagram consists of an IP header followed by a message.

IP Address

Network addresses are usually of two types: (1) the physical or hardware address of a network interface card; for Ethernet this 48-bit address might be 0260.8C00.7666. The hardware address is used to forward packets within a physical network. (2) The logical or IP Address is used to facilitate moving data between physical networks and is made up of a network number, a sub-network number, and a host number. All Internet addresses at SDSU have a network number of 130.191, a subnet number in the range of 1-254, and a host number in the range of 1-254.

IT HAS ALSO THE FAILED
ASSESSMENT SYSTEM?

THE SUMMATIVE ASSESSMENT

THE DROPOUT RATE WAS
DIRECTLY AFFECTED BY THE
FAILED ASSESSMENT SYSTEM?

Codewitz Needs Analysis

Literature Study

PROBLEMS IN LEARNING AND TEACHING PROGRAMMING

- a literature study for developing visualizations in the Codewitz-Minerva project

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Learning to program is generally considered hard, and programming courses often have high dropout rates. It has even been said, that it takes about 10 years for a novice to become an expert programmer [32]. Educational research has been carried out to recognize the characteristics of novice programmers and to study the learning process and its connections to the different aspects of programming. Lately also differences between procedural and object-oriented education approaches have been studied, as Java and C++ have become common educational languages. We will now explore these issues more closely.

Even when they know how to solve the problem by hand, they have trouble translating it into an equivalent computer program. To help students to learn both concept knowledge and strategies for their use, Deck et al. [7] developed a problem-solving approach for a programming course. In their approach, language features were introduced to students only in the context to specific problems, little by little. This was shown to have positive effects on students' course results and their programming confidence. More discussion on problem-based learning in computer science courses can be found in Kay et al. [13].

Topic 198

Programming 2

Continuing with the research paper it tells us: “It takes quite a long time to learn the relationship between a program on a page and the mechanism it describes. Students have difficulties in understanding that each instruction is executed in a state that has been created by the previous instructions. There should be a simple notational machine that simplifies the language so all the programme transformations can be visible, for example logo???. It is based on the fact that all instructions concern moving a turtle that is drawing on the display thus making the program state and transformation clearly visible. However most of the universities are required to teach the programming languages that are close to those used in the industry and only rarely is it possible to select the approach for an introductory programming course based only on pedagogical reasons.

Now the entire problem should be clear; we should not be starting with languages like C++, instead use languages which are easy to teach and understand. Despite the fact that C++ is a powerful language, it is pedagogically difficult to understand. As a teacher this conflict needs to be addressed. “There is often little correspondence between the ability to write a program and ability to read one. Programming courses should include both, in addition some basic tests and debugging strategies should be taught.” Something we do not do. “There are often mistakes in the design and bugs in the code, this requires an ability to track code and build a mental model of the program and predicts its behavior. This is one of the skills that could be developed by emphasizing program comprehension and debugging strategies in the programming courses.

Give them examples of programs that will not run properly, that may give compilation errors that do not run according to our expectations. Allow them to make mistakes in class and

start with the wrong programs.” The labs should be a place of trial and error and learning. This was all from the book “Mental Mindsets,” a hardcopy of which I should be getting soon. In this the author emphasizes, the power of mistakes and struggle. This is not only applicable to programming but applies to all learning.

IP Datagram

It is the basic unit of information passed across the Internet. An IP Datagram is to the Internet as a hardware packet is to a physical network. It contains a source and destination address along with data. Large messages are broken down into a sequence of IP Datagrams.

IRC

This is abbreviation of Internet Relay Chat. It is a program that allows you to carry on "live" conversations with people all over the world by typing messages back and forth across the Internet.

Students have often great difficulties in understanding all the issues relating to the execution of a program. Du Boulay [10] stated that "... it takes quite a long time to learn the relation between a program on the page and the mechanism it describes." Students have difficulties in understanding that each instruction is executed in the state that has been created by the previous instructions. He stated that there should be a simple "notional machine", that simplifies the language and the machine so that all the program transformations can be visible. For example, LOGO language is an example of this kind of approach [24], and was commonly used in 80's in introductory programming education.

It is based on the fact that all the instructions concern moving a turtle that is drawing on a display, thus making the program state and transformations clearly visible. However, nowadays most of the universities are required to teach programming languages that are at least close to those used in the industry, and only rarely it is possible to select the approach for an introductory programming course based only on pedagogical reasons. **Pedagogical concerns requires not to use standard Languages**

Topic 199

Programming 3

In our days we relied on punch cards for our IBM 1130. This was followed by interactive computers which used to provide you with feedback concerning your program instantaneously, unlike punch card based computer which used to provide feedback the next day. Despite the fact that interactive computers addressed one aspect, instantaneous feedback, but no pedagogy was involved. There was no formative assessment and entirely based on the summative model.

Programming is much like an apprenticeship in acquiring a particular skill, which cannot be taught in a conventional manner. How does one learn to ride a bike or drive a car, make a turn or change lanes? Usually, we can do actually by doing it and making mistakes and not through lectures. Tell them how technology - interactive computers have gone forward while the pedagogy is still the same?

ISAAC

This is abbreviation of Information System for Advanced Academic Computing. It serves as a clearinghouse for information about the use of IBM-compatible hardware and software as aids to instruction and research in higher education. Membership is free to all students, faculty, and staff at these institutions.

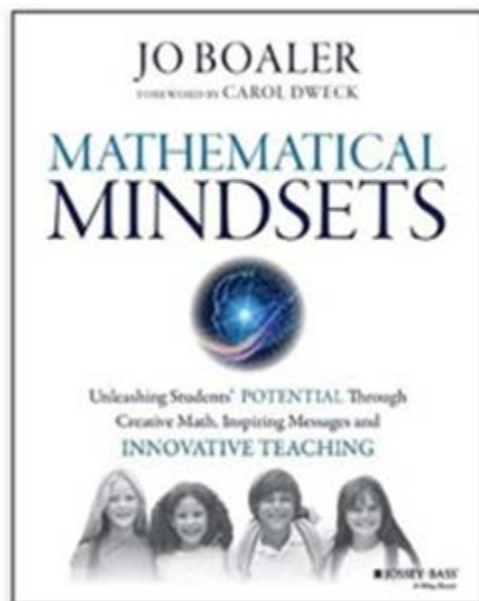
ISO

It is abbreviation of International Organization for Standardization, the group that developed the OSI protocols.

Provision of equipment for ICTs through Public Private partnership

Due to the large costs involved in equipping schools with and maintaining ICTs, innovative public-private-community partnerships are being explored to help reduce costs.

There are often mistakes in the design and bugs in the code. Also in working life, programmers face daily the need to understand a program that is running in an unexpected way. This requires an ability to track code to build a mental model of the program and predict its behavior. This is one of skills that could be developed by emphasizing program comprehension and debugging strategies in the programming courses. Give them examples of a program that is Not running according to expectations - Teach them Mistakes?



The Power of Mistakes and Struggle

I started teaching workshops on how to teach mathematics for a growth mindset with my graduate students from Stanford (Sarah Kate Selling, Kathy Sun, and Holly Pope) after principals of schools in California told me that their teachers had read Dweck's books and were "totally on board" with the ideas but didn't know what it meant for their mathematics teaching. The first workshop took place on Stanford's campus, in the light and airy Li Ka Shing center. For me, one of the highlights of that first workshop was when Carol Dweck met with the teachers and said something that amazed them: "Every time a student makes a mistake in math, they grow a synapse." There was an audible gasp in the room as teachers realized the significance of this statement. One reason it is so significant is that it speaks to the huge power and value of mistakes, although students everywhere think that when they make a mistake it means that they are not a "math person" or worse, that they are not smart. Many good teachers have told students for years that mistakes are useful and they show that we are learning, but the new evidence on the brain and mistakes says something much more significant.

Topic 200

Mathematical Mindsets

As you know we are attempting to study this interesting course. I use the word interesting since keeping abreast of technology and it uses, that we are studying, is a truly interesting exercise. Last time we were looking at a case study of a failed classroom and laboratory system in terms of assessment. In the computer science department a dropout rate of 70% is alarming and entails issues of pedagogy which need to be addressed through technology. An internationally recognized research fact is that commence teaching utilizing programs like C++, Java or object oriented which work? Naturally, what else would a teacher teach? In actuality we should be teaching programs that do not work!

Traditionally, students attempt programs, after being lectured in theory and are unable to get them to work. We should initially allow them to understand the nature and solutions to basic programming mistakes; syntax or designed based errors. In the slide in front of you it says: “This is one of the skills that could not be developed by students emphasizing program compression and debugging strategies in program courses. Give them examples of a program that is not running according to expectations. If you have programmed you would realize how complex and frustrating a task it is; problems can occur due to multiple reasons. The importance of mistakes; an entire course can be dedicated to this. In our conventional educational system mistakes are considered bad: a student’s marks and grades are adversely affected. In the book, mentioned earlier, “the power of mistakes and struggles,” the basic deals with this subject. When a student commits an intelligent mistake his knowledge increases. How this transpires is a big challenge. We now turn towards a problem, addressed earlier. Despite it being simple it

needs to be reiterated since it addresses our real world problem of why 70% of computer science students at our university fail.

Topic 201

Chasing the Rabbit 2

In the book “Chasing the Rabbit” by J. Spear, a doctorate from MIT, he addressed the question of Japanese challenge to the American car industry. In our days at the UET, I remember only two professor, graduate from the States, possessed cars – both Datsuns. This was the mid-60s. In actuality the industry was incepted in the States with the Ford Model-T. As Japanese cars entered in the US market the middle classes soon increasingly turned towards them. Naturally, this was a cause of consternation for the American car manufacturers; who protested since their business was being adversely effected. The two main demands were to impose quotas on Japanese cars and have them build their factories in the States, where all employees would be Americans.

The thinking was that since the same manufacturing facilities (in most cases American facilities more advanced), same workers and quality control, the difference between the products would be eliminated and consumers would turn back to American cars. A similar experiment was tried in education, American universities were replicated exactly in Saudi Arabia, even down to the building yet, the quality of education imparted was not the same. Anyway, once the Japanese manufacturing commenced in the US and imports almost eliminated, the consumer demand for Japanese cars remained the same. To cope with this challenge the car manufacturers turned towards leading educational institutions: a critical function of educational research in the developing world. A graduate engineering student Stephen J. Spear, from MIT, was assigned the task to visit both competing factories and observe what was happening and why the Japanese car preferred. When I was in the States the rule was that only American cars could be rented, to encourage their business. What Spear discovered has great implications for the real life example

we are addressing. It is a simple, small difference which if applied in our classroom can change the face of education.

Topic 202

Chasing the Rabbit 3

On visiting the factories, instead of looking at the quality control function or speaking to management he went directly to the production assembly line and asked for work. He requested for the simplest task, which turned out to be mounting of the first seat. For the assigned task he had two to three for completion, failing which the entire assembly line would have to be halted and production decline. Upon his enquiry as to how the task would be accomplished, he was given a demonstration by the instructor; consisting of mounting, checking for alignment and screwing the seat in place. As an engineer, from MIT, he assumed this would prove no difficult task. When the actual assembly line started he had difficulty in the proper setting of the seat and exceeded the allocated time – failing the task. He was told a worker like him would have to be retrained or let go.

He next visited the Japanese factory. Immediately, he enquired to be assigned to mounting the front seat. This raised some eyebrows; a qualified engineer from MIT requesting for this assignment? Upon hiring he was demonstrated the process by an instructor according to stringent SOP's developed in Japan. The process was based on five steps, similar to the American model, with the first step in fixing the seat and ensuring its alignment. The real difference commenced from the second step, a difference

which if able to be implemented in our classrooms, would make our education world class. Upon enquiring about the second step he was told that he would be shown this only after he acquired proficiency in the first step. The assembly line would start and he would have 15 seconds or so for his allocated task. This was surprising for Spear's but soon was able to complete his assigned task successfully. He was next given the next task, but upon enquiring

about the other steps he was told he would only be given these after acquiring proficiency in the second task. As you can probably now see how such a minor difference would make such an impact upon financial and other variables.

If this philosophy and process could be brought into the classroom: that children only advance in a lesson based upon learning and testing of preceding knowledge, where we would be? We will address this in pedagogical terms and see why this is not being applied, why this system prevails and when it was applied, a century or so earlier. A case of creating requisite foundation upon which a sturdy building can be built; with technology this can be addressed and we can replace our old Ford with a Lexus.

This may seem like a small step but can lead to a huge growth for an economy or conversely a meltdown. So as with a manufacturing education needs a fixed outcome, time can and must be variable. **Testing an assessment at every step is essential.** What will be the results? Stephen was able to master the first step and subsequent steps correctly in the first attempt at the assembly line. This was the difference between Toyota and American car - quality. "Assessment or inspection is interdependently woven into content delivery with fixed results. This minimizes the repair work at the end of the pipeline and there were essentially no dropouts. Without this the results would be catastrophic for Ford motors or a educational

Topic 203

Teaching at Home

We turn to another story: as I had told you earlier we had a great rambling house in Multan. The vast grounds were surrounded by low-income dwellers. Since my father and grandfather had passed away and I lived with my mother and aunts, who would go to teach, my grandmother was alone in the house. She would gather young girls from the adjoining houses and hold a regular class in basic Urdu, Mathematics and recitation of the Quran, out in the veranda. Each girl was of a different age, differing intellectual levels, motivational levels and leaning styles. Her mode of teaching did not consist of lecturing she paid individual attention to each girl. Reminding each one their mistakes the previous day and based upon that what further learning was required. She tried to ensure that the earlier mistakes were clarified before proceeding further. Her teaching maxim was to teach according to their need.

What seemed strange to me was that after a while all girls achieved the basic level of proficiency, in what they were studying. Please consider, for each student; the pedagogy and learning styles are different, each is prescribed different time limits, they are not being assessed through an exam, their prior knowledge is considered. Each factor is variable apart from the output, minimum proficiency in subject matter being handled. Does this not appear strange to you; despite everything being variable the output being constant. What happens in our school, the old industrial model prevails. When giving a lecture it is the same for all students. When they are assessed it is the same. The same preparation and time is allocated for all students. The fees they pay are the same. All are treated equally. The only difference is their grades, i.e. the output? In our school, with its high dropout rate, we are continuously told that these kids do not want to study. Strange that my grandmother who was not even a graduate can lead all students to

success. She learned English from her husband, so that she was able to read stories. In Urdu she had not even cleared the tenth standard. Yet, she managed to teach students who had no previous schooling. In our case we have faculty trained in the States, England, and Europe but cannot teach these students; and 70% of our students feel that they cannot learn computer and leave.

Topic 204

A Story of UET

Now I want to narrate story, which took place in 1988. As a professor at UET, despite the fact that our salaries were quite meager, we were allocated rather large houses which also contained a few servant quarters. The custom was that large families would move into these quarters, rent free, and in exchange they would provide a few free services. The stability provided led to a large number of children, they outnumbered all other residents. Since they had nothing much to do and were not going to schools, they created quite a bit of ruckus. My wife, who is quite energetic in these things, told me that we should open a garage school for these children. She gathered all the students, some entirely there because of curiosity. She had books and stationery donated for free. The one room school was set in motion. All the students were of different ages, but my wife along with other neighbors and volunteers paid individual attention to each student. Again no lecture was given, no exams conducted. The funny thing was that the word of this school reached the vice- chancellor.

At that time the IT bubble had not burst and the authorities thought that we had opened an IT school to make money on the side. Two people came to interview me, upon enquiry as to who they were they told me that they were from intelligence. At that time the students were learning multiplication tables, much like we did in our days – reciting in rhythm. They said that I had opened up a school and was making serious money from it. This was happening while the Kargil war was taking place. I told them the computer school was right in front of them, where servant's children were being taught free of charge. While there is a war going on the intelligence services are engaged in this petty investigation.

Topic 205

Formative Assessment

We now turn towards definitions of pedagogy which explain the differences between what was occurring in my grandmothers and wife's school, in comparison to what was happening at proper surrounding schools. The main difference was that the students were learning. This is what was happening at my grandmother's improvised school and my wife's; at Toyota and in our Madressas a hundred years ago. Please, observe that I have not yet rightly delved into the realm of pedagogical. In the recent language of pedagogical research this is referred to as Formative Assessment. I have used Wikipedia as a reference since, another example of which is now available to us through our cells. This refers to it as 'formative assessment including diagnostic assessment.' From the prefix (formative) you should be able to grasp that this refers to early development; something being formed. In context of our examples: on the assembly line a car being formed; and in my grandmother's school grasping the basic concepts of schooling. Naturally, this does not occur in isolation since any development entails assessment. What does this imply?

It is logical that once a process is completed and something built, i.e. a car; it needs to be tested. Similarly, once taught a student tested. Yet, here we do not refer to this. Formative assessment implies that a thing is tested while being developed or built; this falls under the realm of diagnostic testing. In medical terms this would mean we are diagnosing for disease while it has not formed. "Diagnostic testing is a range of formal and informal assessment procedures conducted by teachers during the learning process, in order to modify teaching and learning activities to improve student attainment." In simpler terms testing becomes an inherent part of the teaching process. In our case, the students have just enrolled and it is the first week; little has

been taught, no topic completed. Yet, the assessment commences; “conducted by teachers during the learning process, in order to modify teaching and learning abilities to improve student attainment.” This allows an assessment of prior knowledge and the current status of the student with respect to where he stands and what should be taught. Pedagogically, while a student is being taught a feedback loop is created while being taught new knowledge.. It is imperative for a teacher to modify his/her teaching accordingly; a continuous process of adjustment.

If this is not done the final product would be flawed and it would be too late to rectify the problem. In terms of car manufacturing this is referred total quality management: a process which was not there for American car manufacturers and one which led to their decline. “It typically involves qualitative feedback rather than scores for both teachers and students that focus on the details of the content and performance.” This is very important, our emphasis is on grades. In my grandmother’s or my wife’s school there were no scores, yet the children were able acquire proficiency is what they were being taught, which was their objective. This also happens in a vocational setting where children act as apprentices in various crafts. Some will acquire the requisite skills in earlier than others, but almost all will eventually acquire these skills; no grading takes place. During the process continuous feedback is provided by the master, another case of formative assessment. “This assessment is commonly contrasted with summative assessment.” By summative we refer to the existing system which is currently in place, consisting of periodic testing. “The assessment seeks monitor outcomes often for purposes of external accountability.” Michael Skriver in 1967, had this to say about evaluation: “there are differences both in terms of goals of the information they seek and how the information is used for striven formative evaluation gathered to assess the effectiveness of curriculum and guide school system choices to which curriculum to adopt and how to improve it.”

“Benjamin Bloom took up the term in 1988 in the book, ‘learning for mastery’; formative assessment as a different tool for improving the teaching-learning process for students. His subsequent handbook on formative and summative evaluation written with Thomas and George showed how formative assessment could be linked to instructional units in a variety of content areas. It is this approach that reflects the generally accepted meaning of the term today. For both these authors, assessment, whatever its other uses, is only formative; if it is used to alter subsequent educational decisions.” This implies changing the entire methodology or content if required in moving forward. “Practice in a classroom is formative to the extent that evidence about student achievement is provided.” Once understanding is ensured further progress is made. In our schools and those around the world the inverse is true. “Evidence about student achievement is interpreted and used by teachers, learners or their peers to make decisions about the next steps in instruction that are likely to be better or better founded than they would have taken in the absence of evidence that was provided.” This not being the case, in our schools there is tremendous pressure to promote children, which results in high dropout rates once transition is made from schools to colleges based on summative evaluation; at which point it is too late to rectify the situation. The existing sad situation of our education is entirely our responsibility and not that of the children.

This leads us to technology and interactive learning applications; these contain the provision for instantaneous feedback. In the existing format through which I am addressing you, there is no room for feedback – a model based on summative assessment. Why are we so interested in game, technology based or actual sports? The element of instant feedback, arising from a player’s actions creates the requisite interest, for players and spectators; a fact which has great repercussion for interactive game design and learning applications, content design criteria.

Formative assessment, including *diagnostic testing*, is a range of formal and informal assessment procedures conducted by teachers during the learning process in order to modify teaching and learning activities to improve student attainment.^[1] It typically involves qualitative feedback (rather than scores) for both student and teacher that focuses on the details of content and performance.^[2] It is commonly contrasted with **summative assessment**, which seeks to monitor educational outcomes, often for purposes of external accountability

	Summative Assessment	Formative Assessment
Time	At the end of a learning activity	During a learning activity
Goal	To make a decision	To improve learning
Feedback	Final judgement	Return to material
Frame of Reference	Sometimes normative (comparing each student against all others); sometimes criterion	Always criterion (evaluating students according to the same criteria)

Topic 206

Formative or Summative Assessment

Again turning to Wikipedia, a comparative summation of formative and summative assessments are provided through a table. The key distinguishing criteria are; time, when employed, goals, availability of feedback and frame of reference. Summative assessment takes place at end of learning activity; which at most time is too late. Comparatively, formative assessment contains instantaneous feedback. In terms of goals in summative assessment a final judgment, based on grades, has to be made. While in formative no judgment is made rather the learning process continues: to improve learning and not grade students. Learning should not be grade driven rather the process itself should be enjoyable and its own reward. The elements of frustration, conflict, fun, excitement and cooperation are necessary constituents. In summative assessment feedback is periodic, while in formative continual.

Summative assessment often leads to final results which are less than desirable and have huge adverse connotations for college if they go on to it. When a teacher at our college narrated that 70% of the students had failed, and I asked why he had failed them, he said they had failed themselves. I enquired how many quizzes were given, none was the answer. In classrooms they were no interactive question and answer sessions. Only the final judgment at the end of the year was in place. We will continue to revert back to these assessments, since critical in terms of pedagogy. Since becoming dean I have tried to bring in some reforms. A course in algorithms design course which had a very high dropout rate, we have created refresher courses for the summer. There are four sections consisting of fifty students each. Summative assessment no longer holds sway, students are encouraged to engage small groups and see what they comprehend. Similarly, students are encouraged to engage more and think rather than sitting

mutely and taking notes. Classroom response systems are being deployed. We need to learn our lessons from the earlier stories and lecture no longer.

Topic 207

God is my Only Helper

In the earlier modules we were looking at a case study of a university where I am the dean. We are particularly analyzing the high dropout rate and its reasons; in context of theories of learning, technology and pedagogy. As you know the monsoon season has begun and the weather milder. While lecturing at another place and trying to convey the necessity of passion for teaching and learning. Some topical verses penned by me which related to the weather and passion:

Khuda he hai wahid madadgar mera

Kay dekhay hai mai nay uske jazbo kay toofan

Kay dekhi hain mai nay uske zulfain pereshan

Kay dekhi hain mai nay uske chahatt kay saman

Vo jazbo kay toofan, vo zulfain pereshan, vo chaat kay saman

Vo jazbay gin ki hararat say jal kar uthi hai sawan ki uudi ghattain

Vo zulfain gin say samhi hoi hai shumali samandur ki thandi hawai

Vo chaat jis ki baho mai aakar, uthi nahi hai uppar nighain

God is my only helper

That I have seen her emotions in a storm

That I have seen her troubled tresses

That I have seen her state of love

Her emotions in a storm, her troubled tresses, her state of love

On seeing these, envious purple storms clouds arise

The cold northern winds fearful of her tresses

The love in whose embrace entwined, the eyes unable to rise

We further look at the differences between formative and summative assessment and how, in our case, attempted to transform the summative system to a formative one. In the slide presented here the rationale for formative assessment is given:

- “To provide feedback for teachers to modify subsequent learning activities and experiences;
- To identify and remediate individual and group deficiencies: here the lack of learning is not due to any lack of ability rather due to different learning styles;
- Focus away from achieving grades and onto learning processes: this is critical and a thing which I continually reiterate to my students. Do not go after grades, take some risks and do not be afraid of failure.
- In order to increase self-efficacy and reduce the negative impact of extrinsic motivation: a passion for learning has to be developed where intrinsic motivation is present.
- To improve students meta-cognitive awareness of how they learn and frequent ongoing assessment, which allows for fine- tuning of instruction and student focus on progress: learning, teaching and testing should go side by side.”

Topic 208

Instant Feedback and Summative Assessment

Instant feedback is the central function of Formative assessment. It typically involves a focus on the detailed content of what is being learned rather than simply a test score or other measurement of how far a student is falling short of expected standards. Nicole and Rick?, synthesizing from the literature list seven principle of good feedback practice:

- It clarifies what good performance is; goals, criteria and expected standards - this function cannot be replicated by technology and still requires human intervention.
- It facilitates the development of self-assessment in learning – as I narrated earlier the computer science students at our university has serious deficiencies in the English language, in terms of comprehension, vocabulary, writing and speaking. Teaching them in medium is terribly difficult. Self-assessment is required, which is done through technology based interactive learning application which asses their level of proficiency and gradually build upon it, through instant feedback.
- It provides high quality information to students about their learning – this is done both for the teachers and students.
- It encourages teacher and peer dialogue around learning.
- It encourages positive motivational beliefs and self-esteem.
- It provides opportunities to close the gap between current and desired performance – this cannot be achieved through the traditional lecturing style.
- It provides information to the teachers that can help shape teaching and it can also help them assess – the system allows for all to be certified at a certain proficiency level, despite varying times contingent upon effort put in.

We had dealt specifically with English education since crucial for collegiate education, whatever the subject. Similarly, in mathematics there has been an enormous amount of research in terms of technology and pedagogy; especially in context of information technology in education. Globally, it is understood that whatever the discipline a certain basic proficiency in mathematics is required; probability, risk analysis, statistics, etc. In the slide you see that mathematical knowledge is essential: “it is important for teachers to see how their students approach the problems.” The criticality of learning from one’s mistakes cannot be emphasized. Also, the onus is upon teachers to discover what approach they are adopting; the traditional bookish approach or one which takes into account real problem solving based upon their individual and innovative approaches. “How much mathematical knowledge and at what levels students use when solving problems.”

Grades should not be the sole criteria rather learning in itself should be an objective. “That is knowing how students think in the process of learning or problem solving and that makes it possible for teachers to keep their students overcome conceptual difficulties and in return improve learning in that sense formative assessment is diagnostic .” One approach is to make students make mistakes, which allows for greater conceptual understanding. “In math’s education to employ formative assessment in the classroom a teacher has to make sure that each student participates in the learning process by expressing their idea – there is a trustful environment.” Again a critical point which is related to our actual case studies and one the grievances students expressed at our college.

“A friendly environment in which students can provide each other feedback: the teacher also provides feedback. All the instruction is modified according to student needs.” A fundamental change since the teachers is forced to change his teaching style in conformity to

student needs – a dynamic change. The inherent feedback provides the requisite clues in adaptation of teaching styles. “In math’s classes thought revealing activity such as modern eliciting activities and generative activities provide good opportunities for covering these aspects of formative assessment.” Turning to an example of formative assessment and instant feedback, which as you know run in parallel, but for formative assessment a better term might be formative learning, since we need to look at it from a higher perspective. This is real example based upon my experience with the science of learning for the past fifteen years and also leads to the use of technology.

Topic 209

Winter and Summer in Lahore 1

In the example we assume a dialogue between a teacher, or should we say a facilitator, and students. There is formative assessment and learning along with feedback taking place. There is also discussion taking place between the two in clarifying the questions put across. The facilitator instigates a discussion by introducing a question and then observes what discussion occurs. A small incident needs to be narrated before going further with the example. In our summer school sessions there are a lot of failed students, particularly in the course which I teach, graph theory and analysis of algorithms. In an interesting and exciting course there are more than 200 failed students over the past couple of years. We had to create four sections for them, for which we found three teachers. We had difficulty finding the fourth teacher. At that time there was woman who had applied for a job and was qualified but had no prior teaching experience. I decided to test her out much to the reluctance of our head of the department, who was not quite sure about this. Even the teacher herself was hesitant and told me she had studied the subject but never taught. I told her she did not have to teach, only ask a question and observe what the students discuss and to facilitate this discussion.

She was still reluctant so I coached her a bit. With hesitation she went into the class but followed instructions. Comparatively, in the other section senior teachers with five to ten years of experience, in this course, were available. Two of these teachers were my past students in this subject and were familiar with the pedagogy required. Can you guess what occurred after a ten day or so? Complaints from all three section started pouring in along with requests for transfer to the inexperienced teacher's section. The main reason cited was the students in her group were enjoying themselves. What was the difference? The vast experience for the senior teachers was

standing in their way. While they were teaching the lady teachers was discussing. Since she did not know her subject very well she empowered the student and instilled a sense of self-belief in them. I deliberately employ this methodology, by feigning ignorance in a subject and not giving any answers to the students, rather letting them arrive at it on their own.



Ashraf Iqbal How one of the most boring - complex - deadly with highest failure rate - subject in FOIT can become exciting - interesting - challenging - enjoyable - by an instructor who does not have any teaching experience of this subject - is incredible? This magic became possible by changing the pedagogy of the said class room? She had very poor knowledge of Algorithms - but she had the most powerful weapon in her hands - the right pedagogy - and that makes all the difference? I challenge that any famous instructor in Pakistan with an experience of teaching Algorithms can not compete with her? Not because she is super but because a pistol can not stand in front of an automatic machine gun - and she had the machine gun?

Like · Reply ·  5 · 21 mins · Edited



Karrar Haider That's right, in order to create a truly conducive environment where learning can change ends; the teacher and the taught must sail in the same boat!



Ashraf Iqbal Samina Akram - please share your experiences and the environment that you create in one of the most hated subjects in FOIT?

Like · Reply · 1 · 3 hrs · Edited



Muhammad Ali Sir 'AA is well known as one of the most difficult subject of the CS course outline. But i must admit that with the change in teaching styles and the student teacher relationship AA has changed the entire course of the CS dicipline. What was once expected as the most difficult course has now proven to be the most 'fun learning' and 'practically important course' toward the completion of the degree. And a hats off to Maam Samina Akram's approach toward teaching the course, without which this course would'nt have been such a success. And thanks to you sir for providing us with such an enviroment .

Like · Reply · 1 hr



Mujadad Rao This has been a really good and a new experience for us... We never dared to discuss anything in the class (talking about two years of experience in UCP) but Mam Samina forced us to discuss and share problems with each other in class. In whis way we were able to solve almost every problem thrown to us. This has been a brand new experience for me and every classmate of mine in UCP 😊

Like · Reply · 1 hr

Topic 210

Winter and Summer in Lahore 2

Returning to our example, the discussion occurring between a facilitator and student; the facilitator put across a question which should be comprehensible to any field of study. Why is there winter and summer in Lahore? Student A gives the following response, an intuitive and common sense response which the teacher should seek to elicit from all students. Also, an answer which we have all provided at one point in time but since no friction was created in mind and no linkages between prior and new knowledge created, most have forgotten. “Because the earth comes closer to the sun in summers and goes further away in the winters.” This answer is naturally based on observation that as you come closer to a heat source, i.e. heater, you feel more heat and further away less heat. The summer interject; “when there is summer in Lahore, is there summer in all parts of the earth?”

In actuality this should not have been said by a student but rather one of the students. As a facilitator the most pertinent job is keep quite as far as possible. The second student’s response to this question is that; “yes it should be because the whole earth is closer to the sun and it is also rotating on its axis.” Student C has this to say; “when it summer in Lahore it is winter in Australia, South America and South Africa.” Naturally, he must have some acquaintances in these palaces and thus his answer. Student D reply to this was, “that means winter and summer do not occur because of the distance of the earth from the sun.” From you see the original basis of the argument, distance, is refuted. An essential step in the process of learning where you build an argument and then try to test it by attempting to refute it. Based on negation student E says; “why argue about distance, when facts can be checked on the internet through your cell phone.

Student F immediately does this and says; “it is the other way around, in the winter the earth is closer to the sun and vice versa.”



Topic 211

Winter and Summer in Lahore 3

Seeing this argument the facilitator says we have arrived at deadlock. Where he again interjects another question based upon student curiosity which is created by the opposite conclusion and which causes friction in their minds; “have you ever seen the sun setting in the evening, you can see it with your bare eyes.” Despite the fact this has no direct connection with the earlier question; many questions are raised which are divergent in nature but their purpose is to allow students to create new linkages based upon their prior knowledge, by stepping out of their comfort zones.

We have studied this earlier but here we are trying to attempt the same problem from formative assessment/learning perspective- foundation knowledge of constructivist theory. The question put across is based on an observable phenomenon which the students can see for themselves. Student A takes the argument further; “but you cannot see the sun when it is at its zenith.” What is the significance of this? You have to realize that answers are not as important as the questions. Student C states; “does the distance of the sun increase when it is setting as compared to midday?” Earlier we were discussing the sun’s distance during the seasons and now we have come to days. In both propositions the distance variable is confusing the students. Student D again refers them to the internet to check for facts. Student F does this and finds out that the distances are the same. Since the earth’s diameter is negligible compared to the distance from the sun, no real difference. Student G adds that summer and winter are also not contingent upon distance; there may be a common reason but which we do not know. At this stage two processes are simultaneously running in the student’s minds which are linked to distance. Now that the distance factor has been clarified their minds are receptive. The facilitator again reverts to a

question which apparently has no link to the earlier question; “we use multiple x-ray plates to see the sun, especially during an eclipse.” Student A immediately tries to connect this comment to the earlier process in his mind; “but when the sun is setting we can see the sun without multiple x-ray plates.” Student further develops this argument; “it seems that x-ray plates get in the way viewing a setting sun.”

Now the students have started to make connections and learning has started. Student C comments; “the sun is not as hot when it is setting because as you suggested some sort of x-ray plates appear to come between us which also reduces its heat and light.” Despite the distance being same, how does this occur? A fact which is confusing the students and this is as it should be. If they are agitated they will learn. Student D says; “the winter may also be like the setting sun.” Again observe how linkage to previous thought is attempted. Many types of superstitions are like this; for example if a black cat crosses my path and I have an accident, my mind creates a connection, despite it being wrong. The concept of x-ray plates apparently has solved the student dilemma. Student E further reinforces this saying that these plates disappear during the summer and when the sun is at its zenith. At this student F enquires in frustration, as to where these plates come from. As you see now multiple processes are running in the students minds, the conflict created is a necessary and an important point of learning.





Topic 212

Winter and Summer in Lahore 4

Now if we were to attempt this through technology. What you see before you see a scene from winters, where there is snow. In contrast now you see a picture by Van Gogh, which illustrates the summers. We continue showing these series of pictures to the students. First the sun, if its size increases naturally there will be a significant impact. If the distance changes a significant impact occurs; while the diameter of the earth does not count, since negligible compared to the distance from the sun. Here we are not discussing anything only showing the students pictures. As you know the earth is tilted and revolving on its axis, this too is shown. Next we show a picture where the tilt increases, for students to think about it. Could this be the reason? Now we reverse the tilt, to show another aspect to the students. Pictures narrate a thousand words but also raise many questions; addressing our curiosity. Having provided this slide show we now start discussing it. The learning application begins with the same, earlier, question; “why is there summer in Lahore.”

Next it shows you multiple options/answers which need to be ticked by the students. The first that the sun comes closer to the earth is usually selected by a significant majority. As designers and teachers we should be aware of all the wrong answers to a particular question; so that we can lead the students to make all the possible mistakes which ultimately lead them to the right answer. If you have read the Sherlock Homes novels, you would appreciate this deductive process. The second is the earth spins about its axis. The third choice is that the earth spins and rotates about the sun. Next the earth has a tilted axis. Some other reason (we do not know).

Whatever answers they provide a new question would be put forth, through technology. In the first case the program would go this slide. The sun coming closer has some ramification in

terms of the question. If the earth is rotating about a fixed axis around the sun there will be no changes in the distance. The new questions include; the earth moves in a circle with the sun at its center. If they click on this option they will be taken to a new slide where they would be shown that since the distance cannot be changed this is not possible. The second choice is that the earth moves in an ellipse with earth at its exact center. Here the distance can be altered. If they tick on this they will be taken a scenario with an animated ellipse, where the earth moves closer or more distance as it orbits. At this stage most of the students would consider the problem resolved. But this too has certain ramification which are essential for them to understand even if wrong.

When we demonstrated this learning application, the student had an interesting thing to say, of which we were not aware. They said according to this there would be two winters and two summers in a year. What was this based on? This needs to be understood and incorporated into our formative assessment in terms of interactive learning application design. Next we remove the sun from the center of earth's orbit and assume it is at a corner of the earth. This is another possibility which needs to be exhausted. All mistakes they make have a certain rationale and ramifications, which need to be addressed. The earth spins only is another assumption. The ramifications for this are too illustrated. Similarly, what happen if the earth spins and rotates? Can it be due to a tilted axis? All these questions need to be addressed, which creates a design tree, through which a successful application can be created. Let the question of the absence or presence of x-ray plates keep you agitated, till next time.



Why there is summer?

- The sun comes closer to Earth?
- The Earth spins about its axis?
- The Earth spins & rotates around the sun?
- The Earth has a tilted axis?
- Some other SECRET?

The earth comes closer

- The Earth moves in a circle with sun at the center?
- The Earth moves in an ellipse with earth at its exact center?
- The Earth moves in an ellipse with sun at a far corner?
- Some other SECRET?

Topic 213

Variation in Seasons

In this exciting course we are studying, I sometimes feel a deficiency. Instead of facing students, I face a camera and am unable to see the glimmer of interest in their eyes; nor their reactions to what I am teaching. At the end of each session I usually show the recording to my students, who are working with me in pedagogy and technology. Their comments, questions and answers are truly interesting and invaluable for me. An occurrence which can only take place in a lively classroom setting. If you remember in the last module we were discussing the variations in seasons; why is there summer and winter. Why is a day hotter in the afternoon as compared to the evening and why we can observe it at that time with our naked eyes? We were also looking at a supposition by students of x-ray plate like mechanism intervening to blunt the heat and allow for us to observe the sun. Based on this a strange question was raised; “why is the sun at its zenith during the summer month cast its rays directly at on top of a person’s head when outside.” From a pedagogical perspective all questions are valid. Perhaps the student had studied some material somewhere of which only a portion was retained. Despite, its phrasing the question is valid. Recently observing a drawing class at our university, who are doing sketches outside in the morning, around 10:00 am or so, they were drenched in sweat. The essential underlying point was that the in the afternoon the sun rays are traveling in a straight line, while in evening striking us at an angle. Is this what determines the difference in temperature and brightness? Does this point hold some resonance for you? Let us now look at a picture; of the sun and earth the distance between the two is noticeable. There is a person stationed on the earth at around 12:00 pm and the sun is directly overhead with its rays falling straight on him.

Conversely in the other picture the sun is setting and the sun rays are falling almost at a ninety degree angle. Based on this difference a student came up with a possible answer. To prove his point he assumed that the man in second picture in made to lie prone on the earth. This implies that the sun rays are also falling directly on his head and the distance of the sun in both scenarios is also equal. The actual positioning on earth does not really matter since diameter of earth negligible compared to distance from the sun. This clearly refutes the earlier point raised by a student despite the fact that it was right to a certain degree but incomplete.

In our earlier actual example of a college where the computer dropout rate is almost 70% we were looking at the reasons for this; while we were looking at examples of seasons in terms of formative assessment. Despite varying examples the context remains the same: how do we rectify this situation at our college.



Topic 214

Assessment of Learning

Returning to our college case study we again refer to a book we had earlier looked at; Mathematical Mindsets by Joe Boaler with foreword by Carol Dweck. The book deals with unleashing student potential through creative mathematics, inspiring message and innovative teaching. In this course we used books from experts in different disciplines; pedagogy, technology and content; some from experts in multiple disciplines, i.e. all three. Till now we have somewhat tentatively established summative assessment to be the culprit and formative assessment as the right methodology. Now we look at what experts with mathematical pedagogy and content knowledge have to contribute. It is stated in the book that all mathematical teachers should be made to watch a movie; 'Race to Nowhere', which is about the use of information technology for education. "Part of the power of Race to Nowhere is in its personal story. The film carries into discovering that pressure of school; homework and tutoring were making her middle school daughter physically sick.

The notion that her daughter, a seemingly normal teenager had been diagnosed with stress induced illness: it was a painful wakeup call and a catalyst for change. Race to Nowhere is a documentary highlighting the stress students are placed under US school, colleges and universities." It was released a couple of years ago and highly acclaimed. The New York Times said it was a must watch movie for all the instructors in the world. It ran to packed houses showing the damaging effects of testing, grading homework and over scheduling on students health and wellbeing. Race to Nowhere has continued to gain support from tens of thousands of educators and parents. "When I watched the film I saw mathematics everywhere as the main source of student stress and anxiety." The basic thesis of the book is that in mathematical

education the stress and anxiety need to be replaced by fun, adventure and discovery. “The film Race to Nowhere features the sad story of a high school student, Devon, who had always done well in mathematics. She was a highly motivated young girl who saw maths as part of her identity.

One day she received an F on a maths test and tragically she committed suicide.” As you can imagine what you are studying can be a matter of life and death. “For Devon and many other students the grade she received did not communicate a message about an area of maths she heeded to work on in her growth learning path.” This growth learning part is the foremost responsibility of an instructor, whatever the pedagogy or technology utilized. “It gave her a message about who she was as a person; she was now an F student. The idea was so crushing to her that she decided to take her life.” From the pictures in front of you, you can see that apart from the girl’s death the headlines state that this was the second one.

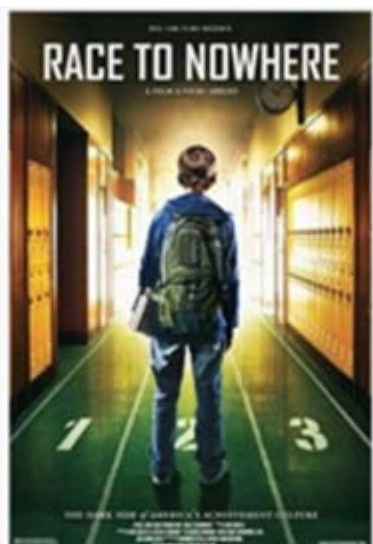
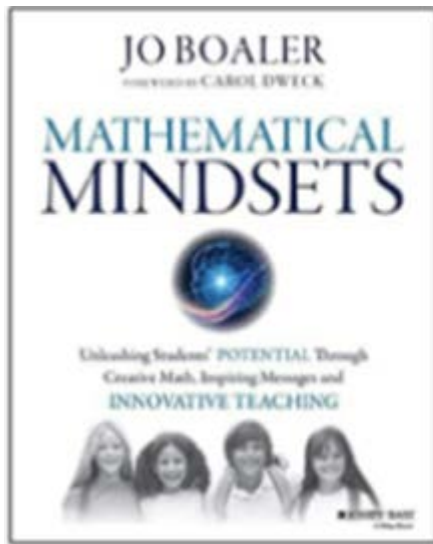
In a typical lecture setting the learning potential is dramatically reduced, especially these days with the help of multimedia. Earlier I had told you about a survey carried out at LUMS, my former university, concerning calculus learning where only 20% learning was established. It was suggested that we do away entirely with multimedia teaching. “When we give assessment to students we create an important opportunity for well-crafted tasks and questions accompanied by clear feedback. Offer students a clear growth mindset, pathway that helps them to know that they can learn to high levels and critically how they can get there. Unfortunately most systems of assessment in classrooms in the United States do the opposite; communicating information to students that cause many of them to think that they are a failure in mathematics and they can never learn mathematics. I have worked with teachers in recent years who shied their methods of assessment from standard tests with grades and scores too. “Based on her findings the author of

the book launched a movement. “This resulted in dramatic changes in their classroom environments.” This was entirely through the incorporation of feedback in the classrooms. “Math’s anxiety formerly common amongst students disappeared and was replaced by student’s self-confidence, which led to higher levels of motivation, engagement and achievement.”

The author is a teacher of mathematics and not pedagogy. Yet, she had to study pedagogy and now is in a position to teach it to us – an interdisciplinary jump. She says: “A little background will be helpful, there are two types of assessments formative and summative. Formative assessment informs learning and is the essence of assessment for learning or assessment *A4* learning. Formative assessments are used to find out where the students are in their learning, so that teachers and students can determine what they need to know next. The purpose summative assessment in contrast is to summarize a student’s learning to give a final account of how far a student has gotten as an end point. One problem in the United States is that many teachers use summative assessment formatively that is to give students an end score or grade, when they are still learning the material. In mathematics classroom teachers use summative tests weekly and then move onto the next subject without waiting to see what the tests reveal.” Things were pretty much the same in our college; only four quizzes were required given. If marked the students were not told what their weaknesses were. The quizzes were examined for underlying patterns, a necessity which is quite impossible without the employment of technology. I hope you can follow the parallel with American schools and our college. “In *A4* learning students become knowledgeable about what they know and what they need to know and ways to close the gap between the two. Students are given information about their flexible and growing learning pathways that contributes to their development of a growth mathematical mindset. Assessment for learning can be thought of as having three parts:

- Clearly communicating to students what they have learned – not grading but rather assessing;
- Helping students become aware of where they are in their learning journey and where they need to reach;
- Giving information to students in ways of closing the gap between where they are now and where they need to be.

The approach is called assessment for learning rather than of learning, because the information teachers and students get from assessment for learning helps teachers make their instruction more effective and helps students learn to the greatest extent possible.” This, I cannot emphasize enough, is the paramount job of all levels of instructors in ensuring that it is taking place in their classrooms.



Topic 215

Rethinking Education

Having discussed mathematical teaching we take you in another related but separate direction; ‘Rethinking Education in the Age of Technology’ a book you have seen earlier. It states that prior to the industrial revolution, when schools had only three to four pupils for each instructor the format was that of formative assessment. After the industrial revolution the population of students increased dramatically and this type of assessment was no longer possible and schools shifted to summative assessment. We are now at a point that technology has progressed so much that we can reapply the earlier model. “We are trying to go into the future and the future is in our past.”

The author also talks about the one room school in the old days, where each student was taught according to his ability; grading was not the sole criteria. Another interesting thing was that the students and instructors are intermingled- a very modern concept in terms of pedagogy. A situation in which there is collaboration and senior students are helping, junior students, since more familiar with how their minds work. The author goes onto narrate about Norman Fredrick; who came up with idea on how to clear navy guns during the Second World War. During that time summative assessment was considered very modern and he was asked to provide a lecture in training other navy personnel after which they would be tested. According the gunner this was not the way to teach them, and this would imply unclean guns which could lead to the destruction of the ships. The navy personnel would need to be trained on the job; a form of formative assessment.

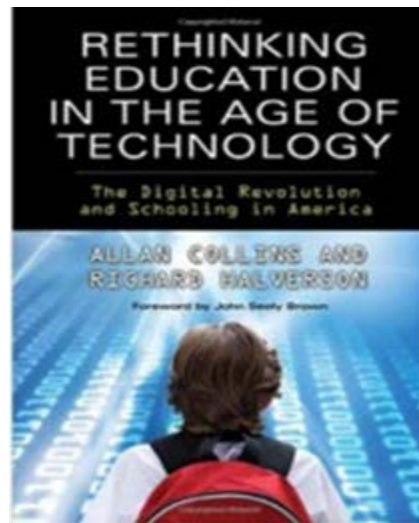
Returning to pedagogy we refer back to an earlier quotation “The educational process as two sides, one physiological and one sociological. It is essential that students be actively engaged

in the process of inquiry. When action is divorced from thought, teaching becomes information transmission. So for this reason higher education experiences are best conceived as communities of learning. According to Dewey educational enquiry is a process to investigate problems and issues and not to memorize solutions. Inquiry within the educational community focuses on intended goals and learning outcomes.

It is a systematic process to define relevant questions, search for relevant information, formulate solutions and apply those solutions and not to memorize formulas. A community of enquiry depends upon sustained communications and collaboration, where the participants share experiences and insights. Education defined as a process of enquiry goes beyond even accessing or assimilating information; enquiry joins process and outcomes. Means and ends in a unified iterative cycle, it links reflection and content by encouraging students to collaboratively explore and reasonably question the organization and meaning of the subject matter. Inquiry is both a reflective and collaborative experience.”

Before you are a table which highlights the need for three presences: Social presence – a collaborative arrange where there is open discussion and students are comfortable with each other. This will enable risk free expressions, encouraging collaboration and expressing emotions.

- Cognitive presence – Is a recursive process that: encompasses stages of puzzlement, information exchange, connection of ideas, creation of concepts and the testing the viability of solutions. This is not to suggest however that the actual practice of enquiry is linear.
- Teaching presence



John Dewey strongly rejected dualism and argued that the value of the educative experience is in unifying the internal and external worlds. Dewey stated, “the educational process has two sides—one psychological and one sociological; and that neither can be subordinated to the other or neglected without evil results following”

According to Dewey, educational inquiry is a process to investigate problems and issues—not to memorize solutions. Inquiry within the educational community focuses on intended goals and learning outcomes. It is a systematic process to define relevant questions, search for relevant information, formulate solutions, and apply those solutions.

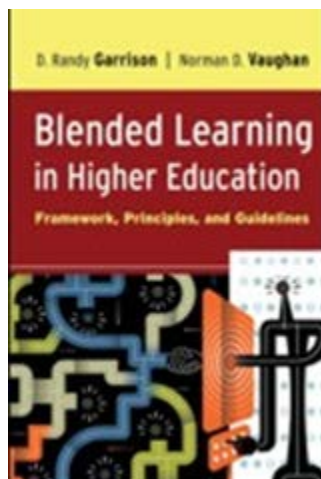


Table 2.1 Community of Inquiry Categories and Indicators

Elements	Categories	Indicators (examples only)
Social presence	Open communication	Enabling risk-free expression
	Group cohesion	Encouraging collaboration
	Affective/personal	Expressing emotions, camaraderie
Cognitive presence	Triggering event	Having sense of puzzlement
	Exploration	Exchanging information
	Integration	Connecting ideas
	Resolution	Applying new ideas
Teaching presence	Design & organization	Setting curriculum and methods
	Facilitation of discourse	Sharing personal meaning
	Direct instruction	Focusing discussion

Topic 216

Schools & Assessment

Four pictures are shown here, where a teacher is teaching in a formative assessment format. This could be any country historically. A teacher is teaching three or four students in a class. One student has understood the first principle of geometry and is ready to progress to the second principle. While another child is still stuck on the first principle and will stay on till an innate understanding is acquired. All gaps in disparate knowledge are closed and right learning pathways developed. In comparison a modern classroom, air-conditioned and with modern technology, the teacher tells the students that their homework has been marked but was unable to bring their copies to school. We now move to new lecture on a different subject. In this setting the teacher has no idea where the students stand in terms of knowledge, all she knows is that she has to take them to a certain point in terms of the course work. If an exam is taken at this point this will only lead to a summative assessment focused solely on grades.

We now turn to a school of the future. As you probably realize the most expensive component of the school is the building, this here has been removed from the equation. Instead of very qualified teachers we have a facilitator. This would in actuality happen since with an exponential population growth qualified teachers would be hardly available. Here computers and technology are playing the same role which teachers used to do in the olden days – formative assessment framework. If a particular student has read and understood the first principle of geometry, a fact verified by the computer then he is allowed to progress. Through technology we can determine where a particular student stands, where he needs to be and how. While another student who is not fully conversant in the first principle is held back by the computer till he thoroughly understands this concept. The future and past in this respect remain similar. Through

the IT revolution and low cost technology we can now impart world class education to students. So as you see we have moved from the one classroom containing a form of apprenticeship in a formative assessment framework; to modern classroom post industrial revolution where conventional teaching and summative assessment took place; to post IT revolution, technology driven formative assessment based learning. Another advantage of technology over the earlier formative system is the number of students which can study in comparison to an instructor.

If you observe the state of our existing schools and colleges, they are entirely driven by a grade based system. Now if you resort to summative assessment in the classrooms, as was happening in our college, and use periodic tests like the midterms to assess them; without letting them know where they are and have to be and how to fill these gaps – only 20% to 30% would pass a systematic failure. Considering this failure rate unacceptable we turned to relative marking; comprising on quality. Next we turned to interactive learning and a formative assessment setup but retained absolute grading. In a real life example, an apprenticeship usually falls in this. A master is not bothered about grades but rather if the apprentice can successfully carryout the assigned task. You can further improve or align the summative system but it would create tremendous difficulties and conflict. Comparatively, formative assessment would require little effort, yet the results would be disproportionately positive. This shift requires little cost.

Let us now turn towards a comparative summary which you see before you. In the Madressas, one classroom model the following activities took place – absorb, do and connect or select, organize and integrate. In the technology model the same holds true. These fall under the realm of formative assessment. While in summative assessment there is only one activity taking place; that of absorb. Now it must be clear to you why there was such a high dropout rate at our college? A major contributing factor was summative assessment. The fault did not lie with the

students or teachers but a fault with the system. How are we trying to rectify the system? We have increased the number of quizzes by threefold, i.e. 12 in a semester, but this has been done, without putting any additional burden on the teachers. The quizzes are automated; they are marked instantaneously and students are able to acquire immediate feedback. There are no grades and students are not penalized for mistakes, rather they are told where they lag behind and helped to overcome this deficiency. The teachers only task is to design these quizzes along with built in feedback; what mistake points to what deficiency in terms of subject knowledge. These gaps are filled through tutorials where no lectures are given but actual conceptual problem solving occurs. The classroom response system is designed with the aid of technology and pedagogy.







Topic 217

Recap

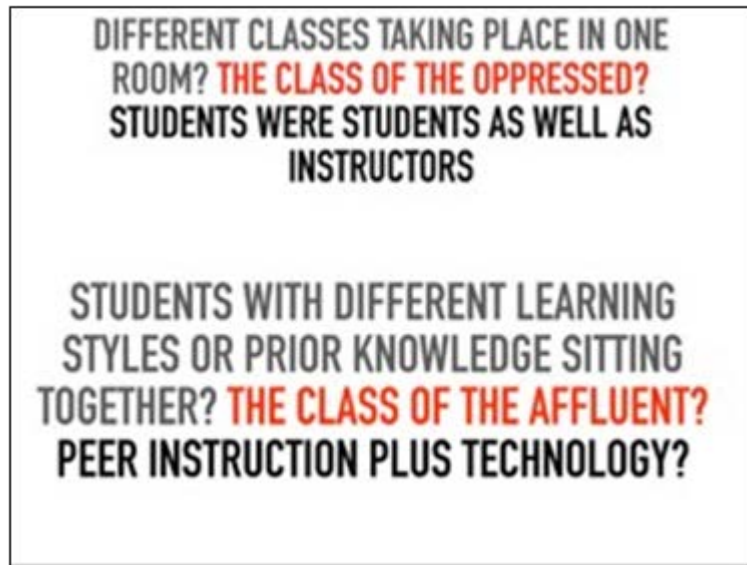
One hopes that you are finding this course interesting and its inherent evolving nature constantly creates more questions and ideas in your minds. If you remember we were discussing our college where, in the computer sciences department, there was a 70% dropout rate. We had looked at some reasons and arrived at some conclusions. In terms of technology we had discovered apparent advantages for two types of students:

- For the affluent, who go to well recognized schools, yet have different learning styles and prior knowledge, there is very little peer instruction, deployment of technology or formative assessment. Technology can easily enable an environment where this takes place.
- In comparison those who belong the less affluent class, where most students have to also work in addition to their studies, technology is the sole means to providing them a quality education.

Some concepts which we would like to reiterate consist of; constructivist pedagogy, formative assessment and instant feedback.

These are especially relevant in terms of classroom response systems: an automation which we have deployed in our classrooms to counter the high dropout rate. Some of the reasons for these high rates were the lack of formative assessment and feedback. Most students only discovered where they stood academically during periodic testing, which was rather late. Apart from the classroom response system, we will also look at the design of multiple choice questionnaires and they are used to provide feedback. While earlier there were four quizzes in the entire semester we have increased these to twelve out of which four take place prior to the midterm. These quizzes are graded immediately and requisite feedback provided to students, in

terms of where the learning gaps are. In parallel we also creating an intelligent database based on the results. In an actual case study on deployment of classroom response system we look at rivers with the title ‘the river flows with music’.



Topic 218

River of Fire

As always we attempt to incorporate art with science, so turning to a famous novel: ‘Aag ka darya or River of Fire’ by Quratulan Haider, a highly regarded author of the Indo-Pak subcontinent. “I do not know much about gods, but I consider the river is a mighty mud caked god; capricious, angry, incharge of its season and moods, destructive. It keeps reminding people of what they do not want to remember. It waits watches and waits. The river is within us and we are surrounded by the sea. Where is the end to these voiceless screams? The silently withering flowers in the fall; that without sound shed their petals? What end to the drifting defeated pieces of a ship? There is no end only gain. Of continually dragging hours and days. Yet we managed to find moments of closeness. The question is not that this closeness was mistaken.”

Returning to classroom response system, I had highlighted the fact that with any discipline a multidisciplinary approach is required. Omar Khayam is famous for his verse yet he was an outstanding mathematician. Steve Jobs for his computers in addition to calligraphy. Leonardo De Vinci was not only a painter, architect and a scientist but also a teacher. His pedagogical principles hold to this day; curiosity, experiment, refinement of the senses (observation), research, development of a balance between science and art, logic and imagination, whole brain thinking, cultivation of grace, fitness and poise and an appreciation and recognition of the interconnectedness of all things. By whole brain thinking the allusion was to understanding the deeper meaning of things, especially in the sciences where a dry formulaic approach is usually prescribed. Similarly, by grace and poise things of design have to go beyond being sufficient, the work has to be a thing of beauty. This beauty, an integrals part of sciences earlier, needs to be rediscovered to make the subjects more interesting and fun to learn. By

interconnectedness he implied a system approach to things. If the prior knowledge within you or rather concepts are not interlinked then no real learning takes place; what is learned is by rote.

These connections have to be created or rather discovered by you to create true learning.

Concepts such as why the seasons change, the price differential of land in urban areas compared to rural, the flow of rivers towards the west, the reason for the wind blowing require a rationale and connection to truly absorb. Curiosity leading to observation and then creating connections, thinking about these and creating new connections, is essentially learning.

Before introducing the classroom response system a tutorial was provided for faculty members. With over 50 sections and cover 26 subjects, the system is responsible for generating quizzes and their immediate grading –instant feedback. It points out their mistakes and why these were made. This is essential, as we keep reiterating, since the principal task of any educational system is to assess where each students stands in terms of learning, where he needs to be and how he will get there. Some pictures, our faculty lounge, where you see a screen with questions on it.

All the teachers have their laptops and tablets, which are also used in conjunction with a classroom response system; they are all given a certain time period to answer individual questions. The answers in the form of multiple choices are displayed and once an answer is clicked upon, on the other screen a histogram begins to be formed. The director center for learning and design is explaining to the teachers that the histogram demonstrates the relationships in their answers. It not only quantifies the right and wrong answers but also what underlying concepts, gaps and misunderstandings that need to be redressed.

The Seven Da Vincian Principles are:

Curiosità—An insatiably curious approach to life and an unrelenting quest for continuous learning.

Dimostrazione—A commitment to test knowledge through experience, persistence, and a willingness to learn from mistakes.

Sensazione—The continual refinement of the senses, especially sight, as the means to enliven experience.

Sfumato (literally "Going up in Smoke")—A willingness to embrace ambiguity, paradox, and uncertainty.

Arte/Scienza—The development of the balance between science and art, logic and imagination. "Whole-brain" thinking.

Corporalità—The cultivation of grace, ambidexterity, fitness, and poise.

Connessione—A recognition of and appreciation for the interconnectedness of all things and phenomena. Systems thinking.

WHAT IS LEARNING

- Curiosity & Observation?
- Making Connections
- Thinking about Connections



WHAT IS LEARNING

- Curiosity & Observation?
- Making Connections
- Thinking about Connections

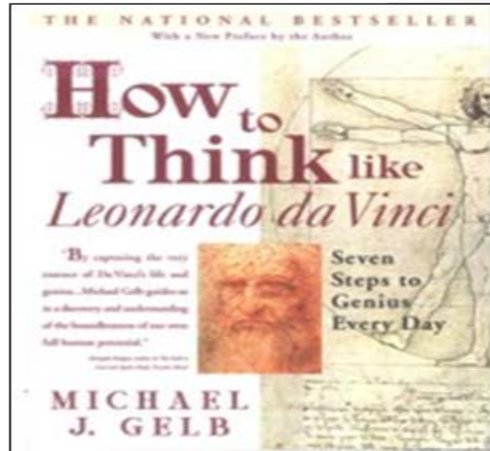


میں دیوتاؤں کے متعلق زیادہ نہیں جانتا ، لیکن میں سمجھتا ہوں کہ دریا

ایک طاقت ور مٹیالا دیوتا ہے ۔ توند مزاج ، غصیلا
اپنے موسموں اور اپنے غیض و غضب کا مالک ، تباہ کن
وہ ان چیزوں کی یاد دلاتا رہتا ہے جنہیں انسان بھول جانا چاہتے
ہیں

وہ منتظر ہے اور دیکھتا ہے اور منتظر ہے
دریا ہمارے اندر ہے ۔ سمندر نے ہمیں گھیر رکھا ہے
خاتمہ کہاں ہے بے آواز چیخوں کا
خزاں میں خاموشی سے مر جھا تے پھولوں کا

خزاں میں خاموشی سے مر جھا تے پھولوں کا
جو چپ چاپ اپنی پنکھڑیاں گراتے ہیں
جہاز کے بہتے ہوئے شکستہ ٹکڑوں کا خاتمہ کہاں ہے ؟
خاتمہ کہیں نہیں ہے ۔ صرف اضافہ ہے ۔
مزید دنوں اور گھنٹوں کا گھسٹتا ہوا تسلسل
ہم نے کرب کے لمحوں کو ڈھونڈ نکالا
(سوال یہ نہیں کہ یہ کرب غلط فہمی کا نتیجہ تھا





Topic 219

Which way the River Flows

In the section we will be looking at the actual responses and statistics. Please, understand that the statistics provided here for a particular faculty are similar to other universities and is in no way a standard of comparison. The point here is to understand the process of learning. Let us then look at the first question. We are asking the teachers to imagine a map of Pakistan, particularly the province of Punjab, where we are currently located. There are probably no faculty members who would have not seen a map of Pakistan since they were students a thousand times; or not studied that the province of Punjab is named after the five rivers which pass through it or not have heard of the historical Indus valley civilization. What we are trying to assess is their ability to observe: the difference between looking and seeing enormous. Getting back to the question there are four possible options as answers: all the rivers located in the province move from East to West; from the mountains in the North to the sea in the South; the rivers move towards the southwest; and the rivers move southeast. Most of you have also seen a map of Pakistan and we let you ponder this question.

Turning to the answers and a histogram of these answers; 27 percent answered selected B, a couple A, four selected C and no one selected D. Sometimes it happens that you may have seen a thing many times yet but your brain does not register it. When asked what you have observed the usual response is what the brain wants to see. If your brain is thinking that rivers should flow from the mountains in the North to the sea in the South, which seems logical, you will always choose this option. In terms of answers I am not in the habit of providing them; I would prefer it if you would figure it out on your own. This discovery is the underlying point behind technology and pedagogy. Having asked you this question now the way you will look at a

map again would be entirely different. The factual and wishful direction of rivers flow would soon be discovered. This dichotomy is often one the major roadblocks to learning.

We again put up the map and repeated our question; what is the direction of our river flows? The faculty immediately clicked on an option, a histogram which was also made. In the new histogram one person persists in saying that the rivers flow from east to west. Two people opt for the north to south option. Twenty seven people, who upon their observation, despite the fact that the logic may escape them, choose southwest; four people choose southeast.

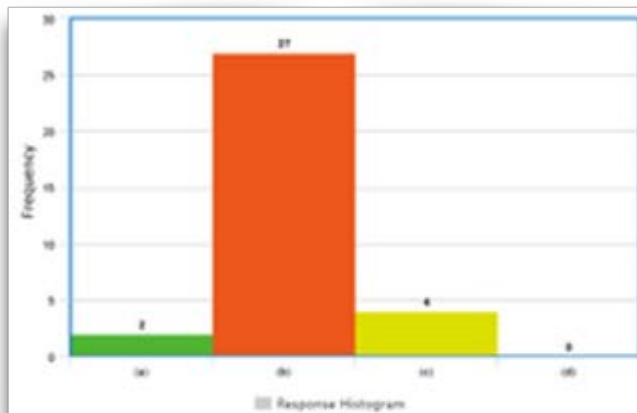
This conflict is a necessary part of the learning process. As you can see the issues seems to be with observation. This is the logic behind a classroom response system. You pose a question and provide numerous options as answers; but these need to be designed to make the students think. This is currently the dilemma I face in terms of our classroom response system, which has just been deployed, the necessity of designing intelligent answers. The wrong answers should also contain some element which leads you to right answer. If all the students are answering correctly then there is something wrong. This is not the point of the test. Rather the whole underlying rationale of this, as of the course, is the determination of where each student in a class stands in terms of knowledge by an instructor and how this disparity needs to be addressed.

Returning to our quiz we have arrived at the conclusion that all rivers run southwest. This was not an easy decision for the teachers intuitively; rather it was based on observation. At this point we could add another dimension and let them discuss the problem – peer instruction

IMAGINE THE MAP OF PAKISTAN (PUNJAB & SIND)

SELECT WHICH
ONE IS
CORRECT?

- (a) The Rivers mostly moves from the East to West
- (b) The Rivers move from North (mountains) to the South(sea)
- (c) The Rivers move towards South West
- (d) The Rivers move towards South East

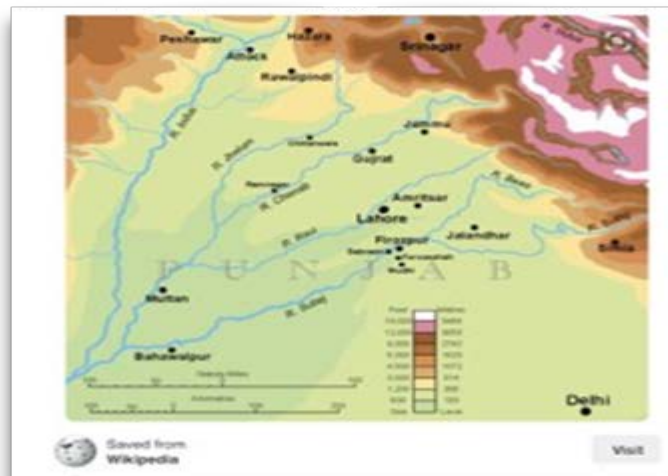




CAREFULLY LOOK AT THE DIRECTION OF THE RIVERS

SELECT WHICH ONE IS CORRECT?

- (a) The Rivers mostly moves from the East to West
- (b) The Rivers move from North (mountains) to the South (sea)
- (c) The Rivers move towards South West
- (d) The Rivers move towards South East



Topic 220

Why South West

We now turn to a more difficult question: Why do most rivers flow towards the southwest? The options are: they are driven by some unseen force; due to gravity; since shortest path to the sea; due to wind. The most popular answer was unknown force. I had personally thought gravity would be selected, since quite popular when replicated earlier at different educational institutions. Three people choose wind. Only two choose gravity. At this point we show them a color coded map, which clearly refutes the gravity option, i.e. upon entering Punjab the effects of gravity are almost nil (plains). We repeat the question again; the same answers are found, even the gravity people. We see some scenes of the faculty lounge while this exercise was being conducted. Perhaps you might have noticed that these answers are interlinked, a point we were trying to achieve.

We again ask them to imagine the map of Pakistan again and specifically ask them to focus on rivers and the large cities. Lahore as an example had Shadarah as a suburb on the other side of the river. The options provided were: some are on the west and some on the east; all of them on east; all of them on the west; no city is near a river. Again we draw a histogram and the answers are; 25 people are saying that the cities are not next to river. A surprising results since a map was shown earlier; yet it seems it was not seen. Next we actually show them these cities on the map and again ask them which side the river is on. This clearly illustrates the fact that most rivers are on the right bank. Why is this so? Apart from this we ask them further question, we cannot discuss all of them, but will show some illustrative examples. When Pakistan was formed Lahore was located towards the Indian side, after which the river

Ravi flowed, followed by the adjoining areas of Lahore. At that time there was a great threat of an attack by India and subsequently the BRP canal was built as a barrier, in which even students volunteered. This is precisely what happened in the 1965 war and the Indian advance was checked at the very canal. Another interesting thing in those days all industry and housing colonies in Lahore were located on the other side of Ravi, away from the border, unlike today's development which adjoins the border. Strangely no one earlier considered developing Lahore on the other side of Ravi, as a natural barrier. Why is the land cheaper in Shadara? I often ask this question. Once I asked an auditorium filled with 200s students and faculty that I can bet no one would be from Shadara. Also I enquired if a marriage proposal came from a family residing in that area, and another from someone residing in Lahore, which in fact is nearer the border, who would be preferred?

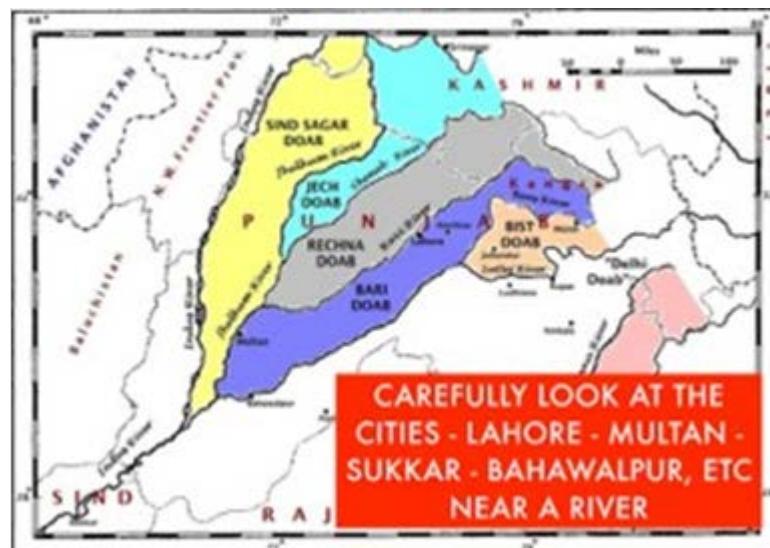
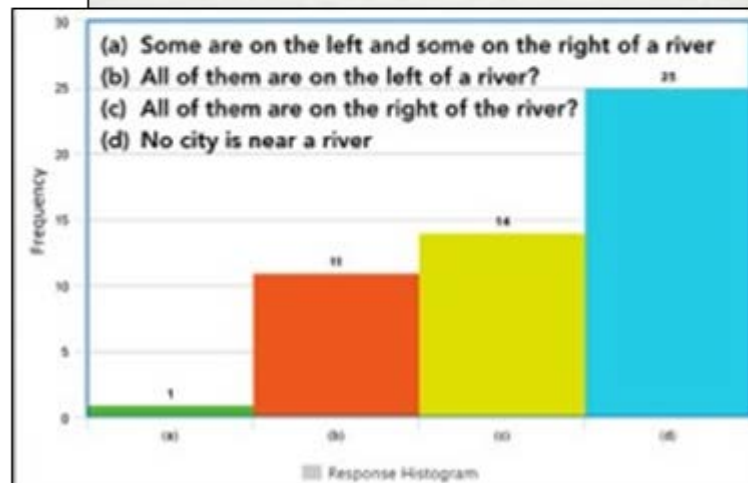
Getting back to our question, people now knew that cities are located on a certain side of the river. The reason for this is that flooding usually occurs first towards the west. Rivers also tend to turn west. Naturally, this is contingent upon the level of flood. Why does this occur? As you can probably start to see all is connected. The teachers having arrived at a conclusion we introduced a 'Do' activity - a fun activity which you can do on your own; one which hints at why the rivers run west and related question.

IMAGINE THE MAP OF PAKISTAN (PUNJAB & SIND - RIVERS AND CITIES)

LOOK AT MOST OF THE CITIES
LIKE MULTAN - LAHORE -
BAHAWALPUR - SUKKAR, ETC.

SELECT WHICH ONE IS CORRECT?

- (a) Some are on the west and some on the east of a river
- (b) All of them are on the east of a river?
- (c) All of them are on the west of the river?
- (d) No city is near a river





SOME OBSERVATIONS

- Most of the cities are planned on the right bank of a river?
- Flooding Initially Takes Place in **Shahdara** - It does not take place in Lahore initially - or it takes place in Lahore when it is very high FLOOD?
- The Prices of Land in **Shahdara** is less than in Lahore?
- **Shahdara** is much safer than Lahore in case of an Indian Attack
- Almost all the rivers - all turn towards South West?
- What force drives the rivers towards WEST?
- All rivers initially floods towards one bank first?

Topic 221

Rivers and Cities

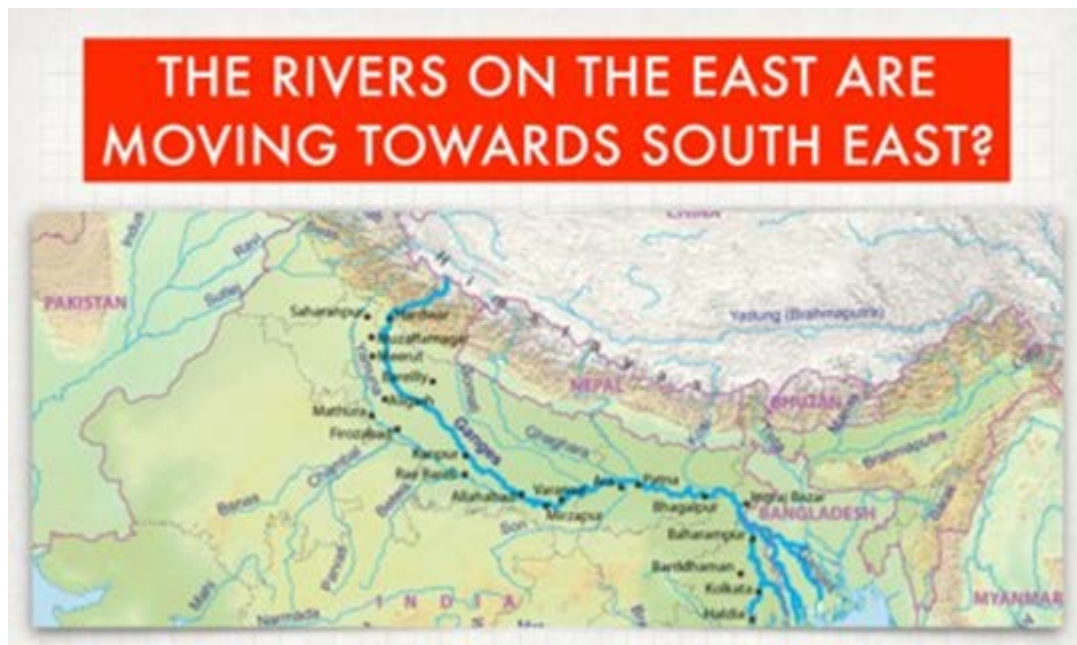
You must have all travelled by Lahore's ring road. You would have also noticed that there are a lot of flyovers/bridges and many sharp turns. Try this, next time you take a turn do this at a slightly higher speed and spray your windshield with water. You will observe that the water drops on the windshield would be moving downwards due to gravity but also turning in a particular direction. Note the direction they move. Moving to the next experiment; clean your windshield and again spray it, turning in the opposite direction at a high speed. You will see the same phenomena as earlier but the drops turn in an opposite direction. Why is this and how is it linked to the rivers flowing west? Is it the earth's spin? All these questions should be arising in their mind.

Due to the interconnectivity of these questions it is quite possible that they arrive at the right conclusion: the flow due to Earth's spin. Despite the fact that this might be the correct answer we will not let them know this too easily. We would want them to go through the frustration, friction and conflict. Again, a map of the subcontinent is shown from which it can be seen that the rivers originate from the North and their flow through India and Pakistan. The origination of these rivers is almost the same. Those that flow into Pakistan turn towards the west. If they were traveling south they would have cut across India and not turn towards Pakistan. The surprising thing for you might be the fact that the rivers flowing into India (Ganga and Jamna) turn towards Bangladesh in the east rather than the south. Due to these facts the theory seems to stand disproved. The right answer is not as critical as creating the requisite linkages in discovering knowledge.

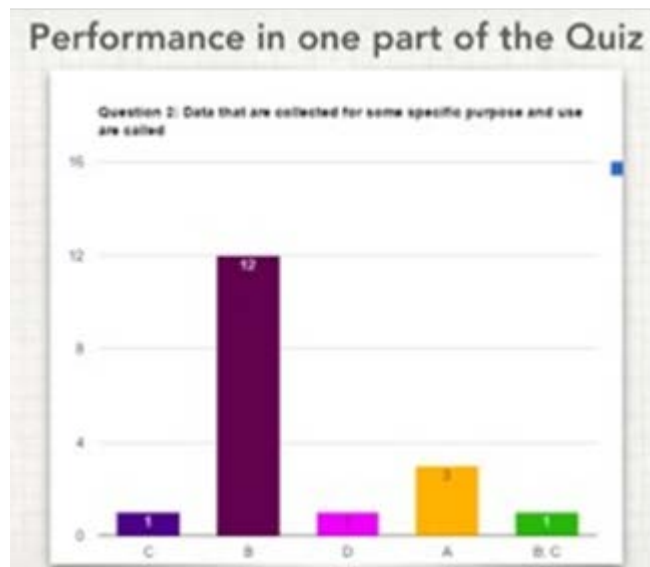
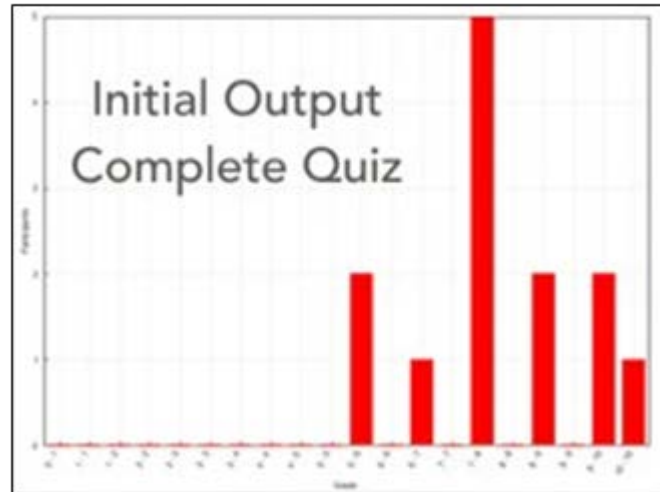
Returning to our classroom response system, we look at some pictures. As you can see the students are really enjoying it. Why is that? The answer lies in instant feedback, as you can see from the interfaces of their tablets. The technologies we are developing at our center of design of learning is not solely for the well-off rather focused towards the less privileged: people in small villages (remote areas), those who work for a living, etc. We tested this technology based on intranet, i.e. no internet connections were required. You will also observe that while travelling there the system was also utilized in the bus. What does this mean; imagine a mobile school which can move around the entire day focusing on provision of education to those in need. Lessons can be uploaded or down loaded and assessment also provided; allowing us to understand where each students stands and where he needs to be taken and how to accomplish this. In the next scene, returning to the classroom, when given permission to discuss the more difficult questions; you can see how engaged they are.

Nobody is using Facebook or gossiping. Now we look at the output of these quizzes, based upon four to five questions with multiple choice based answers. On analysis an integrated picture comprising of individual questions is formed. Only one student scored 10 out of 10, two children got 9 out of 10, between 7 or 8 out of ten there were approximately five children and so on. As soon the quizzes were over I started getting emails from fifty sections giving me the results immediately. If 70% to 80% students are getting full marks, it would imply that the skew would turn towards the right. Reflecting a fact that our quiz was not properly designed, which I could see right away and point out to the relevant faculty. From my perspective it is more pertinent to see mistakes – what mistakes they are making and why. Where they stand and where they need to be. After this we started creating histograms for each individual question. From these we started getting the relevant information. What part of the quiz was found to be the most

problematic by the most students? As you can see this is clearly illustrated by the histograms of different subjects for one section. The histogram at the bottom seems to be problematic since all correct, the red lines the largest. Similarly, the results for same subject but different sections, here more realistic and the red lines the smallest. As you can see this is a treasure trove of useful data which can be critically analyzed. The concepts where help is required can be scheduled for tutorials. In the center of design for learning there is collaboration from the pedagogy, content, quality control, and technology. The main focus of all these areas is towards designing of an appropriate quiz which can discern problem areas for students. The questions have to clear in terms of reflecting what underlying concept or concepts are captured and are clearly understood or not.







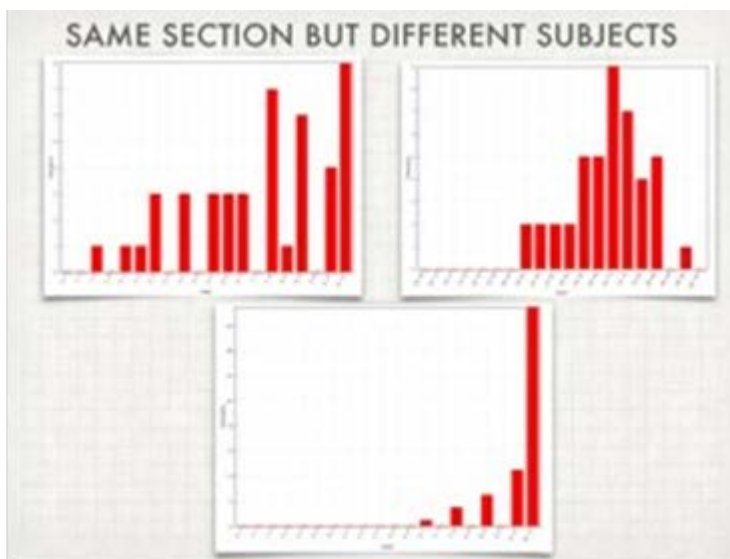
Topic 222

A Visit to the Doctor

Today we attempt to wrap up the course. I could not be with you last week since unfortunately I was suffering from sciatica pain and consequently immobile. Having visited with the physician I was prescribed quite a few tests. The reports of these tests were used for diagnosing the ailment. But my concern was greater; what corrective measures were needed, i.e. prescriptive and corrective. You must be wondering how this is related to information technology and our course? The parallel lies in when students come to us, faculty, or in my capacity as Dean, concerning educational issues. At our university the fees for one course amount to Rs.25,000, which is quite substantial. In a typical semester, consisting of four courses, the fees amount to Rs. 100,000.

Against these fees, the goal of these students is to acquire a bachelor's degree in computing with good grades; leading to well-paying jobs. During their course of studies they are assessed through quizzes, midterms, assignments, projects and naturally final exams. These examinations during their journey, especially the quizzes and midterms, since earlier, are good indicators of their performance. In most instances in all universities across our country; most of the quizzes undertaken by students are hardly graded. If graded, no feedback is provided in terms of their performance. By the midterms arrive, where students stand in terms of performance, becomes quite clear. What knowledge he/she has acquired, what should he have acquired, what gap remains and how this can be filled? Little attention is paid to this diagnostic process. Especially in informing students of the situation and what final grades they can expect – A closed loop feedback. Each student should be able to take advantage of these diagnostic tests and provided individualized feedback. There is no need to take him through the lecture process

again; rather underlying concepts where there is a problem need to be clarified. To do this, initially we thought of incorporating a weekly automated quiz for each semester. If we had resorted to manual methodology the tendency with faculty is that delay marking or retuning the quizzes to the students. This implies that students receive neither feedback nor the faculty where their class stands, i.e. has what was attempted to be taught - learned. This also allow for the appropriate prescriptive- corrective action. With an automated system we can even detect what underlying concepts require clarity.



Topic 223

Automated Response System

As soon as a quiz takes place, I receive an email with analysis of the results, consolidated and for individual questions, including histograms. Here you see the results for one section in different subjects; different sections for the same subjects; and the results for sub problems in a quiz. The red line indicates the number of correct answers. If the line is too high or too low, it is a worrying fact. You can have a brilliant technological solution in the form of a classroom response system; yet the quality of the quiz is critical. Although a time consuming process yet needs to be undertaken initially with great care. Once correctly done all it requires is continual refinements and well worth the initial time investment. The system should be accessible to all the stakeholders in an educational system, and able to keep them in the loop of what is transpiring. Any gaps in technology and pedagogy need to be addressed as soon as they arise- diagnostic, prescriptive methodology which is evolutionary in nature.

In the slide here you see a histogram of our student results. The results reflect a summation and individual question, i.e. which student attempted what. What is the linkage here? The chain of logic which needs to be followed entails a linkage of entire quiz results to individual concepts and most importantly the underlying concepts. Usually, when forming a quiz and breaking it down into four component parts, you do not usually consider what concepts these are derived from. At time it can only be singular concept while at others multiple options. This diagnostic approach is critical as with a medical practitioner. As an illustrative example in class of forty students 10 choose option A, twenty B and five C. We begin our analysis from bottom up. In the first slide we try to determine the appropriate underlying concepts. We see that five question are linked to three underlying concepts. It easy to observe what questions are not

answered correctly but harder to determine which underlying concept is not clear. Suppose we had designed the quiz with sole reliance upon concept A; only the people familiar with this concept would answer correctly, i.e. 10 in our example. About twenty went for option which relied upon concept B.

Thus at least thirty correct answers tell us that students are familiar with these concepts. We could further change the question to only rely on one concept and figure out what concepts needed to be clarified for students. Similarly, question relying upon multiple concepts can be analyzed for intersecting concept, sets and bottlenecks.



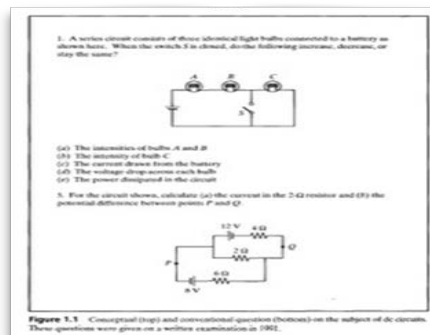
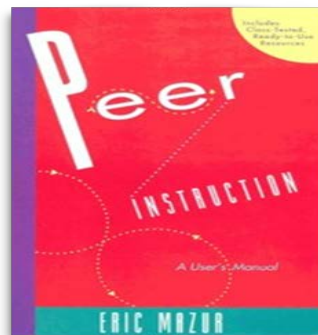


Figure 1.1: Conceptual (top) and conventional question (bottom) on the subject of dc circuits. These questions were given on a written examination in 1982.

Topic 224

Peer Instructions

Here you can see the book by Eric Mazur, who was instrumental in popularity of classroom response systems. His experiments paved the way for improved learning for students in the classroom. In one of his experiments he presented a very innovative problem, presented here, to his students. The upper part you see, is conceptual, while the lower conventional.

The conventional problems are usually found in textbooks at the end of each topic. By attempting these most students feel that they have successfully mastered the topic; an assumption which is incorrect. Histograms for both types of problems are created. Next these are correlated. What were the results? You have a soft copy of this book in which you can discover this for yourself.

When I joined as Dean at my present university, in the introductory course on computing, offered in fall 2015, out 44 or so students in one section around 21 failed, 18 withdrew and only five passed. If you make a histogram of these results it would be entirely skewed towards failure. With these results and being an introductory course we manage to completely quash their interest or emphasize their inability in this subject. If you look at further results for other sections the results are as grim. In another course design and analysis of algorithms, which we had discussed earlier, out of 37 students, only three failed, thirty three withdrew, despite being taught by a very experienced and senior faculty. The course has acquired a reputation in which students fear to enroll since unlikely to pass. The course programming fundamentals was offered in spring 2016, in one section there were 36 students. Thirteen failed, three passed and the remainder withdrew; in all probability from the program too. This was the prevailing situation when I joined the university.

As narrated earlier the course design of algorithms was used as a test case. Four sections were made and we had trouble finding the fourth faculty member to teach. A female teacher with no experience was hired. She was told to follow certain pedagogical principles and asked to take the students on a journey of discovery, where the teacher would only act as facilitator. This created an interesting immersive experience for the students. And in comparison to the other sections where senior faculty was teaching, students started asking for transfer to the other section. The major comparative difference was the pedagogical tools which were provided to the fresh instructor.

Despite, the criticality of content knowledge, these tools still allowed her to overcome this obstacle. Her major contribution, apart from the pedagogy, was the instilling of self-belief and self-discovery in the students. This in addition to regular quizzes with in-depth feedback will allow you to replicate this experience and compete with any highly qualified teacher. The interesting thing is that any new teacher without any preconceived notions can be trained to do this. Looking at some of the results the biggest surprise was the midterm. The same midterm was given to all four sections. It was designed by the senior faculty of the three sections and the new female teacher had no input. Since she did not know what was going to be in the midterm, she could not revise the topics which were to appear in the test. The results came in recently. As you can see the curve which was skewed towards failure has now become much more balanced. The 70% to 80% percentile tier now forms the largest component. Only two or three students fall in the minimum passing range. What a dramatic turnaround!

The email, after the midterms, which I sent to all the students, emphasized the fact that their job was not over. They had to carefully go through the results and understand what mistakes they made and what concepts still missing; while we too at our end would look at

what conceptual misunderstanding still existed. If the students accomplish their assigned task seriously they would be given five extra marks in their midterms. Prior to this the methodology was that if a student challenged his grades and if the fault of the faculty his marks would be revised otherwise he would be penalized, i.e. marks deducted. We have turned the entire concept on its head. Now those who will not discuss the paper with the instructor would be penalized. We have assigned two people to focus only on analysis of what mistakes were made and which concepts still require clarity. We are also looking at replication of mistakes in comparison to quizzes, i.e. were the same mistakes repeated. Apart from the requisite feedback from students you can see the favorable comments for the new instructor. Also as you can see here the classroom experience; students are having fun, there is little formality, no lectures are taking place, the students are engaged and involved in a “do” activity, they are “connecting” ideas, “reflecting” and “discovering.” This is in marked contrast to what is transpiring in other classes: a situation which needs to be thought about.

Introduction to computing Fall 2015					
Sections	students	Pass	Fail	withdrawal	Teacher
D	48	5	23	18	Mr. Shafiq-ur-Rehman
E	43	11	22	10	
F	48	9	16	19	
G	45	8	25	8	
J	50	10	27	13	
Total	226	43	115	68	
Design and Analysis of Algorithms Fall 2015					
Sections	students	Pass	Fail	withdrawal	Teacher
A	37	1	3	33	Mr. Shafiq-ur-Rehman
E	49	17	11	21	Mr. M. Amjad Iqbal
C	49	31	6	12	Mr. M. Amjad Iqbal
D	53	22	11	20	Mr. Sumran Shabbir
Total	188	71	31	86	
Programming Fundamentals Spring 2016					
Sections	students	Pass	Fail	withdrawal	Teacher
A	32	19	2	11	Mr. Shafiq-ur-Rehman
B	38	18	1	19	
C	34	13	2	19	
D	20	10	2	8	
Total	126	60	5	61	



Ashraf Iqbal

August 15 at 9:45am · 🌐 ▼

My dear FOIT students

Your midterm examinations are over? But they will not be over unless you people - each one of you - look at your midterm answer sheet and find out the mistakes that you have committed? Then we need to know what concepts were weak because of which you committed those mistakes - so that we make corrective actions in making you understand and you people realize what is missing and do your final in a better way? I repeat each one of you should discuss your midterm with your respective instructor?



Ashraf Iqbal

22 hrs · 🌐 ▼

My dear students of FOIT in the summer semester

Let us start with AA? I shall increase 5 marks in your midterm if you do the following to my complete satisfaction?

Tell me in writing - typed soft copy- the following?

1. What mistakes you have committed?
2. What concepts were necessary to attempt those problems that you did wrong?
3. Why you have committed this mistake - did you not understand the problem?
4. Did you commit the same mistakes in the quizzes conducted?
5. Now did you understand this problem?
6. We shall send you the question paper
7. Correctly solve the problem - underlying those parts which you have done wrong and understood after the midterm?
8. I need this write up to me mailed to Syeda Beenish by tomorrow before 10pm? After 10pm tomorrow you can not avail this facility?

Chaudhry Waleed mentioned you in a comment.



Chaudhry Waleed

August 14 at 11:27pm

Ma'am Samina Akram is one of the best teacher ever. Thankyou ma'am for being there for us and thank you sir Ashraf Iqbal for teachers like her. :)



Topic 225

Learning Management System

We now look at the interface which is used in our quizzes. As you see from the slide different grades for each student for each individual quizzes provided as list. This allows us to compare various quizzes with the same subject or between courses. After logging into the program they are provided with the relevant question. Initially, the multiple choice answers are not provided rather we let them study the question and context. As they understanding the question is 60% of the answer. Next the options are presented. As soon as a student's selects an option in the prescribed time, they are provided with the right answer instantaneously, along with what mistake they had committed. This is formative assessment. Upon implementation we discovered a loophole, which was communicated by the students themselves and for which we did not penalize them.

Due to the instantaneous correct response being provided to each student, the class collectively selected one person to go through the entire quiz in the first ten second, the prescribed time for the first question. Upon receipt of the right answers these would be passed to remaining class. Based on this we have modified the system to only generate the right answers once the prescribed time is over for all students and they have provided the answer. A classroom response system is not static as with most things there is an evolutionary process involved. A teacher came to me the other day and enquired if I had looked at the data from the system. Once a student receives the answer options there are two variables which we can look at: the time involved in answering the question; and how many times does a student change his option during that time.

First Name	Surname	Quiz 1	Quiz 2	Quiz 3
U1100000001	CHALERS STINE	1.00	1.00	1.00
U1100000002	CHALERS STINE	1.00	1.00	1.00
U1100000003	CHALERS STINE	1.00	1.00	1.00
U1100000004	CHALERS STINE	1.00	1.00	1.00
U1100000005	CHALERS STINE	1.00	1.00	1.00
U1100000006	CHALERS STINE	1.00	1.00	1.00
U1100000007	CHALERS STINE	1.00	1.00	1.00
U1100000008	CHALERS STINE	1.00	1.00	1.00
U1100000009	CHALERS STINE	1.00	1.00	1.00
U1100000010	CHALERS STINE	1.00	1.00	1.00
U1100000011	CHALERS STINE	1.00	1.00	1.00
U1100000012	CHALERS STINE	1.00	1.00	1.00
U1100000013	CHALERS STINE	1.00	1.00	1.00
U1100000014	CHALERS STINE	1.00	1.00	1.00
U1100000015	CHALERS STINE	1.00	1.00	1.00
U1100000016	CHALERS STINE	1.00	1.00	1.00
U1100000017	CHALERS STINE	1.00	1.00	1.00
U1100000018	CHALERS STINE	1.00	1.00	1.00
U1100000019	CHALERS STINE	1.00	1.00	1.00
U1100000020	CHALERS STINE	1.00	1.00	1.00
U1100000021	CHALERS STINE	1.00	1.00	1.00
U1100000022	CHALERS STINE	1.00	1.00	1.00
U1100000023	CHALERS STINE	1.00	1.00	1.00
U1100000024	CHALERS STINE	1.00	1.00	1.00
U1100000025	CHALERS STINE	1.00	1.00	1.00
U1100000026	CHALERS STINE	1.00	1.00	1.00
U1100000027	CHALERS STINE	1.00	1.00	1.00
U1100000028	CHALERS STINE	1.00	1.00	1.00
U1100000029	CHALERS STINE	1.00	1.00	1.00
U1100000030	CHALERS STINE	1.00	1.00	1.00
U1100000031	CHALERS STINE	1.00	1.00	1.00
U1100000032	CHALERS STINE	1.00	1.00	1.00
U1100000033	CHALERS STINE	1.00	1.00	1.00
U1100000034	CHALERS STINE	1.00	1.00	1.00
U1100000035	CHALERS STINE	1.00	1.00	1.00
U1100000036	CHALERS STINE	1.00	1.00	1.00
U1100000037	CHALERS STINE	1.00	1.00	1.00
U1100000038	CHALERS STINE	1.00	1.00	1.00
U1100000039	CHALERS STINE	1.00	1.00	1.00
U1100000040	CHALERS STINE	1.00	1.00	1.00
U1100000041	CHALERS STINE	1.00	1.00	1.00
U1100000042	CHALERS STINE	1.00	1.00	1.00
U1100000043	CHALERS STINE	1.00	1.00	1.00
U1100000044	CHALERS STINE	1.00	1.00	1.00
U1100000045	CHALERS STINE	1.00	1.00	1.00
U1100000046	CHALERS STINE	1.00	1.00	1.00
U1100000047	CHALERS STINE	1.00	1.00	1.00
U1100000048	CHALERS STINE	1.00	1.00	1.00
U1100000049	CHALERS STINE	1.00	1.00	1.00
U1100000050	CHALERS STINE	1.00	1.00	1.00

GRADE OF EACH
STUDENT IN THE
SAME SUBJECT IN
DIFFERENT
QUIZZES

Topic 226

Formative Assessment &MCQs

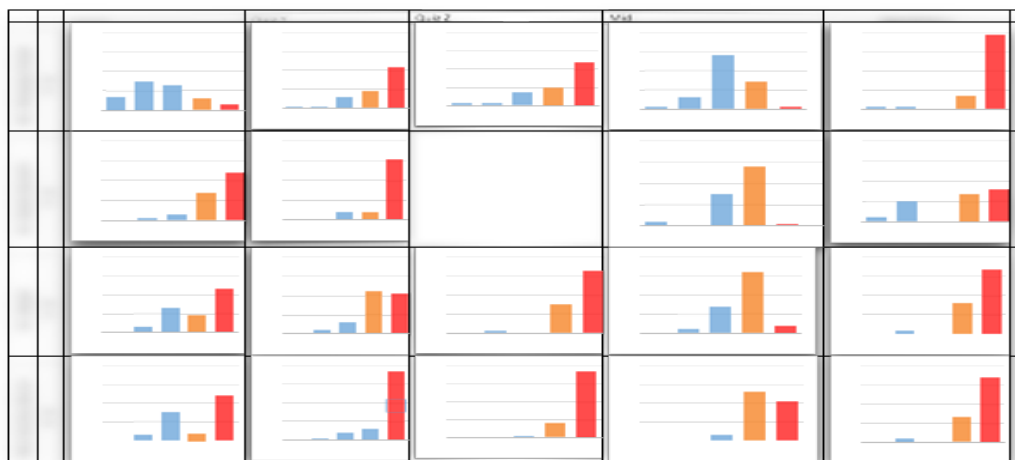
How can we use a multiple choice format in assessing formative assessment? Teachers at our college, who taught programming and algorithms, were initially hesitant since a multiple choice format could not be applied for their discipline. I had to explain that conceptual knowledge could be broken down. A challenging task albeit one that can be done.

Interactive learning and formative assessment needs the classroom response system to be used several times in the classroom. Another issue is the quality of the quizzes. Naturally, this can be problematic for teachers in terms of assessment, based upon the large number of students. An automated solution is required. With technology the quality of the quizzes can be gauged. If all the students are getting good grades and are faring well or vice versa; technology would immediately point this out. Similarly, how do we quizzes for containing problems conceptually? Are these just capturing rote learning in a conventional framework? Also, do these into account any element of cheating? Along with us other people are also working at conceptual soundness of these quizzes. This is a problem which you too will face at the design stage. One method is to introduce a conceptual problems or a rote learning problem and asses their correlation to other answers. Depending on what is correlated, intelligent conceptual question or rote learning, you can discover how the quiz is skewed. Similarly, the time spent upon questions by students can be used to gauge the element of cheating. By looking at the performance of a specific student in three subjects, can we predict their performance in a fourth subject? If our grading system is correct then the answer is yes; if not the fault lies with our design. All these checks can be accomplished with the help of technology. Teachers cannot do this; they are overburdened as it is. Turning to correlation between different subjects: in our computer science department we are

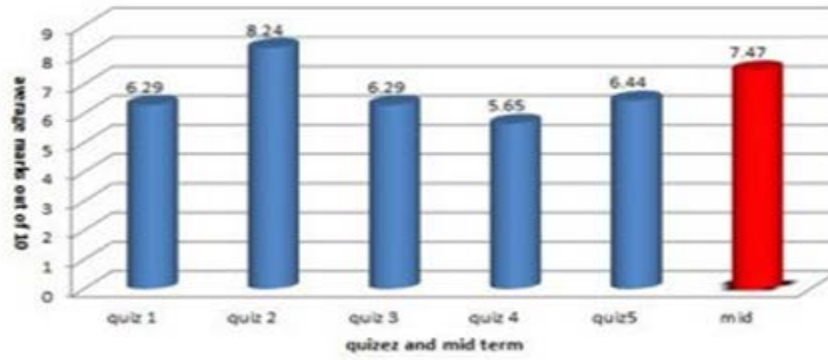
offering two seemingly unrelated courses; calligraphy and free hand drawing. How are these related to the sciences and other subjects? Unfortunately, especially in Pakistan, the nexus between arts, sciences and mathematics is little realized. As De Vinci and Omar Khayam pointed out this is a must. The left and right hemispheres of the brain need to be equally engaged. In the sciences and mathematics there lies great internal beauty, elegance and equilibrium. These aspects need to be illustrated to students.

Other considerations which need to be kept in mind include: learning styles and multiple ways to solve a problem; types of mistake with highest frequency; future database and intelligent data gathering. It is our desire that students make mistakes since this would provide us with the requisite data for future refinement of the classroom response system. How do quizzes predict outcome of major tests such as the midterm? If there is correlation between the two then there is utility to the quizzes, otherwise not. Feedback to each question individually is a key requirement. This can only be achieved with the help of an automated technological solution. The system should generate an individualized email for each student containing a summary and analysis of their results, in addition to one for instructors which allows them to take remedial steps for areas in which the student is lacking. Misconceptions have to be cleared at each step before moving on in a progressive manner. Apart from correlation to next levels of learning the present data also provides insights to many conceptual errors, answers and further questions. This feedback loop, which you see here, is essential since it allows for a continual upgrading, refinement and creation of an intelligent system. There is an essential utility for making intelligent mistakes and these need to be encouraged and not discouraged, as is usually the case in our context. We dealt with this subject in the book, "Running to nowhere." The idea is assist a student, much like a doctor

does to a patient, through prescriptive and remedial measures and not to penalize them. Without a proper feedback loop learning cannot be improved.



Quizzes VS Midterm (Average marks)



Topic 227

Roman English

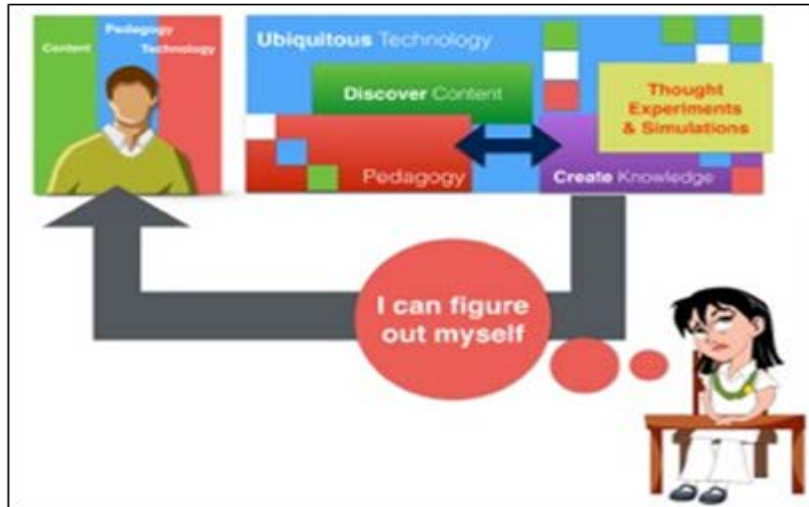
Another serious problem which I noted at our university was the lack of English skills among the students. A serious concern since the medium of curriculum being taught is naturally English. This led to many of our graduates not being looked at in a favorable light when in a working environment and a high rate of dropouts. In comparison to other major universities our student body was very socio-economically diverse, which further exacerbated the language issue. The differing proficiencies created a great hurdle for teacher in terms of teaching. Despite introducing three English proficiency courses, which are segregated into short, simple sections; some of the text or email message I receive from students are mostly in roman Urdu. What was the solution? Our thought was that we are not the only university which suffers from the same problem. If this disparity exists then a summative assessment system and lecture style teaching is not possible. Of all the main problems with language, speaking and writing are the main ones. Without these skills the students are unable to communicate. To alleviate this you would require a one tutor one learner model. A solution which is only possible with technology!

Looking around for solutions we looked at what MIT, in the States, which surprisingly were facing the same problem. How you may ask? About 30% to 40% of students are from China, who are brilliant in their fields but have issues with English. To make these students productive, a language software system was developed. Students registered on this system (Byki) showed remarkable results in terms of acquiring language proficiency. The system did not rely on traditional pedagogical methods. Seeing the system success we decided to purchase it. Apart from computers this system also has provision for mobile device connectivity. To test the system, I had my driver who is sixth standard dropout, to register on the system. The results

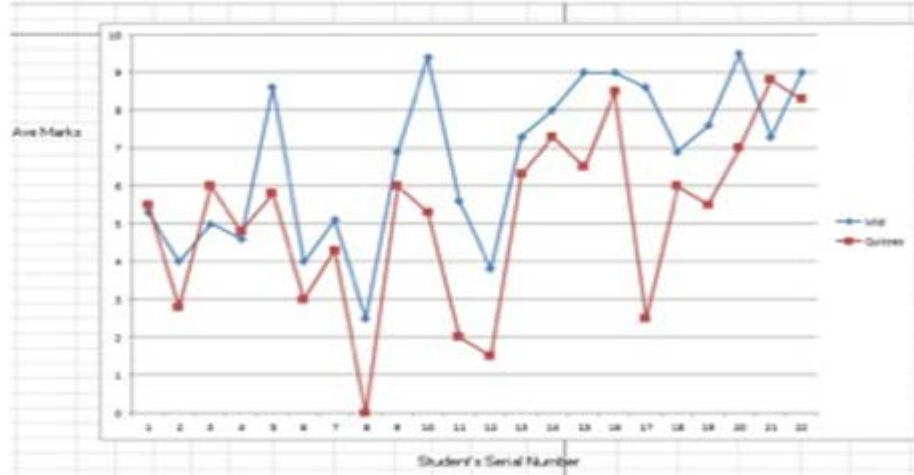
based on declarative acceleration methodology were truly remarkable. From the slides presented here you can gauge what a right blend of technology and pedagogy can achieve? These illustrate the five principles entailed in the system design.

With an annual fee of ten to twelve thousand dollars, which comes to around a dollar per students annually, we can now register our entire student body and resolve our serious language problem. Without technology this is impossible. We do not have enough trained teachers in Pakistan to teach English to all students who require this. To handle this we are trying to develop an indigenous system by reverse engineering the imported software. We are not entirely conversant in all the background pedagogy involved; since based on input by numerous neuroscientists, psychologists, linguists, pedagogists and other experts. We might not be able to replicate the entire thing but can mirror most of the processes. This can be rolled out to all students in the country, even has utility for those not formally studying. In this illustration you can see my driver updating me about his progress without the help of an instructor.

Here you can see some of the interfaces; the questions and choices provided. The beauty of the system is that, instead of isolated elements of language, it focuses on development of all four areas simultaneously; listening, speaking, reading and writing. An example of formative assessment based on interactive learning and incorporating instant feedback. We hope you would also be part of our future designers who will develop these systems; so that no child is left behind in terms of learning. We hope this course has provided you with the insights to do exactly this. With the course there will also be accompanying book which will greatly assist you in playing a part in provision of quality education to each and every child in Pakistan.



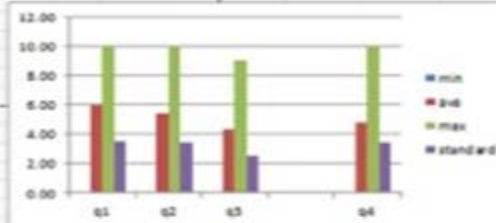
STUDENT WISE RELATION OF QUIZZES AND MIDS



ANALYSIS OF QUIZZES BEFORE MID AND MID EXAM.

	q1	q2	q3	q4
min	0.00	0.00	0.00	0.00
ave	5.95	5.45	4.95	4.82
max	10.00	10.00	9.00	10.00
standard	3.51	3.39	2.52	3.42

QUIZ DATA



	Q1	Q2	Q3	Q4
min	0.00	0.00	2.50	4.00
Ave	5.00	7.23	7.71	6.77
Max	10.00	10.00	10.00	9.00
Stdev	3.87	3.57	1.95	1.51

MID TERM DATA

